

**Researches
on the
Visual Organs
of the
Trilobites**

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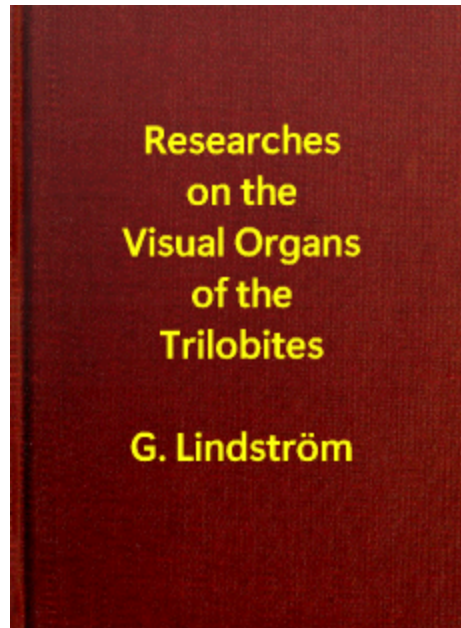
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Transcriber Note

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RESEARCHES

ON

THE VISUAL ORGANS

OF

THE TRILOBITES

BY

G. LINDSTRÖM.

WITH SIX PLATS

COMMUNICATED TO THE ROYAL SWEDISH ACADEMY OF
SCIENCES

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At the outset of this memoir it must be said, that it is in fact a joint work by the first discoverer of the remarkable hypostomic structure to be described, Herr G. LILJEVALL, and by myself. A considerable portion of this work belongs to him as he with great discernment searched for a large amount of the material, prepared it as to be fit for the many microscopic sections which he executed, and at last with his well known artistic skill drew the figures. My share, again, in the common work consists in the supervision of the figures, in the arrangement of the observations and details in a coherent form, in which suggestions and remarks of Herr LILJEVALL have been of great use, further to write the descriptions and to make all the necessary researches in literature, so whatever may be faulty in these respects must be laid to my charge.

September 12th 1900.

G. LINDSTRÖM.

INTRODUCTION.

It is well known amongst palæontologists that the hypostoma of the trilobites has been the particular object for research and description by a few authors, as BARRANDE, NOVÁK and BRÖGGER and some passing notes concerning it are given in the descriptive works of other authors. In the following pages reference is often made to them and I may here give a list of the chief works and the abbreviations with which they are cited below in this paper.

ANGELIN *Palæontologia Scandinavica* = ANG.

BARRANDE *Système silurien de Bohême* = BARR.

BRÖGGER *Die silurischen Etagen 2 & 3 im Christianiagebiet* = BR.
1.

—— *Die Ausbildung des Hypostoms bei einigen Asaphiden* BR.
II.

HALL, JAMES, *Palæontology of N. York*, vol. I, II, VII = H.

HOLM *Trilobiter ur Dalarnes graptolitskiffer* = HM I.

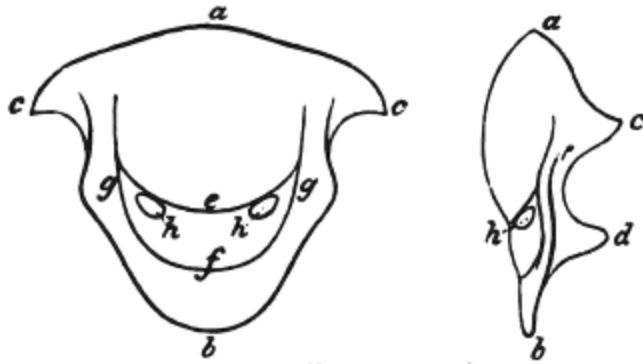
—— *Ostbaltische Trilobiten, Illæniden*, = HM II.

NOVÁK, *Studien an Hypostomen Böhmischer Trilobiten*. Three
different papers between 1884-1886. = N. II, N. III, N. IV.

SALTER *Monograph British Trilobites*.

SCHMIDT *Ostbaltische Trilobiten* I, II, IV. = SCHM.

WOODWARD, HENRY, *Carboniferous Trilobites*. Pal. Soc. 1883-84.



Hypostoma of Bronteus, front and side views.

a. anterior margin. *b.* posterior margin. *c.* anterior pair of wings. *d.* posterior pair of wings. *e.* anterior groove. *f.* posterior groove. *g.* lateral grooves, *h.* maculae.

Before proceeding further I shall give a general representation of the shape of the hypostoma. This enigmatic part of the trilobite skeleton resembles as to its outline a heraldic shield being broad at its anterior margin, gradually tapering towards the posterior extremity, which generally is acuminate or, as in the Asaphidæ, bifid. In a few, as Deiphon, Remopleurides, it is nearly square. Its exterior surface may consist of a single field or be divided transversally into two or three fields, according to the presence of one or two shallow grooves. These are transverse, curved forwards near the lateral margins and may be designated as the anterior groove and the posterior groove. Beside these grooves there are in some instances others, one on each of the lateral margins, parallel with these. On both sides of the anterior margin a protuberance, sometimes of considerable length, stands out. They are bent backwards towards the interior side of the hypostoma and in some genera, as Dysplanus, assume the shape of long, acuminate, halfways hollow horns. These protuberances were called »*ailles*» by BARRANDE and it is best to retain this term, in english »wings», in spite of its being not quite expressive. A smaller posterior pair lies nearer to the posterior margin and these wings are so much bent backwards, that they are seldom visible from the outside. They are prolongations from the »duplicature» or the narrow fold of the posterior margins round the interior side of the hypostoma.

For the rest the whole exterior surface is covered with terrace lines running concentrically and varying according to the different genera and even species. Many other genera, again, as *Acidaspis*, *Calymmene*, *Homalonotus*, *Dalmanites* etc., are devoid of them and instead granulated or smooth.

In the plurality of species there are two tiny patches or maculæ, sometimes elevated above the surrounding surface like tubercles and so they have also been called by some authors. But I have preferred to use the name »macula» for them, as the plurality does not form tubercles. They are generally entirely smooth and glossy and situated next to the anterior groove, either above it or in it, at a regular distance from each other and the lateral margins. They may form a sunk spot or, as commonly, an ovoid or elliptic area surrounded by a linear elevated border. Thus amongst the *Asaphidæ*. In others, again, (*Bronteus*, *Illænus* etc.) they form a moderately elevated tubercle. The direction in which they are oriented in relation to the longitudinal axis of the hypostoma is quite as variable. It is to the closer consideration of the nature of these macula that this memoir will be chiefly devoted. They have been delineated several times, but very seldom has any accurate attention been paid to them by previous authors. It is highly doubtful if BARRANDE ever alluded to them, when he in his »*Système Silurien de Bohême*» vol. I, p. 156, in describing the hypostoma, says: »Cette plaque bombée porte souvent des saillies et des empreintes creuses, dans le voisinage de la bouche. Leurs formes varient suivant les espèces mais en conservant toujours les caractères génériques. Nous les considérons comme les points d'attaches des muscles et des mâchoires.» In the numerous specific descriptions in his grand work there is never any mention made of the macula and in contemplating the figures and reading the explanations of these, we shall find that he meant something more comprehensive with his statement, to wit, the entire groove where these tubercular macula are situated.

In NOVÁK's third paper, p. 4, we see the figures of two *Phillipsiæ* with tubercles on the hypostoma and he says only that in the anterior groove there are two such symmetrically placed ... glandular intumescences.^[1]

[1] »In der Mittelfurche sitzen 2 kleine symmetrisch gelegene, nicht immer dentliche drüsenartige Anschwellungen.»

BRÖGGER again in his paper on the »Ausbildung des Hypostoms» devotes a page (p. 19) to explain the nature of these maculæ and regards them as the exterior marks of two muscular impressions or points of attachment on the inside for muscles which have attached the hypostoma to the walls of the glabella or the head. The real nature of these so variable tubercular maculæ is, however, a quite different one, as we hope to be able to demonstrate in describing all their varying shapes in several genera.

The interior surface of the hypostoma shows all the protuberances of the exterior surface (tubercular maculæ, spines and granulations) as shallow pits, the exterior grooves being on the contrary elevated ridges. For the rest the interior surface is smooth and its margins are posteriorly covered with a more or less broad duplicature, from the lateral margins of which the posterior wings project.

The manner in which LILJEVALL became aware of the interesting structural features of which I am going to give an account, is as follows. We were about to describe and delineate some new or not sufficiently known Upper Silurian trilobites from the island of Gotland. In order to draw a specimen of *Bronteus polyactin* ANGELIN, which is rarely found in the Wenlock and Ludlow strata of Gotland, LILJEVALL was preparing and cleaning its hypostoma. The shape of this hypostoma, as represented in [pl. II](#) fig. 22, is clypeiform, tapering towards the posterior margin, which is regularly tongue shaped. The anterior margin is faintly curved and at both sides elongated in a broad, pointed wing. Its exterior surface is divided in three transverse fields, defined through two shallow grooves, the anterior and largest field occupying more than the moiety, the median one is crescent-shaped and narrow, and the posterior one is nearly of the same size and confluent with the lateral borders of the hypostoma, which near the median transverse axis of the latter project in a blunt angle on each side. The surface, for the rest smooth, is covered by concentric, irregular terrace lines. On the inferior edge of the upper groove two small tubercles are situated, one on each side closer to the lateral margins than to the central axis. The size of their longitudinal axis amounts to 0,96 mm. in the shortest and to 1,07 mm. in the longest specimens. This axis is parallel with the groove and consequently oblique to the longitudinal axis of the hypostoma. They are oblong or ellipsoid, their inferior apices bluntly pointed or rounded, varying a little as seen in the three specimens figured, (Pl. II figs. 23, 24, 25). For

about two thirds their surface is perfectly smooth or rather glossy, and the lower third is covered with a compact accumulation of small granules. The extension of this peculiar accumulation varies, being somewhat larger in some specimens, and the limit which runs oblique towards the glossy part is more or less curved. Such a granule taken single is of the extremest minuteness, measuring 0,055 millimeter at the highest. Their outline is perfectly circular and semiglobose.

When LILJEVALL saw this group of granules he was at once struck with the perfect, outward similarity between them and the facets of the compound eyes on the upper surface of the head of this same species, and I fully concurred in the same view. It now became of importance to ascertain whether the intimate structure also corresponded in both, and microscopic sections were accordingly prepared. Although not so clear as in other species of this genus a vertical section of a cephalic eye (Pl. II fig. 21) shows a median row of black spots which compared with other sections (Pl. II fig. 9) must be the centres of the badly preserved lenses. This is confirmed by the horizontal section (Pl. II fig. 20) where the well known image of the polyedric facets appeared. In the same manner the granulated spot of the hypostoma shows in a vertical section even more clearly ([pl. II fig. 27](#)) the elongate, coherent lenses with a black centre and the polyedric form of the lenses in a horizontal section (Pl. II fig. 26). The assumption of a pair of small adventive eyes on the exterior side of the hypostoma became thus based on the clear evidence of the perfect structural agreement between them and the eyes of the head. But it was necessary to strengthen this evidence through extended researches on the hypostoma of as many trilobite genera as possible and also to study the intimate structure of the cephalic eyes in as many genera as could be accessible.

We shall now give first a general account of the structure of the eyes in the trilobites as far as we have been able to study them, and then proceed to describe the maculæ of the various genera, which we have observed.

With regard to the presence or non existence of visual organs, we must remark that a considerably greater number of genera than those which unanimously have been regarded as blind, also are devoid of eyes though they still by many authors are ranked amongst the oculate ones and we then had better first to make a review of them and thus to eliminate them from the number of the oculate species.

Blind Trilobites.

In the chaos of generic forms and in the great disagreement which prevails as to the systematizing of the trilobites of the Cambrian time, there is a thorough revision of them highly needed by a person having access not only to the literature, but also to the original specimens. It is almost impossible in the present state of things to tell with any degree of certainty how many well established genera had been living during that period. Hence the difficulty of fixing the systematic names of many specimens the visual organs of which are to be described.

My researches on the visual organs of the Cambrian trilobites are founded on the Angelinian Collection in the Swedish State Museum together with collections of foreign species, but also largely on the waste European and American literature, though we have to deplore the often occurring inexactitude of the figures, especially in the older works, and constructed or schematized figures in some of the newer ones, which give a quite false notion of the structure. There is no lack of figures to show how it ought to be, according to preconceived notions and, on the other hand, a great scarcity of representations, to show how it really is. In spite of all this there is a sufficiently great number of well established facts to demonstrate the organization of the Cambrian genera.

The trilobites of this division may be called blind only in so far as they have no eyes on the upper surface of the head, but they may have been provided with visual organs, though more imperfect, on the hypostoma as really seems to have been the case with some of them.

According to the structure of the head shield the blind trilobites may be subdivided into the following well characterized groups. These are:

I. Without facial ridge:

1) The Archæan Trilobites.

II. With facial ridge:

- 1) The Olenellidæ.
- 2) The Olenidæ and related.

I. Blind trilobites without facial ridge (= »eye lobe«).

Group 1. The Archæan Trilobites.

In these the head shield is in one piece without any facial suture and facial ridge, and without the least trace of anything that might be called a visual organ and they must consequently be considered as totally blind. In contradistinction to the following groups, excepting the oldest Olenellidæ, the head consists of only three parts, 1) glabella, and 2-3) the two fixed cheeks. These genera range from the oldest zone in which hitherto trilobites have been found, that of Olenellus (Holmia) Kjerulfi, to the zone of Paradoxides Forchhammeri, and some, as Agnostus, even continue as high in the Lower Silurian series as in the Brachiopod schists. Beside Agnostus the other genera are Conocoryphe (seven species in the Swedish Cambrium), Toxotis, Ctenocephalus (?), Elyx, Aneuacanthus, Conophrys and Microdiscus.

»Harpides» breviceps ANG., also belongs here. Anopocare of ANGELIN should also be regarded as one of this group. But it cannot be retained any longer because it is founded on two other, well known forms, being, according to LINNARSSON, Peltura scarabæoides (pl. 27 fig. 1, a in Pal. Scand.) and young specimens of Sphærophthalmus alatus (ibid. figs. 1 & 2).

It is remarkable that some of the Conocoryphidæ have an imperfect facial ridge, to be compared with the commenced one in Sao BARR. I (pl. 7 fig. 9). So the American Con. trilineata and reticulate. WALCOTT U. S. Geol. Survey 10th Rep. pt. I, pl. XCV f. 5 & 6. It is, as it were, arrested in the development and these adult trilobites had stopped, where the larva of Sao was proceeding in its second stage. They are the forerunners of the blind trilobites with facial suture, belonging to the third group. It needs scarcely be mentioned that the genera now enumerated have hardly anything in common, beside the general character of the head, and that real affinity

exists only between *Agnostus* and *Microdiscus*, and probably also between *Conocoryphe*, *Ctenocephalus* and *Elyx*.

BEECHER^[2] asserts that there is a suture in *Agnostus*, but in vain we have searched for it in numerous well preserved specimens and Dr HOLM also denies its presence. Nor are there any signs of closed up sutures, which also could not possibly be expected in so early a stage of evolution. It may then be taken as well settled that a fundamental character in these the oldest^[3] of all known trilobites is the total want of a facial suture and a compactness of the whole head shield which later is broken up in several parts through the disjunction of the free cheeks. In the Lower Silurian formation there are a few genera sharing in the same structure of the head shield, though by no means else related. Such are *Dindymene*, *Areia*, *Carmon* and in the U. Silurian *Cromus*. The two species forming BARRANDE'S genus *Dindymene* are so dissimilar that *Dind. Friderici Augusti* had better to be removed to a new genus and the first described one to be retained as type of the genus *Dindymene*. The same is the case with *Carmon*, where the type species *C. mutilus* is blind and without free cheeks while the other species, known only by its fixed cheeks and glabella is one of the *Olenidæ*.

[2] Nat. Classification of the Trilobites, p. 183.

[3] Oldest in that sense that they are the descendants of an archaic precambrian stock, the chief characteristics of which they have retained in the main unchanged and persisting long ages after the close of the Cambrian times, some, as *Agnostus*, continuing high up in the Lower Silurian.

II. Blind trilobites with facial ridge.

This large division embraces the second and third groups or, with a few exceptions, all the rest of the Cambrian trilobites on account of a feature in the cephalic sculpture common to them all, though widely different as to its first origin in both. What forms the prominent and common characteristic of these two groups is the presence of the *facial ridge*, which emanates from the basis or the front of the foremost segment of the glabella and in a great variety of different shapes continues backwards near to the posterior border of the head. It has received several names as *eye-line*, *palpebral lobe*, *ocular ridge*, *eye-lobe*, *ocular fillet* (MATTHEW). In German it is named *Augen-leiste*, in French *filet* (BARRANDE) and in Swedish *ögonlist*.^[4] Some

authors make a difference between the more narrow part, calling it eye line, and the thicker posterior node, which they name the palpebral lobe proper.

[4] That name is the most current amongst the swedish authors, together with »palpebrallob»; HOLM says *ögonlob* and *frontallob*.

As this peculiar ridge exists before any facial suture has made its appearance and separated the head shield in five parts, viz. the median glabellar part, the two fixed cheeks and the two free cheeks, and as it occurs in genera which never possessed any facial suture, and where no eye ever was formed, it is not adequate to call it an ocular ridge etc. the more so, as it, at least during a long series of genera succeeding each other, has had no connection whatever with any eye. I therefore propose to call it *facial ridge* (in swedish *faciallist*). It occurs on the head of almost all Cambrian trilobites, excepting the archaic ones, and it is retained in the later Cambrian *Peltura*, *Sphærophthalmus* etc., which have real, compound eyes, as well as in a few Lower Silurian genera as *Triarthrus*, *Pliomera*, [5] *Euloma*, in the Upper Silurian *Arethusina* and *Acidaspis* and most persisting in *Harpes*, ranging from the Lower Silurian Lower Red Orthoceratite Limestone into the middle Devonian beds.

[5] Pl. Törnquisti HOLM.

It presents itself in the most variable shapes, and as it in fact can be followed through its development in the oldest known species, it is suitable to begin its description together with the characteristics peculiar to the oldest or second group in this large subdivision, that of the *Olenellidæ*.

Group 1. The *Olenellidæ*.

In this group we have two families of different age, the older, less developed the *Olenellidæ* proper, and the younger the *Paradoxidæ*. The former consists of the genera *Olenellus*, *Holmia*, *Mesonacis* and *Schmidtia*, and we shall attend to them first. They have no facial suture [6] and consequently a tripartite head shield like the Archaic ones, no eyes, but there is that strongly developed and most characteristic facial ridge. As seen in *Olenellus Thompsoni* [7] the crescent like ridge starts as a direct outflow from the base of the first segment of the glabella, and is in direct continuation with it as an integral part. It is roundbacked, regularly curved

and at its starting point as broad as one of the posterior glabellar segments. It is regularly semilunate, tapers posteriorly and ends near the occipital segment. From the narrow second segment of the glabella, inside the just described larger ridge, a smaller ridge emanates, broader and flatter than the former, slightly curved, and ends between the third and fourth glabellar segment. Already in *Holmia Kjerulfi* the ridge is modified. There it is only one ridge, the anterior one, nearly as doubled through a shallow groove running along its back. The second one has dwindled away so as to be seen only as a narrow stripe near the occipital segment and ending outside this in a point. The anterior ridge, represented by HOLM as consisting of two nearly independent parts, is indeed in one piece, though its dorsal groove sometimes is deep, and it is with its total breadth joint to the first glabellar segment. Along its outer edge, where it lies close to the cheek, a narrow slit runs, and I suppose that it is to be considered as the first indication of the forming of a facial suture, which however does not reach longer than the ridge. As in *Paradoxides* it is so tight to the cheek, that there has been no place for an eye. It is no accidental break, its edges being too regular and unbroken in all specimens. This ridge is by far much shorter than in *Olenellus* and terminates opposite the third segment.

[6] BEECHER Natural Classification of the Trilobites, p. 191, pretends that in *Olenellus* and *Holmia* real sutures »in a condition of symphysis» occur. He seems to deny the facial sutures and to accept as »real sutures» the »internal sutures» described by HOLM in *Olenellus Kjerulfi*. It is highly doubtful if these interior elevated lines are to be regarded as sutures. They are indeed no sutures, but in reality elevated linear ridges, inclosing, as it were, narrow canals. The real sutures known, the facial sutures, never form elevated lines, be it on the outside of the head or on the inside. Probably these lines are derived from some now unknown interior organization and it may be fit to remind of the somewhat similar though more numerous linear canals of the branchiæ on the interior surface of the great head shield of *Apus*. (ZADDACH De Apodis cancriformis Anatome ... pl. II fig. 1) or what HUXLEY (Anatomy Invertebr. Animals p. 281) calls the convoluted »shell gland» in the carapace. A quite different structure is what I suppose to be the incipient facial suture in *Holmia*.

[7] WALCOTT in 10th Ann. Rept. U. S. Geol. Survey, pl. 82 f. 2.

Larva of *Olenellus asaphoides*, chiefly according to FORD and WALCOTT.



1. Hypothetical figure, based on fig. 2. represents a stage preceding the next. There are three pair of pleura and three segments of the rhachis.



2. Copy of WALCOTT's fig. 5 in »Bullet. 1886«, pl. XVII. Three pleura at right, two at left, the larger formed through fusion of the two hindmost in fig. 1.



3. Copy of WALCOTT's fig. 6: the second and third pleura have been united into one.



4. Hypothetical figure as a further development, following upon the stage represented in fig. 3. Fusion completed between all posterior plectra (2, 3, 4), thus forming a single large pleuron composed of the three mentioned.



5. Copy of FORD's fig. 1, pl. IV, Amer. Journ. Science 1877, enlarged to the same size as Walcott's. First sign of the pygidium.



6. Copy of FORD's fig. 2 (1877) enlarged. The pygidium has been added to the head shield.



7. Copy of FORD's fig. 3, enlarged.



8. Copy of FORD's fig. 1 page 251 in Am. Journ. Oct. 1881, enlarged.



9. Copy of FORD's fig. 2 page 251. Amer. Journ. Science 1881. Slightly enlarged.



10. Copy of FORD's figure 5, pl. IV, Amer. Journ. Sci. 1877, slightly enlarged.

Already LINNARSSON observed the second ridge as he tells in a paper where he describes *Olen. Kjerulfi* for the first time.^[8] It has been called an »ornamental» spine, but in the following we shall learn what it really is. This ridge in connection with its spine has not been observed in any other of the *Olenellidæ*, at least not in their adult stage.

^[8] Öfversigt Vet. Ak. Förhandl. 1871, tafl. XVI fig. 1.

Thanks to the discovery of very early larval stages of the American *Olenellus asaphoides*, which FORD^[9] and WALCOTT^[10] have described and figured, we can through combination of these decipher the development and signification of the facial ridges. To facilitate my explanation I here join a series of cuts from the earliest stage to the more developed, with addition of

two schematic stages to complete in a certain degree the series of WALCOTT and FORD.^[11] See figures [p. 13](#).

[9] In American Journ. of Science 1877 p. 265 and 1881 p. 250.

[10] Bullet. U. S. Geol. Survey N:o 30, 1886, pl. 17 fig. 5-6.—Tenth Annual Rept. U. S. Geol. Survey, 1888-89, printed 1890, plate 88, fig. 1, 1 a.

[11] In his paper on »The larval stages of Trilobites» p. 175 BEECHER gives a new figure (f. 6), original from FORD's collection, of the larva of *Ol. asaphoides*, but it is so sketchy that I cannot with certainty make out what it means. I cannot agree with him when he speaks of free cheeks and eyes in these and he is completely wrong when he says that the outer pair of spines belong to the free cheeks etc. (p. 176).

The [figure 1](#) has been hypothetically composed as a deduction from [fig. 2](#), which presupposes an earlier stage of development like that in [fig. 1](#), when there existed three or four pair of lateral appendages in the larva. This then consists of a central portion of five segments. The large anterior crescentlike segment does not, however, show any distinction between a central part and lateral appendages, it is nearly as large as the three next taken together and its backwards bent side horns embrace the next two posterior segments and attain with their narrow pointed tips the back of the fourth segment. The central portion consists of five segments, when the somewhat not well definite posterior marginal segment is taken in account. Each of these segments excepting the fifth one has lateral appendages, those of the second and third segment being quite as broad as the central part and bent backwards in a curve ending in a small pointed spine. The lateral appendages of the fourth segment are largest of all, more than double the length of the two next in front, triangular and standing out beyond the posterior border of the shell as a broad spine.

In [fig. 2](#) a great change has set in. There is no distinction between the lateral appendages of the third and fourth segments at left. These two have been fused together, they have united, so as to make the left triangular spine look larger than it was originally. The appendages of the right side are still in the same state as before. But this fusion of the lateral appendages also takes place in another direction, as shown in another specimen ([fig. 3](#)), so that the second and third appendages on both sides coalesce into one piece. Now it is easy to imagine that at last a complete fusion has set in between all lateral appendages and that instead of the original three on each side, there is only

one large piece, reaching beyond the shell as a broad spine, as represented in the hypothetical [figure 4](#).

In the [figure 3](#) a progressive change is also seen in the transformation of the first central segment. From occupying the whole foremost space of the shell it has been lessened in size, more distantiated from the anterior border of the shell and rounded off, more prominent and definite from the lateral appendages, which have become narrow, though of the same length as before. The two first central segments seem to have been united into one.

Between the hypothetical [figure 4](#) and [fig. 5](#), there is evidently a great lacuna, not yet filled up. In the interval of time the two appendages, which we saw in [fig. 4](#), have been much modified, the posterior one having lost so much in bulk, and the anterior being lengthened and stretching out beyond the border of the shell in a narrow spine alongside the posterior one. The central segmented part has now assumed a shape, which on comparing it with the following stages of development makes its true nature evident and that it indeed is what in the adult animal becomes the glabella. The meaning of the previous stages then also is easily understood. The central segmental piece in them is the glabella or we may, as BERNARD already has done, call it for rhachis and the side appendages for pleura, as this little larva represents the whole body of the future trilobite, and embodies all its parts *in nuce*. Through the great changes which these pleura undergo, it results, as we have seen, that two pair vanish, being incorporated with the large fourth pair and that only two rest for a while, the anterior one being the so much renowned frontal-lob or eye lobe and the second one the so called »ornamental spine», which in fact is a compound of the original second third and fourth pleura of the corresponding segments.

It is to be borne in mind that this larva, which represents the future trilobite in its earliest stage, is nothing but the head, or what in the adult takes the place of the head, and especially its dorsal surface and that it thus solely consists of the future head.

In the [figure 5](#) (FORD's fig. 1.) there are the first signs of the pygidium coming and in the [fig. 6](#) it is well developed, but the thorax is still non apparent. In the stage [fig. 6](#) both pleura have increased in length and the compound one also in bulk. They project with spiny points beside the beginning pygidium and the anterior pleura have united across the first

segment of the glabella through a narrow ridge, which seems to cut that segment in two. The sequence in the order of development or growth thus is first the head, then the pygidium and last the thorax. At least it is so in these the oldest of all trilobites. But in nearly all trilobites of which there are good data, the head is the part first developed.

Evidently a large hiatus exists between the stages represented in figs. [6](#) and [7](#) (FORD's figs. 2 & 3, 1877), in the latter of which the animal, though not adult, has had the thorax and pygidium added to the head. The modifications in the size of the pleura are the chief changes. The anterior pair is reduced and retired within the posterior border of the head forming a semilunar arch joined with the occipital ridge in an angular bend. The posterior pair is enlarged and its spine is by and by reduced (figs. [8](#), [9](#), [10](#)) till it quite disappears and only the wide semicircular field between the first pleuron and the glabella is left behind as a remnant of its dilated body. It is to be remarked, that while in the plurality of the adult Olenellidæ all traces of the spiny projections of the second pleuron have vanished, they are still retained in the adult *Holmia Kjerulfi*, though not in the American *Holmiæ*, and thus give it at the same time a more ancient and a more larval stamp. The shallow groove along the back of its first pleuron indicates strongly its pleural nature, as the thoracic pleura commonly are divided through such grooves. The same peculiarity is also observable in several of the American Olenellidæ. It is much the same with the posterior pleuron, the pleural nature of which is revealed through its spine, that is homologous and identical with the spiny terminations of the thoracic pleura. We have thus through the remarkable finds of FORD and WALCOTT combined received an explanation of the morphological origin and nature of the facial ridge, the so called eye lobe and found that it has nothing whatever of the character of a visual organ. But it must be borne in mind, that these developmental changes are peculiar only to the Olenellidæ, the origin of the facial ridge in the later trilobites is, as we shall see, quite a different one.

The Olenellidæ belong chiefly to the oldest of the Cambrian beds with trilobites, and none of them has as yet been found higher up.

Next in order of evolution we have the important tribe of the Paradoxidæ. These are preeminently distinguished from the Olenellidæ through the well developed facial suture, which without exception in them all runs outside the facial ridge and separates this from the free cheek.^[12] This is a great

step forwards in the evolution and establishes the fact, demonstrable also by other evidence, that the formation of the facial suture is subsequent to the appearance of the facial ridge. This preexisting ridge seems to have had no small influence on the development of the suture, it checked its progress from the front or from the sides toward the fixed cheeks and directed its course against the genal angles. It lay as a protection for the glabella against this dissection, causing the separating line to run along its outside.

[12] They have thus a quinquepartite cephalic shield, as the later trilobites.

This group consists of the genus *Paradoxides* proper, as well as of *Centroleura*, *Metadoxides* and *Hydrocephalus*, if this is an adult form and not the larva of an unknown *Paradoxides*. Perhaps such forms as »*Conocephalites*» *Emmrichi* BARRANDE, as well as *Anomocare limbatum*, *An. excavatum*, *Bathyriscus* and *Dolichometopus* may on account of the shape of their facial ridges be considered as related to the *Paradoxidæ*. But this must be left for coming researches to decide. Some American Cambrian forms also share in this characteristic and may upon closer inspection be ranged here. So *Zacanthoides*. In these as in the true *Paradoxidæ* the facial suture follows the ridge along its whole length, while in the trilobites of the third group the facial suture is in contact only with the posterior end of the ridge, the so called eye lobe. *Remopleurides* does not show characters, that as BEECHER thinks, could unite it with the *Paradoxidæ*. These are blind and *Remopleurides* has well developed eyes and an organization that gives it an isolated position in the system.

The facial ridge continues in a great variety of shapes, short or long, but always forming the fraction of a circle, always of nearly equal thickness, only slightly tapering towards one of the extremities, and always when in direct connection with the glabella, starting from the base of its foremost, largest segment. As a rule the ridge is more developed in the young or larval individuals, continuing from the glabella to near the posterior cephalic border in an uninterrupted arch^[13] quite as in several of the adult *Olenellidæ* of the oldest Cambrian. It can be taken as granted that its origin is the same as in the *Olenellidæ* though at present the only evidence at hand is the small larva of *Paradoxides oelandicus*, which LINNARSSON called *Parad. aculeatus*.^[14] In this we see the anterior pleuron or the facial ridge alone present, elongated downwards like the same pleuron in the figure 6 of *Olenellus asaphoides* and terminating like this in a fine spine stretching

backwards outside the posterior border. Of the second pair of pleura there is nothing to be seen. This must then have been aborted at an earlier stage than in the Olenellidæ.

[13] G. F. MATTHEW had before me, as I now find, pointed out this distinction in his memoir »Illustrations of the Fauna of St. John» N:o IV, p. 163. When he speaks of »the embryonic stage» in this and other passages he evidently means »larval stages», as the embryonic stages of necessity must remain unknown to us. (Later remark.)

[14] Om faunan i lagren med Paradoxides ölandicus (1877), p. 359, pl. 14, f. 11.

The connexion between the free cheeks and the middle part of the head has been very lax not only in the Paradoxidæ, but on the whole in nearly all Cambrian trilobites with free cheeks. When the free cheek is dissevered it shows no trace of the ridge, there is only a large scallop on the spot where it embraced the ridge. In the Paradoxidæ the rim of the indenture and the ridge are in so close contact that there is not the least place for an eye between them, as can be seen in the few specimens with a complete head. In all oculate trilobites again without any exception the facial suture separates that part of the eye which is the real visual organ with corneal facets, from the interior often elevated portion, opposite it, the so called palpebral lobe. The eye is *always* placed on the free cheek,^[15] the lobe again always on the fixed cheek of the head shield. No real eye exists without the palpebral lobe, and, on the other hand, that part of the facial ridge which later in the development changes to a palpebral lobe, occurs alone without any eye in a great number of Cambrian trilobites, and consequently these are blind and such is the case with the Paradoxidæ and a great number of the succeeding.

[15] Excepting in Harpes, which has no free cheeks.

There is not the least evidence to support the suggestion that the »ocular ridge» is homologous, with the eye of Apus^[16] and that the real crystal cones lay sunk beneath the surface in a »water sac». As we, for instance, in Peltura have an »ocular ridge» (= facial ridge *mihi*) on the fixed cheek and opposite its posterior extremity, the »eye lobe», a real eye with facets on the free cheek it is not likely to suppose that the »ocular ridge» nor the »eye lobe» ever functioned as a visual organ or that two widely different sorts of eyes were placed in closest vicinity opposite each other.

[16] BERNARD, The systematic position of the Trilobites. Qu. Journ. Geol. Soc. 1894, p. 411.

It may be worth while here to remind of the great similarities, whether analogous or homologous, in the formation of the superior surface of the head in the trilobites and the embryos and the newly hatched larva of *Limulus*. The latter have the head shield separated into five portions, partly through a facial suture which, as in *Peltura*, *Dalmanites* and others, forms a continuous line around the glabella. This suture divides the eye node in a similar way as in the trilobites, that is, sectioning it in two parts, of which one adheres to the central fixed cheeks and the other to the free cheeks.^[17] And to judge by the figures of KINGSLEY the former, the whitish moiety is the first developed and sometimes for a while quite alone as the facial ridge of the trilobites and probably also anterior to the suture, as this is not complete at this stage. This white node reminds of the small facial ridge in *Arionellus ceticephalus* BARR.

^[17] PACKARD, Development of *Limulus polyphemus*. Memoirs Boston Soc. Nat. H. vol. II, pl. V, fig. 25. Nothing is said about the exterior structure of the eyes.

DOHRN, Zur Embryologie und Morphologie des *Limulus Polyphemus*. Jenaische Zeitschrift 1871. A very good figure (pl. XIV, f. 4) shows clearly the two parts of the eyes, the interior one being larger.

KINGSLEY also (Devel. of *Limulus*, Journal of Morphology, vol. 7, 1892, pl. VI, fig. 34) has in the last larval stage the suture and the eye in two parts, of which one is white lying inside the suture and the eye proper, black, outside.

As KISHINOYA^[18] has already pointed out and as I have anticipated above the head of the most developed trilobites in their adult state, and the head of the larval *Limulus* consists of five parts, viz. 1) the glabella in the centre, 2 & 3) the fixed cheeks, 4 & 5) the free cheeks. An elevated ridge in the adult *Limulus* shows where the suture once lay and it is on the outside thereof that the eye of the adult is placed. What other authors call the ridge or the eye ridge in *Limulus*, KISHINOYA rightly names a suture.

^[18] Journal College of Science Tokio, vol. V, p. 53, 1892.

It was the renowned Swedish naturalist WAHLENBERG, who first recognized the importance of the facial suture, which he called »linea ocularis»,^[19] but to another Swedish palæontologist DALMAN^[20] the exact definition of this suture is due, to which he gave the name still in use. He expressly remarks that the suture crosses the ocular node and limits the outside of the »lobus palpebralis» and he makes a clear distinction between that lobe and the »tuberculi and eminentiæ oculares» (= facial ridge) of which he says that they are the more or less evident elevations situated in the *blind* Palæades

on the place of the eyes (in which he is wrong) »and which perhaps are an indication of such organs». But then he says doubtfully (p. 255) »tuberculorum ocularium veram naturam determinare haud ausi sumus, etsi oculorum formam sat bene exhibere videantur» and he adds concerning Paradoxides »oculi nulli, eorum loco autem tuberculi duo».

[19] Additamenta quædam ad petrificata telluris Suecana, in Acta Upsaliensia, vol VIII, 1821, p. 294.

[20] Vet.-Akad. Handl. 1826 p. 126.

Group 2. The Olenidæ and related families.

Next in the ascending order we have the largest group of trilobites in the Cambrian, of which the greatest part is formed by the Olenidæ. A facial ridge different in shape and different in development from that in the former group characterizes them. Unlike the semilunate ridge of the Olenellidæ it issues mostly from the front of the first segment of the glabella and goes generally backwards till it meets the facial suture. It is narrow and fine as a thread, but for the rest assumes a great variety of forms. It may be curved as a circle segment as in *Sao*, *Liostracus*, it may be long and straight, standing out in a right angle to the glabella as in *Eurycare*, it may be short and straight in an acute angle to the glabella as in *Parabolina* and so forth. In the same genus, as for instance *Olenus*, straight and curved ridges occur in the various species, and consequently the form of this sort of ridge cannot be used as a generic character. It moreover differs from the ridge in the Olenellidæ and the Paradoxidæ in having the posterior extremity widened as a tubercle or node, which commonly has been called the »palpebral lobe», while in the older groups the ridge is only at the most a little pointed in both ends or of equal thickness.

Some persistent archaic genera as *Conocoryphe* and *Elyx* inform us how this ridge has originated and how widely different it is from that in the Olenellidæ. On the surface of their head shield no ridge is visible, but on the interior side of it we behold, as the figures show ([Pl. VI, figs. 43 & 44](#)) on each side of the glabella a ramifying system as of the most minute vessels, which spread and cover the whole surface of the cheeks. This reticulation issues from a main trunk that goes nearly straight in an oblique direction from the foremost segment of the glabella and emits narrow branches forwards and backwards on its both sides. And these branches go on dividing till they occupy the entire interior surface of the fixed cheeks.

The figures representing this are taken from *Elyx laticeps* [fig. 43](#), and an almost similar image has been obtained from a species of *Conocoryphe* [fig. 44](#).

Now the question may be asked, what does this network of branching canals mean? I think we cannot gain a more plausible answer than that given us through the inspection of the soft parts that lie hidden behind the glabella of *Limulus* and its fixed cheeks, the only living crustacean which offers the greatest homologies with the head shield of the trilobites. Next below the shell of the glabella and the cheeks of *Limulus* there is a complicated stratum of muscles and behind this the heart and the great central circulatory system spreading from that centrum towards the sides of the head shield, the vessels being the more fine and minute, the more they are elongated from the centre. I now suppose that likewise in these trilobites the narrow and prominent glabella has been the receptacle not only of a strong mass of muscles, but also for the central part of the circulation. This centre has there sent out two relatively strong ducts or canals, one on each side of the glabella feeding all soft tissues inside and near the cheeks and probably also other important parts of the body. It is also clearly seen in *Elyx* how the orifice of the main trunc opens in the hollow of the glabellar apex.

In all crustaceans, as far as is now known, their more or less hard calcareous or chitinous skeleton is moulded by the subjacent tissues and glands. The sculpture of the surface consequently is an outcome of the fashioning procedures showing what has been going on below and what is still going on. If we then on the exterior surface of the head shield of the trilobites see the radiating lines in relief we must conclude that they are due to a subjacent system of almost capillary vessels causing hollow impressions on the inside and elevated ridges exteriorly. In *Elyx* the vessels have made no strong impression as to be visible on the outside. In a couple of American *Conocoryphæ* again (*C. reticulata* and *C. trilineata*) the main trunc of the vessels has formed a short faintly elevated ridge. In *Solenopleura* ([Pl. VI, f. 45](#)) the facial ridge has been fully developed, and by the casts of the inside it is clearly seen that the main trunc of the vessels makes the inside of the ridge and has been much incrassated. On the other hand the smaller branches issuing from it have been reduced in size and number, but are still visible. Thus, if I am right, the two, the facial ridge and

the main trunc, are in causal connection and the former has been moulded by the later, when it had gained in size sufficiently, and when a richer affluxion of nutritive fluid was directed backwards towards the point where the eye at last originated. The further changes in this ridge thus are related to the development of the eye. As far as I am aware, there are as yet no data to tell us whether the appendages or pleura of the glabella in the larval (or primordial) Olenellidæ have the same origin as the ridge of the Olenidæ or are homologous with it. Perhaps it may be so in respect to the foremost one, which also is persisting. It is, however, not known how the pleura in the olenellid glabella have been developed, whereas it is well ascertained by the growth of the glabella in *Sao*, *Liostracus* and others that such pleura never have been developed in them.

What the phylogenetic evolution has taught us concerning the formation of the facial ridge is confirmed by the ontogenetic development of individuals of some species as for instance *Sao hirsuta* BARR., *Liostracus* sp. BRÖG. and others.

This development can be followed in detail through the excellent figures of the larval stages of *Sao hirsuta* which BARRANDE has given in his magnificent work, vol. I, pl. 7. In the *first* stage figured (figs. 1 d-e) the whole animal consists only of the head shield which is completely smooth, the glabella scarcely segmented, no facial ridge, no facial suture. In the *second* stage, according to me (figs. 2-4 a), the glabella has become distinctly segmented, and the pygidium and partly the thorax have been added. In the *third* stage, 4 c, d-9, one small narrow string exits from each side of the front of the glabella, making the first faint beginning of the facial ridge. They form a right angle with the glabella. In continued growth they become by degrees a little more curved (figs. 5 c, 6 b) and the lengthened ridge bends parallel to the outer lateral margin of the head (fig. 9 b). In fig. 9 d it has become so far complete that it reaches nearly back to the posterior cephalic margin, but is still of the same narrowness all along. In the *fourth* stage, in a specimen (fig. 10 a) of 3 mm. in length, the facial suture makes its first appearance, setting in from the posterior margin of the head and meeting in its forward progress the posterior extremity of the facial ridge which now begins to swell out. It seems that both have a mutual influence on each other, the suture being deviated from a straight course^[21] to take a bend outside the ridge, and the extremity of the ridge again at this contact to

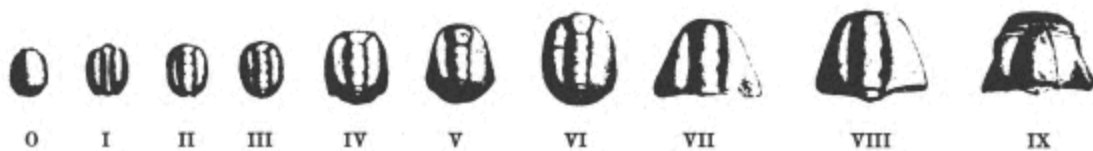
increase in size so as to form the elongated tubercle, often called palpebral lobe (fig. 14 b). This now augments in the same rate as the whole body. The characters of the four stages of the development of the larva consequently are: 1:o The archaic stage, only head shield with ridgelike glabella. 2:o The coming and growth of the pygidium and the complete segmentation of the glabella. 3:o The coming and growth of the facial ridge. 4:o The coming and development of the facial suture.

[21] In Trinucleus and Ampyx where there is no facial ridge, the suture has a straight direction along the lateral margins.

BARRANDE regards the whole ridge as a prolongation of the eye and the tubercle at its posterior extremity as the eye itself.^[22] But, again, in page 399 he says »Sa surface (of the eye) est toujours mal conservée, pour nous permettre de voir si elle était réticulée.» And he also confounds the ridge, »filet» as he names it, with the eye itself, and the first faint beginnings of this ridge in his fourth stage he considers as the eye.^[23] By a partly schematical figure of the free cheek (fig. 25) he places the eye on this cheek, and in the same manner in fig. 29, »restaurée d'après divers fragments» he figures a reticulated surface of the eye on the free cheek, outside the tubercle. I have sought for a reticulated surface on sufficiently good specimens, but never found any, and I must consider Sao as one of the blind genera. BARRANDE himself also in the table on the eyes of the trilobites places Sao in the group with »Surface visuelle inconnue», p. 131.

[22] Page 383 »l'œil argue est prolongé par un filet en relief, vers le front de la glabelle».

[23] p. 389.



(0 is new, added here to supplement fig. I.)

BRÖGGER has also succeeded^[24] in finding a series of small larva, which he considers as belonging to a species of Liostracus. As the figures drawn by BRÖGGER twenty five years ago may now be very little known, I here reproduce them with the kind permission of the Editors of »Geologiska Föreningens Förhandlingar», where they were published in 1875. This development proceeds nearly upon the same plan as in Sao. The first stage,

however, (I) seems to be much earlier than any of Sao, the rhachis or glabella consisting of an unsegmented ridge of more primitive appearance. Before this first stage of BRÖGGER's larva a still older phase of development can be imagined, (0) a simple rounded, smooth head shield without any indication of a glabella at all. This stage might correspond to the head of certain species of Agnostus, as *A. glandiformis*, *A. nudus*, which have no glabella. This stage 0 is also valid for the larva of Sao. Several stages are evidently wanting between I and II, in the later of which the thickened glabella is divided in four segments. In III we have six segments, all these three stages consisting only of the ovate head with narrow fixed cheeks. In IV the pygidium has been added to the primitive head, but the segments of the glabella have been reduced to four and in V slightly altered in shape. In VI, again, we see the head with five glabellar segments and scarcely the first sign of the facial ridge. Between VI and VII there must be links missing, as the change can not be so abrupt, and likewise between VIII and IX as in VIII there are still no free cheeks nor any facial ridge. This interesting discovery of BRÖGGER confirms, together with those of several other authors^[25], the supposition that the development of the later Cambrian and older Silurian forms is a quite different one from that of the Olenellidæ and the Paradoxidæ. They have a rhachis, but no pleura proper, as the single facial ridge has a quite different signification and appears at a comparatively much later stage than the facial ridge of the Olenellidæ, which is present from the earliest stages known.^[26]

[24] Fossiler fra Öxna og Klettna, Geol. För. Förhandl. 1875, p. 572, pl. 25, fig. I-X.

[25] Foremost among these stands MATTHEW in »Illustrations of the Fauna of St. John» IV, where he, p. 143, pl. II figs. 1 f. etc., describes a few stages very like those given by BRÖGGER, so the glabella as an unsegmented, narrow ridge etc.

[26] A deviating form of the ridges is shown by »Liostracus» tener HARTT, Acad. Geol. 2d Ed. p. 652 (see also MATTHEW Illustrations of the Fauna of St. John [1887] p. 132 p. 1 f. 3 a-3 c). Beside the usual facial ridge there is a second pair of ridges between the first and the glabella, arching in an opposite direction. See also WALCOTT, Bulletin U. S. Geol. Survey, N:o 10, 1884 pl. V, figs 6, 6 a, 6 b, new figures and copy of HARTT's description.

This form of facial ridge, although prevalent in the middle and later Cambrian times, dates back so far that genera coeval with *Holmia*, viz. *Ellipsocephalus* and *Arionellus* show it along the facial suture. This early occurrence of the ridge coeval with the less developed Olenellidæ leads to

the assumption of a different origin of these trilobites as a branch, which already far back in the oldest Cambrian or precambrian times had deviated from a common ancestor and I have tried to give a view in tables of these two coordinated lines of evolution further on, at pages 24 and 25. The most remarkable genera which, as far as I have found, belong to this fourth group are *Ellipsocephalus*, *Arionellus*, *Liostracus*, *Olenus*, *Leptoplastus*, *Parabolina*, *Corynexochus*, *Parabolinella*, *Sao*, *Ptychoparia*, *Doropyge*, *Oryctocephalus* etc.

I have not been able with an absolute degree of certainty to recognize whether some of these genera now enumerated, have been oculate or blind like *Sao*, like the *Paradoxidæ* and similar. The precarious state of preservation prevents all definite conclusions in that respect. It seems, however, that the evidence gathered through the examination of numerous specimens rather points in a negative direction. As the free cheeks in these old Cambrian trilobites have been in a very loose connection with the fixed cheeks and generally deciduous, contrary to the condition in the Silurian ones, it is in many instances very difficult to tell whether species with facial ridge, especially those from the earlier *Olenus* schists have been blind or provided with eyes.

The order of succession of the genera in the Swedish uppermost Cambrian, in the Olenid slates is, according to S. A. TULLBERG'S researches^[27] at Andrarum in Scania as follows:

1. *Parabolina* (oldest division of the *Olenus* slates).
Olenus.
Liostracus.
2. *Eurycare*.
Leptoplastus.
3. *Peltura*.
Sphærophthalmus.
Ctenopyge.
4. *Cyclognathus* (uppermost).
Acerocare.

^[27] Om *Agnostus*arterna vid Andrarum.

It is already in the second division that we find the earliest oculate genus in *Eurycare* and in the higher strata. *Sphærophthalmus* and *Ctenopyge* with

enormous hemispherical eye balls, while *Olenus* and *Parabolina*, their earliest predecessors, probably were blind. We have not amongst hundreds of specimens of these genera found a single specimen showing an ocular globe covered with facets. *Olenus* and the nearly related genus *Parabolina* are found in innumerable specimens in the thinly laminated alumschists of Scania and other provinces of Sweden. But rarely a perfect head shield, or nearly so, is to be found with the free cheeks in place. If so, the semicircular scallop in the free cheek is entirely filled up by the posterior lobe of the facial ridge and there is no place left for any eye.—If we, again, find a non compressed glabella and fixed cheeks likewise, the facial lobe (-»eye-lobe») is in some elevated so much as to leave a little space between it and the scallop of the free cheek, which space must have been an empty lacuna if not filled up by an eye ball. But it may also be that the free cheek has been somewhat put out of its order and that consequently some space has been left between it and the free cheek. It is quite as much with older genera, as *Solenopleura* especially, in which the posterior lobe of the facial ridge (*vulgo* »eye lobe») has attained a great development, and in which one just could expect to find a sphærical eye resting between the elevated lobe and the scallop. The elevated rim of this scallop does not in fact constitute a proof for its having clasped an eye as the elevated scallop in *Sphærophthalmus* did. In *Paradoxides*, again, in *Ellipsocephalus*, where there is absolutely no trace of an eye ever having been present, whenever you succeed to find the free cheek in juxtaposition with the fixed cheek it is evident that the elevation of the scallop rim is due to the impact of the posterior lobe of the facial ridge. In many specimens of *Dolichometopus* and *Corynexochus* etc. no free cheeks have ever been found and to judge by the shape of the facial ridge it may be concluded that these also were deprived of eyes.

On seeing this great number of trilobites, that on account of their organization must be considered as blind, the first suggestion that strikes the mind, is that they must have lived in abyssal depths of the Cambrian sea, where the most intensive darkness prevailed. Nor does the nature of the strata contradict such an assumption as evidently this fine sediment must have been deposited far beyond the reach of the influence of the wave motion, a depth amounting to more than a thousand metres as now calculated, for else it could not have preserved unbroken such delicate parts of organisms as that free cheek of *Ctenopyge pecten* with its extraordinary

long and delicate horn, figured on plate III fig. 27 and many others. But it seems incredible that such a state of things should have prevailed during the deposition of all the Cambrian strata, although they in Sweden amount to only 160 feet in thickness, according to the evaluation of S. A. TULLBERG, not considering what has been lost through denudation. The length and duration in time can in this instance not be measured by the thickness of the beds, but by the great changes in the faunas which there have succeeded one another. The physical conditions, to judge by the composition of the rocks, seem in the main to have lasted during immeasurable periods and still the fauna has changed in no little degree. During that enormous length of time, embracing in Sweden eight well separated periods, there must, however, have been minor changes in the conditions of depth and consequently in the nature of the depositions. To a certain extent the physical agents must have influenced the organization of the animals, but not essentially. There are sure evidences of another factor being the chief agent and that is the evolution.

We have from the lowest Olenellus beds to the lowest Lower Silurian strata followed the clear traces of the changes from the eyeless Olenellidæ past the Paradoxidæ with the facial suture to the great tribe of the Olenidæ in which the facial ridge with its lobe is so prominent, and to the oculate genera at the top of the formation.

As stated above there seems to be evidence enough for accepting two different lines of evolution in these the oldest trilobites. For the oldest, Olenellus is the type and for the second Sao may be taken as the representative.

These two evolutionary lines could be represented with their phases in chronological succession as follows:

A. The Olenellus line.

1. The trilobite consisting only of the head with rhachis and pleuræ, no sutures, no ridge, no eyes. Adult forms precambrian. Corresponding larval forms (Olenellus asaphoides) in the Lowest Cambrian. Still older and simpler forms may be presupposed as preceding these.

2. Semilunate facial ridge on the tripartite head-shield, no sutures, no eyes, but a hypostoma fused to the rostrum of the head and provided with maculæ. The adult animal with thorax and pygidium in the oldest Cambrium, the Olenellus zone. Genera: Olenellus, Holmia (with beginning suture), Mesonacis, Schmidtia.

3. The head quinquepartite with facial suture, short semilunate ridge in the adult, a long ridge in the young, no eyes. The genera occur in the Paradoxides-zones of the Cambrian formation. These genera are Paradoxides, Centropleura, Metadoxides, Hydrocephalus, which are direct descendants of the Olenellidæ.

This line of evolution, in which the species never acquired eyes, was extinct in the sixth zone of the Swedish Cambrian, Centropleura Lovéni being the last.

B. The Sao line.

I call this so because we see in the development of Sao, as represented by BARRANDE, most of the different' phases in evolution, which the ridge bearing trilobites of this second group have experienced during the Cambrian times.

I.

The body consists only of the head, neither pygidium, nor thorax yet developed.

1. *Psilocephali* (ψιλος, bald), primeval, precambrian, adult stage not found, supposed to be like the head shield of Agnostus glandiformis and A. biplicatus quite round and bare without any glabella nor sutures. The corresponding stage in development also wanted.

2. *Glabellate*. The entire animal only head with a mesial ridge, which at first is entire or unsegmented and later is metamorphosed into the segmented glabella.

a. *unsegmented.*, Precambrian, adult form unknown, supposed to be like the first larval stage of Liostracus according to the figure I of BRÖGGER and also as the head of Agnostus parvifrons LINNARSSON. Corresponding larval stages Liostracus (I).

b. *segmented.* Probably precambrian, not represented as adult in the Cambrian strata. Supposed to be like the larval forms II and III of BRÖGGER's Liostracus, consisting of the round or oval head shield and the segmented glabella. Also BARRANDE's first stage of Sao (BARR., fig. 1 pl. 7).

As a transition to the next phase forms may be imagined having head and the beginning pygidium, nearly as Agnostus atavus, minus the thorax. The corresponding larval stages are Liostracus (BRÖGGER's figures IV, V, VI) and Sao (BARRANDE, pl. 7 fig. 2-4 a, b). Nearly so, though the pygidium is more developed, are the larva of Agn. bibullatus and Agn. nudus (BARR., pl. 49) both without thorax.

II.

These have the three chief parts of the body developed.

3. *The facial ridge.* Cambrian, adult with glabella and facial ridges, short or long, emanating from the top of the glabella, thorax and pygidium. Several American Conocoryphæ. Corresponding larval stage, Sao, stage 3, BARR. pl. 7 figs 4 d-9.

4. *Suture.* A suture dividing the fixed cheeks in two pair viz. two fixed cheeks and two free cheeks. Fully developed facial ridge. Oldest known adult forms are Ellipsocephalus, Arionellus, already in the Lowest Cambrian, the Olenellus beds, what presupposes a long, antecedent lineage far back in the precambrian times. Corresponding larval stage in Sao, BARR. pl. 7 figs 10-13. The plurality of Cambrian trilobites belong here. An intermediate stage leading to the next is seen in such forms in which the ridge posteriorly is widened into the »eye lobe«, which rests in the scallop of the free cheek. So it is in Liostracus and many others besides. Solenopleura possibly oculate.

III.

5. Globular eyes. Cambrian, in the youngest zone, the Olenus schists. The oldest at present known oculate trilobite Eurycare is found in the second division of these schists.

In the lowest Lower Silurian division, the Ceratopyge limestone, Euloma and Ceratopyge occur as the last survivors of the blind, partially ridge bearing genera. The Trinucleidæ and Ampyx belong to another group of trilobites.

Even among other exclusively Lower Silurian genera, in which the plurality of the species is oculate, there are species entirely blind. So with Illænus, in which genus Dr HOLM has not found any eye in Ill. Angelini, I. leptopleura and Ill. cæcus. The free cheek in these three species is much narrow, as the facial suture lies near the margin of the head.

The eyes of the Trilobites.

If, as is probable, to judge by the conformity of their cornea with that of recent crustacea, the trilobites like these were provided with crystalline cones beneath the corneal lenses or facets, only the latter have been preserved in a fossil state. Although the crystalline cones in consequence of their solid consistence might have been petrified as well as the cornea, they must, imbedded as these tiny cones lie, entirely wrapped up in delicate tissues, fall away and be lost, when the dissolution of the dead body had set in. Consequently the curious appendages on the inferior side of the lenses in *Dalmanites vulgaris* ([Pl. III f. 50](#)) or *Phacops quadrilineata* ([Pl. V fig. 38](#)) can noways be considered as belonging to the original structure of the eye, apart from their great dissimilarity with anything appertaining to the eyes of the Arthropoda. The cornea on the contrary cohered with the integument of the body, and it has been well preserved in a great number of trilobites.

I subdivide the trilobites in respect to the form of their eyes in the following manner:

I. Genera with compound eyes.

1. With prismatic plano-convex cornea facets.

Acidaspis, as a transitional form to the next group.

Asaphus.

Bumastus.

Cyphaspis.

Dysplanus.

Encrinurus.

Illænus.

Megalaspis.

Nileus.

Phillipsia.

Niobe.

Proetus.
Ptychopyge.
Symphysurus.

2. *With round or biconvex transversally elongate lenses.*

Acerocare.
Bronteus.
Chirurus.
Ctenopyge.
Cyrtometopus.
Eurycare.
Peltura.
Sphærophthalmus.

II. Genera with aggregate eyes of biconvex lenses.

Acaste.
Chasmops.
Dalmanites.
Phacops.

III. Genera with isolated eyes, one or several stemmata at the extremity of a straight facial ridge.

Harpes.
Harpides.
(?) Trinucleus in the larval state.

Only a few authors have before now occupied themselves with the intimate structure of the trilobite eye. PACKARD gave in 1880, in the »American Naturalist« a note on the structure of the eye of trilobites (p. 503). There are some rough and inexact sketches of the eyes of *Limulus* and *Asaphus*, and although he seems to have known the beautiful researches of GRENACHER he still »claims that the trilobite eye was organized on the same plan as

Limulus». This statement is altogether wrong, and as I hope to show the trilobites have had eyes entirely different from that of Limulus and instead agreeing with those of the Isopoda and perhaps also with a few other Crustacea. In 1889 J. M. CLARKE published an account^[28] on the »Structure and development of the visual area in the trilobite Phacops rana GREEN». The aggregate eye described by him are of the type forming my third group. His holochroal division embraces my first and second groups and the schizochroal my third.

^[28] Quart. Journ. of Morphology, vol. II p. 253.

The latest contribution to the knowledge of these eyes is found in EXNER's »Physiologie der facettirten Augen von Krebsen and Insecten», 1891, where he gives good figures of the lenses of Phacops fecundus, pl. II figs 18, 19. He says that the palpable difference in the structure of these eyes and those of Limulus point to a change in the function of these eyes.

I. Compound eyes.

1. Eyes with prismatic, plano-convex lenses.

A pellucid, smooth and glossy integument, a direct continuation of the common test of the body covers the corneal lenses, quite as is the case in so many of the recent crustacea. In the plurality it is, however, difficult to discern the lenses from the outside.

The lenses, as seen in a vertical section of the eye of *Asaphus expansus*, ([pl. I fig. 12](#)), are columnar prisms, like the pillars of basalt, attaining a length of 0,2 mm and at the point where the eye joins the test of 0,3 mm. At their interior extremity they have a breadth of 0,066 mm. On that point the surface is convex and at the exterior surface plane. They are closely packed and in a transverse section resemble a pavement of regular hexaeders. But they also assume other shapes and become rhombs or even quadrates, as seen in a specimen of *Asaphus fallax* ([pl. I fig. 18](#)), where the hexaeders and quadrates lie side by side without transitional forms. As a rule the lenses become more and more irregular in the vicinity of the surrounding frame or near the suture, nearly blotted out, as it were, and without any definite border line mingled with the confused, spongy mass that like a belt or a frame surrounds the eye in *Asaphus* and is sharply limited from the other part of the free cheek. This remarkable zone which is almost only present amongst the Asaphidæ (*Asaphus*, *Megalaspis*, *Ptychopyge*, *Isotelus*) retains in a confused manner somewhat of the prismatic structure of the eye as shown in the section ([Pl. I, fig. 11, b](#)). The eye of *Bumastus* also is environed by a similar zone, with a structure like that of the eye ([Pl. II fig 35, 41](#)).

In an undetermined species of *Asaphus* the lenses, although somewhat apart, are of an elongated hexaedric outline, which passes into a regular circular one farther away and on the surface of the eye they are slightly convex ([Pl. I figs 27-29](#)). In other genera belonging to this group the shape of the lenses are like those of *Asaphus*, so for instance in *Illænus* (*I. chiron*

and I. Esmarki) and in Niobe. In *Dysplanus centrotus* they are shorter and broad, and their interior or lower surface strongly convex. It is likewise so in *Nileus*, where *Nileus armadillo* has an exceedingly thick exterior integument above the lenses. Such an integument has in a still higher degree increased in *Bumastus sulcatus* so as to exceed in thickness the stratum of the corneal prisms and it may in fact be doubted if the eyes of this species ever were able to function as visual organs. *Proetus* nearly resembles *Bumastus* in the thickness of the integument covering the prismatic lenses, which are interiorly convex, with a diameter of 0,03 mm.

In all genera belonging to this group a horizontal section gives the image of the hexaeders as in *Asaphus* with some change to squares or rhombs.

In scrutinizing a horizontal, somewhat extensive section of an eye in this group of trilobites, it will be perceived, as for instance in the figures ([Pl. V fig. 16, 22](#)) that the regular and evidently homogenous and intact prismatic lenses by and by have been altered and in a part of the section, a little distant from the intact ones disintegrated in their interior, showing various aspects of alteration. I cannot but think that this is a destruction which has set in long after the fossilization. It has revealed certain states of the intimate structure, certain delicate details, that now with an astonishing regularity come in sight and probably also lie hidden in the intact prisms. In the specimens of *Asaphus*, which we have studied, the alteration has taken the shape of a concentric stratification forming the body of the prisms, which is well discernible in a horizontal section, but not easy to detect in the longitudinal one ([Pl. I figs 9-10, 11](#)). It is likewise so in *Niobe*. In the other genera again the decomposition makes the prisms look like empty tubes in which a few irregular traverses and trabecular remains of their solid mass radiate towards the interior. They thus assume the aspect of a composite coral with its septa in the calicles ([Pl. VI fig. 31](#)). This is also evident in *Nileus palpebrosus* and *Dysplanus*.

2. Eyes with biconvex lenses.

The surface of the eye is, as in *Chirurus glaber* ANG., a mass of contiguous hemispherical lenses, probably once covered with a membrane, as is still to be seen in well preserved specimens of *Bronteus laticauda*. Both in

Chirurus and Bronteus the lenses seen vertically are globular and ordinated beside each other either continuous or separated only through a faint dividing line. In a horizontal section passing right through the point of contact they show the common hexaedral shape and when somewhat corroded the interior radiate structure also comes forth, the radii directed towards a little black point in the centre. The lenses of the Brontei have the same stellate structure as in Bumastus ([Pl. II fig. 7](#)). In Cyrtometopus the lenses are in size the fourth of those in Chirurus and they form an extremely thin stratum in strongest contrast with the adjoining cheek, which surpasses them more than six times in thickness ([Pl. III fig. 19](#)). The lenses of Cyrtometopus are more flattened and irregular than in the former genera. The free cheek around the eyes does not form a border zone, somewhat imitating the eye structure as in Asaphus, but is more compact, composed of vertical elements which give to the test of the trilobites in general a tendency to split up in vertical prisms.

Of a peculiar interest are the eyes in the oldest of all oculate trilobites, at present with certainty known, Eurycare, Peltura, Sphærophthalmus and Ctenopyge.

Of these genera Eurycare is the oldest (see table [p. 22](#)). Amongst the many free and detached cheeks only a single, very little one has been found with the eye ball fixed. It seems to be of the same structure as in Sphærophthalmus. In Sphærophthalmus and Ctenopyge the eye globes are enormous, considering the size of the cheek in which they are set and occupy more than a third of the length of the free cheek ([Pl. III f. 26, 31](#)). They are hemispheric, blackish and glossy, more so in the former genus. The spheroidal lenses, projecting on the surface, are in Ctenopyge larger near the facial suture and small at the opposite side where the eye is fixed in the free cheek. For the rest, in both genera ([Pl. III fig. 34](#)) the lenses form a thin stratum, where they in a vertical section lie elongated, flattened and biconvex, slightly joined with each other at the point of contact. The fine form which they exhibit reminds of the lenses of Sphæroma.^[29] They are in diameter thrice as long as they are high. Seen in a horizontal section passing through the point of contact they show hexaeders with a curiously jagged outline ([Pl. III fig. 33](#)).

[29] BELLONCI Atti dei Lincei. Memorie, vol. X, 1881, Sphæroma, pl. II fig. 11.

Peltura which is coeval with these, has a narrow semiglobose visual field ([Pl. III figs 35-41](#)), the superior surface of which is quite smooth and evenly rounded. On its interior side there stand out, somewhat distantiated, in a low relief semiglobular facets, quite as regular incrassations of the cornea, thus not forming free lenses, but rather reminding of the for the rest differently formed quasi-lenses of Limulus. In a vertical section they appear as the inferior moiety of real ovate lenses ([Pl. III f. 40-41](#)).

The much younger Acerocare has a similar cornea. A very little specimen, the head of scarcely more than one millim. in length, retains both eyes, of which one shows the slightly convex lenses and the other a cast of the interior side as in Peltura. These both genera should in consequence of their peculiar limuloid cornea be ranged for themselves apart from the real lenticulate genera, but any material sufficient for doing this properly, is at present not at hand.

II. Aggregate eyes.

These are found solely in the family of the Phacopidæ, unless the Lichadidæ were also provided with this sort of eyes, but we have had no opportunity to study them. It seems, however, not likely that they had aggregate eyes. BARRANDE has represented them quite as finely reticulate as the eyes of any Asaphid. We have sectioned and figured the eyes of *Dalmanites vulgaris* and *D. obtusus* from the Silurian of Gotland and found that these have truly aggregate eyes, each consisting of a regular biconvex lens, lying enclosed in a socket of its own and covered by a cornea of its own. The distance between the eyes is much variable and in a few instances they are nearly contiguous. Extremes are seen on [pl. III figures 43, 47](#). The lenses are comparatively large, and have always had a covering membrane, though this in many instances has been lost. This membrane which is an immediate continuation of the general integument of the body covers the lenses all round their superior moiety. In its prolongation downwards between the lenses ([Pl. VI fig. 3, 4](#)) it is free from the contact with them and hangs alongside and around much incrassated, so as to take in a section a lengthened lancet like shape. It lies thus alongside the other interstitial test, and is like this perforated by longitudinal canals. In a horizontal section taken a little below the surface it encircles the lens as a wall like ring ([Pl. VI fig. 1, 2](#)). In a vertical section the lenses lie in direct contact with the cheek without any intervening zone and the cheek has the structure so common amongst the trilobites, being perforated by vertical tubes going straight down from the surface ([Pl. III fig. 44](#), [Pl. VI f. 5](#)).

In *Dalm. vulgaris* and also in *Phacops quadrilineata* there is as already before mentioned a peculiar structure beneath the lenses, consisting of narrow, threadlike, straight lines, twice as long as the lenses ([Pl. III 49-50](#), [pl. V fig. 38](#)). In a horizontal section they are found to be irregular prisms closely packed. It can not be any structure peculiar to the eyes or the lenses, rather some parasitic growth added since the death of the animal. The lenses are in several specimens composed of clear calcareous spar. In others again they have been filled with a dark muddy calcareous rock excepting in the lower moiety where there is left a residue of the white spar, having in all

lenses assumed a regular shape which I consider as organic ([Pl. VI fig. 5](#)). This spar covers the whole bottom and its upper rim is incrassated and bent inwards. In horizontal sections this residue is a whitish ring close inside the interior ring wall ([Pl. VI f. 2](#)). I would suggest that this curious conformation is due to the original structure of the lens, supposing that it in these crustaceans has been built upon the same plan as in several other Arthropoda. In *Cymothoa*^[30] and in *Sphæroma*^[31] for instance the lenses are built up of thin strata, which are parallel with the convex outside, so that on the inferior surface of the lens they are arched downwards and on the superior side upwards, being not strictly concentric. In the spiders they are constructed upon this same plan^[32] perhaps more evidently. If now in *Phacops* the lens consisted of such semiconcentric strata and the upper moiety has been destroyed, the rest must have taken the shape as we find it. It is moreover peculiar that the destruction has been exactly similar in all lenses of that specimen. Can it be due to the circumstance that the power of resistance in the inferior strata has been greater?

[30] BULLAR Philos. Transact. 1878, pt. II, pl. 46, fig. 12.

[31] BELLONCI Atti dei Lincei, Memorie X 1881, pl. II, fig. 11.

[32] See GRENACHER pl. II f. 18 *Epeira*.

In *Phacops quadrilineata* the lenses are more elliptic than in the former. On their interior surface beneath the spiny tufts mentioned large hexaedral prisms of clear calcareous spar issue, one prism for each lens ([Pl. V fig. 38](#)), having thus a very deceptive appearance, but no doubt of inorganic origin with the lenses as a basis for their crystallization, quite as in the Cystoids where the interior often is converted into a mass of crystalline prisms, issuing from the interior surface of the plates.

III. Genera with stemmata and ocelli.

In a little group that has retained larval or ancestral characters during a great part of the palæozoic period, the genus *Harpes* stands as a type. It has ranged from the oldest Lower Silurian, if we join the related *Harpides*, to the middle Devonian. From near the top of the glabella, though not so much forward as the facial ridge of the blind trilobites (*Olenus*, *Liostracus* etc.) a straight ridge of much varying length stands out on both sides and at its extremity two or three globular stemmata with glossy surface lie encased. Probably this ridge has the same origin as in the *Olenidæ*, the more so, as there are indications of an extensive circulatory system. On [plate IV fig. 18-19](#) the right hand ocelli of *Harpes vittatus* BARR. from Lochkow, Bohemia, are represented. They are two, lying isolated near each other, quite globular with circular outline, smooth and glossy as to exhibit a shining surface. Their size is 0,4 mm in diameter. Being cut vertically in the direction of the longitudinal axis of the head they resemble elongated hemispheres, convex on the exterior surface, slightly concave on the interior. The test of the head lies between them as a saddle and covers them only partially and on the outer sides they lie with their margins encased in the head shield. Seen in thin sections of the right lens and magnified the whitish mass is traversed by vertical, blacker streaks, standing somewhat radiating towards the sides and cancellate. The other lens has a horizontal row of black dots. All this is probably not of any structural value, only due to later changes.

The remarkable genus *Harpides* from the lowest Lower Silurian belongs also to this group and has beside the fixed peduncle a peculiar elongated ridge going from the eyes to the lateral margins of the head^[33], a ridge which is also present in some of the true *Harpes*.

^[33] *Harpides breviceps* cannot belong to this genus and is rather related to *Erinnys* SALTER, as also MATTHEW holds it.

As BEECHER has shown^[34] the larva of *Trinucleus* possesses quite the same transverse ridge with intumescent eyelike extremities, and although the smallness of the specimens has not permitted to ascertain the presence of a true eye, it may be apposite to suppose it on the homology with the eyes of

Harpes, a genus with which Trinucleus is related. But as well known, in the adult Trinuclei there is no trace of these ocular ridges nor of real eyes so that BEECHER in his paper »Blind Trilobites»^[35] numerates Trinucleus amongst these. Out of the nine Scandinavian species of Trinucleus no less than seven have a well marked little tubercle on each side of the glabella placed exactly on the same spot where the larval Trinuclei had their much larger eyes placed. In this case it may be allowed to suppose that the tubercles are the direct successors of the larval eye and that they are true ocelli. REEDE seems to be willing to regard them as possessing a visual function.^[36] BEECHER^[37] holds the eye nodules in the larva and the ocelli in the adult to be identical.

^[34] Structure and appendages of Trinucleus.

^[35] Geol. Magazine 1898. pp. 439, 493, 552.

^[36] l. c. p. 447.

^[37] l. c. p. 309.

But, as BARRANDE has shown,^[38] there is a certain species of Trinucleus, the larva of which wants a facial ridge and eyes, as there also are several adult forms without ocelli. These have remained on a much ancestral stage, while the larva with eyes are more highly developed in such species, where the adult have been subject to a retrograde development. Ampyx and Dionide, though completely blind, evidently belong to this group, and once, as is to be hoped, larval forms may be discovered showing their development. In a certain way Arethusina shows characters proper for this group, in having the straight ocular ridge, quite as in Harpes, but eyes of the reticulate type and probably prismatic. It thus like Harpes conserved an ancestral characteristic long periods since it had disappeared in most of the other genera.

^[38] Sys. Sil. de Bohême I, pl. 30, figs 41-50.

Of the groups, in which JOH: S MÜLLER^[39] long ago classified the Crustacean eyes, his second »Hauptgruppe» (»Aggregate von einfachen linsenhaften Augen») and the fourth »zusammengesetzten Augen ... facettirte Hornhaut» the former corresponds with my third and the latter with my two first divisions. In so far as the cornea and its facets or lenses are to be regarded, there is the greatest analogy with the Isopoda. In vertical sections of Sphæroma we have the same sort of elongated, flattened biconvex lenses as

in *Sphærophthalmus* and others. Since GRENACHER and EXNER and others have published their excellent works on the eyes of the Arthropoda, there can be no foundation for speaking of the resemblance of the trilobitic eye with that of *Limulus*, as this genus stands completely isolated amongst all Arthropoda in that respect. There is, as stated above, a certain resemblance between the cornea of *Peltura* and that of *Limulus*, but this is not yet ripe for a discussion. Nor is there any evidence for correlating the eyes of the trilobites with the eyes of the Phyllopoda. BERNARD thinks that the so called eye of the *Paradoxidæ* has been formed upon the same plan as that of *Apus*. There is nothing to prove this hypothesis that the facial ridge or any part of it ever had been a visual organ, and the evidence at hand rather tends in a contrary direction.

[39] In Merckels Archiv 1829 p. 46 and in Treviranus Zeitschrift für Physiologie Bd IV p. 97.

There are signs of long physiological and anatomical efforts to prepare the development of the eyes on the free cheek, as revealed through the long series of blind trilobites. A system of radiating blood vessels, similar to those described above as covering the inside of the head in some older genera, all issuing from the scallop in the free cheek, where later the eye had to find its place, have left their stamp, their mark on the surface of the free cheek. They attest the great vital activity which was so intense at the point where the eye was to be formed. We give the figures of two such cheeks of different types. One from *Parabolina spinulosa* ([pl. V fig. 31](#)) is the more common, where six or more isolated trunks radiate from the semilunar ridge round the indenture and subdivide in branchlets which cease near the lateral margin of the cheek. It may be that it is an annular vessel near the indenture that feeds them all and that this probably is in connection with the great central circulatory system. In *Olenus* ([pl. V fig. 29](#)) the vessels are partly anastomosing and form a reticulate system and they are studded with minute wartlets. Another sign, which may be taken as a preparation, is the elevated rim around the scallop, which is so prominent in several of the Cambrian genera, but which does not embrace any facet bearing cornea..

From what has been stated above the following conclusions have been arrived at.

1. The plurality of the genera living during the Cambrian period were blind and it was first at the close of that period, in the Olenus schists, that genera with real visual organs appeared. There may have been oculate trilobites earlier, as Solenopleura, but we know nothing of their eyes.

2. The primordial glabellar pleuron which was metamorphosed into a facial ridge is no visual organ. It is in the Olenidæ nothing but the elevated line made in the test by the subjacent main trunk of the circulatory system. It swells out in a node, »palpebral lobe«, but not before the facial suture has been formed. In the genera where there is no facial suture, there is no node. In the Paradoxidæ where the ridge is of a different origin, there is no node, though there is a suture.

The four types of eyes in the trilobites have probably succeeded one another in the following chronological order:

1) with stemmata or ocelli; 2) biconvex or lentiform; 3) prismatic; 4) aggregate. The oldest known representatives for each type are for 1) Harpides rugosus in the Ceratopyge limestone of the Lower Silur., for 2) Eurycare, in the Cambrian Olenid schists, division 2, for 3) Megalaspis, in the Ceratopyge limestone of the Lower Silurian, for 4) Phacops in the Lower gray Orthoceratite limestone. The eyes of the trilobites show the greatest conformity with those of the recent Isopoda.

The most perfect eyes amongst all the trilobitic eyes may be those of the Phacopidæ, which are also geologically the youngest, the least developed again those of the Proetidæ or rather of the Bumasti. The great thickness of the cornea in these must have weakened their power of vision and they had probably only a faint perception of light.

On the maculæ of the hypostoma.

We shall now turn our attention to the visual organs which LILJEVALL discovered on the hypostoma of Bronteus. In doing this, we may bear in mind, that the genera in which we really have found lenses on the maculæ are relatively few, but that we shall review the maculæ in all genera, which we have been able to examine, and try to show that even most of these may, although in an inferior degree, have acted as visual organs. At the same time some more details shall be given about the cephalic eyes for comparing them with the hypostomic ones.

Acidaspis MURCH.

The hypostoma is of a peculiar type, deviating from that of the other genera, squarish or rectangularly transverse, entirely without terrace lines and no maculæ proper. BARRANDE has given no less than nine different samples of these hypostomas.

Acidaspis crenata EMMR.

([Pl. I figs 1-6.](#))

There are certainly no maculæ of the same sort as in so many other trilobites and the interior side of the hypostoma does not bear the least traces of impressions which might be taken as the reverse of the maculæ and still more less as muscular impressions. In fact, the total absence of such in *Acidaspis* militates against the interpretation of the macula impressions as muscular scars. When the maculæ fail, also the so called muscular impressions fail. Moreover we are able to see still clearer in this matter through what we know about the structure and position of the extremely similar hypostoma of *Apus*, of which we have given a description in the end of this memoir. This skeletal part of *Apus* is without any connection with the surroundings, excepting at its anterior margin, and is consequently movable in a direction outwards and upwards. Along that

anterior margin it is fixed to the outside of the ventricle through three pair of muscles, three muscles on each side of the margin. ZADDACH De Apodis cancriformis anatome (pl. II fig. XIV p. 68).

For the rest the marks of the attachment of the muscles are as a rule in the Crustacea, at least those of the head, elevated small platforms, so in the Trilobites, of which I have excellent specimens in a *Bumastus* and others. On the inside of the head of *Limulus* they are faintly elevated patches. The small hollows on the inside of the hypostoma formed by the maculæ are indeed the sockets in which the soft parts of these more or less developed hypostomic eyes were sheltered. But there is still a feature in the hypostoma of *Acidaspis* which merits our attention and which perhaps may have a significance akin to that of the maculæ.

The hypostoma is square, with two short pointed wings, one on each side of the slightly bent anterior margin and likewise two smaller ones at the corners of the posterior margin. A groove follows on a short distance the lateral margins and the posterior margin and disappears a little below the anterior one. In the same direction, distally, though a little more inside and unconnected with them there are two small grooves, the bottom of which consists of a shell substance of different colour and structure than the other parts. Having been a little ground and seen in transmitted light it exhibits the shape of a club and a homogenous yellow spot, tapering posteriorly and swelling out distally ([Pl. I fig. 4](#)). It must be left an open question whether these maculæ share in the nature of visual organs as the quite different maculæ of the other trilobites, but it may be possible that it is so. It must, however, be remarked that there are two types of hypostoma in the genus *Acidaspis* as shown by the illustrations of BARRANDE. One has the small grooves, possibly all sheltering the claviform maculæ, disposed as in the now described *A. crenata*. This group embraces five species of the Bohemian Silurian formation. The other group of three species again has the hypostoma of the same quadratic or rectangular shape, but the two short grooves, which may be expected to contain the maculæ, are placed midway between the anterior and the posterior margins, nearly as the maculæ bearing grooves of other trilobites. We have however not had material for pursuing our researches in this genus, the other species of the Swedish *Acidaspidæ* being unknown as to their hypostoma.

It may here be added an observation concerning the ornamentation of the exterior surface of the hypostoma of *Ac. crenata*. It is covered by a great number of diminutive circular or oblong wartlets occupying the whole surface excepting the lower third of the central field just above the posterior groove which is smooth. These wartlets seen through transmitted light ([Pl. I fig. 5](#)) show in their interior something like a peculiar black spiculum rising from a bifid rootlet and confined within the wartlet and not extruding from it. In a longitudinal section ([Pl. I fig. 6](#)) the spicula perforate the wartlets reaching through their whole length. As in *Calymmene* these interior pseudo-spicula are tubes, filled with iron-pyrites. It is probable that these tubes were once bearing setæ and quite as in *Apus* formed a fur of bristles.

The structure of the cephalic eyes ([Pl. 1 f. 1-2](#)) is prismatic, but the separate prisms are rather short and broad. Their lower or interior end is convex. The separating lines between the single prisms are not always distinct.

Agnostus DALM.

[Pl. 1 fig. 7.](#)

Agnostus glandiformis ANG. Although there is not the slightest evidence of eyes in this the largest of its genus, nor any free hypostoma hitherto has been found, we may here give a little account of our researches into this species. The scantiness of material forms a chief obstacle to our knowledge. Only three entire, rolled up specimens have been found, and it is by sectioning and preparing such that any hope can be entertained to gain reliable results.

A rolled up specimen from Andrarum Scania was sectioned lengthwise. The tail-piece closed tight against the head-shield, so there had been little chance for foreign matter to penetrate into the interior which, however, is filled. Close below the cephalic shield there is a remarkable structure, mostly resembling an elongated intestine with swellings joined by more narrow ducts and anteriorly the coherence is interrupted. As in crustaceans and Arthropods in general the stomach and the intestine are situated on the dorsal side of the body, there is nothing unlikely in assuming that this in reality may be the remnants of the intestine. This may also be compared with the observation made by VOLLBORTH in his memoir »Ueber die mit

glatten Rumpfgliedern versehenen Trilobiten», 1863, p. 46, tab. 1 fig. 12 where the heart-tube probably is delineated. BARRANDE has also given figures of what he considers as the intestine in Trinucleus.

Below this organ in Agnostus, there lies a section of a vaulted calcareous plate with its convexity turned against the dorsal side of the head shield, that is to say quite the reverse what might have been expected if it had been the hypostoma in its true position. This may, however, not be any objection against considering it as a sectioned hypostoma, loosened from its connection with the cephalic shield, disturbed in its original position and turned round, when the shell became filled with mud. At the distal end there is a much distinct duplicature.

Asaphus BRONGN.

The wellknown hypostoma of this genus has the same characteristic shape, though more pronounced, as in Ptychopyge, with its posterior margin deeply indented, so as to form two large, pointed lobes. There is properly only one median field, surrounded by flat, lateral borders, continuing down into the posterior lobes, from which it is separated through a shallow groove. The maculae are situated in this groove, on each side of the inferior border of the central field. They are more or less prominent, but whatever their form may be, their surface is always entirely smooth, lying well circumscribed amidst the surrounding terrace lines.

The following list enumerates all the species in this genus, of which previous authors have delineated the tubercles, though they in the descriptions only in very few instances have mentioned their presence.

A. acuminatus NIECZKOWSKI, »Zusätze zur Monogr. der Trilobiten 1859«, tab. I, f. 6. Maculae most prominent, but no mention made of them in the description.

A. (Isotelus) canalis J. HALL, Pal. N. Y. I, pl. 4 bis, f. 18-19. Though this and a following species probably on account of the deviating conformation of their body, belong to a different genus or subgenus, I mention them along with the Asaphi, as there is the greatest similarity in their hypostoma. Fragment of an interior cast; still more strange is the fragment showing part of the interior surface and the duplicature.

A. expansus L. BR. I pl. VII f. 3, BR. II pl. I f. 2, 2 c. In the two first figures there are no tubercles nor maculae marked, in the last figure there are distinct tubercles.

A. fallax A. pl. XXVIII f. 3, c. BR. II pl. I, f. 3, both nearly congruent.

A. (Isotelus) gigas J. HALL, Pal. N. Y. 1, pl. 60, f. 7 g, pl. 66 f. 5, the inside of an entire, uncommonly large hypostoma gives a general good view with the macula; which are large and evident. He mentions the maculae as »two circular spots». »These probably indicate the points for the attachment of muscles and tendons upon the inside,» he adds.

A. ingens BARR. NOVÁK II, pl. I, f. 7, a good figure showing two semiglobular maculae. BARRANDE's figures pl. 33, f. 7, 8 are not distinct.

A. ludibundus TQT. BR. II, pl. 1, f. 7. Two semilunar sulci on the cast of the interior surface, being only the posterior borders of the maculae which are a little more raised and distinct than the rest.

A. nobilis BARR. pl. 32, f. 6. Indistinct traces of maculae. In the figure 6, pl. 31 representing the hypostoma of a young specimen, there are no tubercles at all. If the figures were to be relied upon, it might be assumed, that the tubercles appear at a more mature age.

A. Powisii SALTER, pl. 23, f. 6. Two semilunar, narrow tubercular maculae, their interior apices converging towards the anterior border of the hypostoma.

A. raniceps A., pl. XXVIII, f. 2 c., BR. II, pl. 1, f. 4. A good figure.

A. raniceps var. *maxima* BR. II, pl. 1, f. 6. Semilunar impressions on a cast and thus far incomplete.

A. striatus BOECK. To this belongs probably *A. expansus* M. SARS »Ueber einige neue oder unvollständig bekannte Trilobiten» in Isis 1835 p. 333, and especially p. 340 and the following (Bemerkungen über die untere Seite von einigen Trilobiten). SARS there gives a very good description of the exterior side, where he also mentions the two maculae, pl. IX, f. 9 a, b. (»2 kleine Knoten»), BR. I, pl. VIII, f. 4 a, and BR. II, pl. I, f. 9. These latter figures differ in so far, that the former has the maculae excavated or rather tubular below, in the latter again they are regular.

A. trinucleorum BR. II, pl. I, f. 16 a large specimen with linear maculae arranged rectangularly in respect to the longitudinal axis of the hypostoma and both on the same level.

A. tyrannus MURCH. SALTER in Mem. Geol. Survey, Brit. Foss. Dec. II, 1849 pl. V, f. 4, excellent figure. On p. 2 he says: »there is an oval circumscribed tubercle at the origin of each (fork) most distinct on the inner surface«. This figure is again reproduced in the »Monograph«, pl. 22, f. 6, where it is said »Two linear tubercles with their interior apices converging towards the posterior margin of the hypostoma«.

Asaphus sp., probably *A. raniceps*, POMPECKJ, Trilobitenfauna Ost- and West-Preussens, taf. VI, f. 7. p. 80 »an seinem Hinterrande liegen zwei kleine, flache Höckerchen, die als Reste des Hinterlappens aufzufassen sind«(!?).

We shall now describe the *Asaphi* which we have been able to examine more in detail.

***Asaphus expansus* L.**

[Pl. I figs. 8-17.](#)

In the numerous specimens of which we have examined the hypostoma, there are always small, nearly circular maculae which by their lighter colour are marked out from the surrounding smooth space of the inferior sinus of the lateral grooves where they are situated. They do not rise at all above the surrounding surface from which they are separated only by a fine, scarcely perceptible line. On the interior surface of the hypostoma they are better seen, and the enclosing line is deeper and more distinct. Their surface is there quite as smooth and even as on the exterior side of the hypostoma. They do not consequently in the least manner resemble muscular impressions as observable on the inside of the glabella of other trilobites. These are on the contrary elevated above the surface and finely striated in various ways.

The peculiar nature of these macula is revealed through the various sections we have made. In [pl. I fig. 15](#) a vertical section across the entire hypostoma is represented. The sectioned, dark maculae (a) lie on both sides of the faintly curved central field, in the sinus of the lateral grooves, and the

strongly developed terrace lines continue sideways just a little on the interior side, where the duplicature turns round. As seen in a magnified vertical section ([fig. 16](#)) the maculae consist of horizontal, whitish, straight lines, probably lines of successive growth, and these are crossed rectangularly by more irregular whitish lines separated from each other through dark spaces. The whole thus gains the aspect of a dark surface cancellated by white lines. This reticulated or spongy macula is enclosed as to its superior as well as to its inferior part in the compact and homogenous test of the hypostoma and occluded from the influence of the light. Its value as a visual organ consequently is insignificant. In a horizontal section ([fig. 17](#)) the structure is still more bewildering. There the whole macula is a confused, spongy white mass with dark spaces between the white meshes, and only at one side, the left one, some obscure indications as of polygons are visible. It would indeed have been impossible to interpret what this means, had not the study of the cephalic eye given a clue thereof.

The eye of this species is covered with a delicate perfectly smooth and glossy integument ([f. 12](#)) which is reposing immediately on the prismatic lenses of the cornea. In a few instances, depending on the colour or the state of preservation of that integument the lenses are indistinctly translucent. Owing to their state of preservation their aspect is greatly variable. They are in many instances, as seen in [fig. 9](#), six-sided or some rhombic, foursided or even quadratic ([f. 8, 10](#)). They are all of the same length, 0,2 mm. along the surface proper of the eye, but are lengthened to 0,3 mm. towards the border of the eyes, [f. 12](#), which will be described further on; they are of equal breadth, amounting to 0,066 mm. Their inferior ends are slightly convex or nearly plane when well preserved, else, when as often is the case, disintegrated as to be scooped out and vaulted. Their interior structure, as revealed by sections, shows a cylindrical core, [f. 8-10](#), composed of concentric strata. This cylinder fills nearly the whole interior space of the prisms, there being, however, in many instances a compact dark mass between the cylinder and the walls of the prisms. There are also sections in which the prisms are filled with a uniformly black mass without any concentric structure. In longitudinal sections the outlines of the individual prisms are not clearly discernible ([f. 11](#)). There are longitudinal, white lines of varying thickness with lateral irregular offshoots, which may join with

those from opposite walls, and give the interior a sort of spongy or cancellate appearance.

Towards both sides of the ocular surface, towards the superior and inferior side, a change sets in as to the shape of the lenses, as best seen in horizontal sections ([figs. 8, 10](#)) they are lengthened and become more and more indistinct, and at last in the upper and lower marginal zones pass over into a reticulate, spongy mass, which seen in a longitudinal section presents almost the same aspect as in the regular prismatic surface of the eye. [Fig. 11 b, a](#) being the visual field. It is, however, more densely reticulate, but a prismatic arrangement is quite as much evident as in the ocular surface proper. In the rule the passage from the prismatic surface to the reticulate is gradual, the prisms becoming by and by irregular in their outline and diminishing in size ([fig. 12, a](#) the eye proper, [b](#) the border zone), but there are also instances where the distinction between these two fields is sharp and without any gradual transition. In the inferior reticulate zone there are generally some oblong, funnel-shaped pits. I am uncertain whether they are to be regarded as regular parts of the eye structure or rather as burrows of some parasite. They do not continue deep down.

If we now compare the reticulate zone of the eye with the maculae of the hypostoma, for instance the vertical section [fig. 16](#) with [fig. 11 b](#), we find the most complete identity in structure. In the same manner the horizontal section of the border zone of the eye, [fig. 8 a](#), [fig. 10 b](#), and of the macula, [fig. 17](#) are similar. The same chaotic, spongy mass in both, with some tendency to form prisms more evident in the longitudinal sections, where the same reticulate structure with predominant white, longitudinal streaks is so palpable. That there is a complete identity in structure between the two, the macula of the hypostoma and the border zones of the eye, is as evident as anything can be, but as to the functional identity or what this function may have been it is difficult to decide anything with certainty, at least it seems to me that the capacity of vision must in both have been far more restricted than in the eye proper. They rather give the impression as of rudimentary visual organs.

Asaphus raniceps DALM.

[Pl. I f. 23-26.](#)

The maculæ of the hypostoma are placed obliquely in the saline manner as in the preceding species. They are more prominent and the oblong macula, with the longest diameter of 1,2 mm. is on the top of a little mound and surrounded by a fine, elevated marginal line ([fig. 23](#)). Its somewhat convex surface is entirely smooth and if sectioned horizontally exhibits the same sort of spongy texture as *A. expansus* ([pl. I f. 24](#)). In a vertical section ([fig. 25](#)) the macula does not occupy so large a space as in *As. expansus*, but rather lies as a lenticular disk in the hypostomic test closer to the superior surface than the inferior. With sufficiently high power the same sort of pillars, divided by horizontal strata is seen. Upon the whole the vertical section is not so clearly developed. The reticulate zone of the eye is more definitely separated front the prismatic zone than in *A. expansus*. There can be no doubt that there is a correspondence in this species between the structure of the macula! and the spongy zone of the eyes.

***Asaphus cornigerus* SCHLOTH. (A. Kowalewskyi LAWROW).**

Of this strange species with its enormous eyestalks, more than 2 centimeters in length I have through the kindness of Akademiker FRIEDR. SCHMIDT in St. Petersburg had occasion to study some specimens.

The rather large maculæ, obovate-circular, are like those of several other Esthonian species oriented inwards and upwards instead of inwards and downwards as in the Swedish species described. As the specimens have suffered through corrosion of the surface there are only faint traces of a marginal line. Their microscopic structure is badly preserved, and they look pale and transparent with only few indications of the spongy texture. The lenses of the cephalic eye are nearly square prisms sometimes with a slight approach to hexaedral pillars. There is no clear transition into a spongy or reticulate border zone. But this may depend upon the bad preservation.

***Asaphus fallax* DALM.?**

[Pl. I fig. 18-22.](#)

It is very difficult to distinguish this species or rather variety from *A. expansus*, but if we have found genuine specimens, there are some points in the shape of eyes and maculae which make it different. The hypostoma ([f. 21](#)) is rather more broad and the duplicature is large with an upturned margin. The little macula is placed on the top of a smooth rounded elevation and enclosed by an elevated rim.

The cephalic eyes are rather short, regular hexaedral prisms ([f. 18-20](#)) and as shown in [fig. 18](#) change into irregular squares near the border of the eye.

***Asaphus* sp.**

from Brunsby kanal, Segerstad parish, isle of Öland. The maculae are large pale whitish and of an uncommonly fine-meshed reticulation, well limited from the surrounding hypostoma.

***Asaphus* sp.**

[Pl. I fig. 27-30.](#)

from the islet of Sandö, north of Gotland. Although we cannot give any account of its hypostoma, not having had sufficient material, the structure of the eyes is so peculiar that it seems worthy of being recorded. The integument is extremely thin and transparent and the subjacent lenses are clearly seen, and through their impact on the thin integument they make this to stand out in a very low relief above them ([fig. 27](#)). These lenses are uncommonly short, forming at the surface rather oblong, slightly hexaedral prisms with a narrow interspace between them. In a section lower down they have the shape of hollow, white rings filled with black mud and in a longitudinal section the white walls of the lenses look like short pointed spikes and interiorly they are completely empty. We here find also the same gradual change from regular cones to the spongy border zone as in the above mentioned species [f. 29](#). The border zone is finely reticulated.

***Asaphus (Isotelus) gigas* J. HALL.**

[Pl. II figs. 1-3.](#)

The enormous hypostoma of this giant resembles in a high degree that of the *Asaphi*, but is at the same time the most evident verification of the experience that the hypostoma *per se* cannot be regarded as the sole criterion for determinating the generic affinities of different species. The whole structure of this trilobite in other respects gives it a quite independent position, distinct, from *Asaphus*.

We have had at our disposal several specimens of the large hypostoma, the dimensions of the largest, fragmentary in its anterior margin, being as follows: breadth 48 mm. length 41 mm., probably 47 mm. when entire, breadth of each of the posterior lobes at their bases 19 mm., length of the same 23 mm. The maculæ which are placed on the flat surface of the hypostoma without being at all elevated, are prominent through their great size and their pale, whitish colour. They are somewhat oblongue having their longest axis directed inwards and downwards. They attain a diameter of four millimeters. Horizontally sectioned, [f. 2](#), they show the spongy texture and vertically the quasi prismatic reticulate structure richly developed, [f. 3](#). The pillars are very distinct and continue without interruption all through the macula and in the interstices there are traverses joining between two pillars or partially filling the darkish interspaces, thus giving the whole the aspect of some »tabulate« coral sectioned. By the inspection of the horizontal section alone, presenting the irregular spongy texture it would have been impossible to imagine the ordinated arrangement which the vertical section reveals to us. The eyes consist of regular hexaedral prisms, and there is a very sharp boundary line between them and the spongy zone, which is very narrow and without distinct separating lines joins with the cephalic test. For the rest there are in all probability at least two different species sent from America under the name of *Isotelus gigas*, of which only the largest, almost like a *Homalonotus*, and the hypostoma of which has been partly delineated in this memoir, seems to be the real one. They differ both in the shape of the hypostoma as well as in other respects.

Barrandia MAC COY.

According to *SALTER Monogr. pl. 19 fig. 9 his B. Portlocki*, of which there is a fragmentary and broken hypostoma, shows feeble traces of oblique

macula, nearly in the fashion of *Asaphus*. In the description at page 139 is told about »the usual pair of tubercles».

Bronteus GOLDFUSS.

BARRANDE delineates the maculæ on the hypostoma of

B. palifer, Tab. 45 fig. 17,

B. planus, Tab. 48 fig. 7, and

in the supplementar volume of

B. rhinoceros, Tab. 9 f. 16,

B. furcifer, Tab. 11 fig. 16, but there is not the slightest indication of granulation on any of them, nor is there in the descriptions, generic or specific, the least mention made of the tubercular maculæ. It is also remarkable that in the works of ANGELIN and NOVÁK where several hypostomas belonging to species of this genus are delineated, not a single one shows these tubercles. We shall now continue the descriptions of the Swedish Brontei, already begun with *Br. polyactin* in the introductory part of this memoir.

Bronteus irradians LINDSTR.

[Pl. II figs. 4-5.](#)

has a hypostoma that much resembles that of *Br. polyactin*. In its general shape it is similar and the two concentric grooves with the two maculæ placed in the same way, just below the superior groove. These maculæ are much larger than in *Br. polyactin*, nearly thrice their size. They are also more ovate or rather like a bean, the smooth surface is larger and the granulated spot restricted to a more narrow space forming an oblique patch. The granules or lenses are also individually larger than in the allied species, double their size or 0,06 millimeters. We have not succeeded in making sections of the cephalic eye nor of the maculæ.

Br. platyactin ANGELIN.

Pl. II f. 14-19.

The hypostoma has a transversally triangular form, and is divided only in two fields through a shallow semicircular groove near the posterior margin. The two maculae are situated above the groove near its superior sinuses. They are elongated, [fig. 17](#), elliptic with the narrow pointed end directed outwards and the broad rounded end inwards. The chief surface is scooped out as a shallow depression. The granulated spot is situated on the broader end and covering it completely. The relatively large lenses are arranged in five regular rows, the uppermost one being the longest. On the interior surface of the hypostoma there are the corresponding sockets of both maculae with smooth surface. The horizontal sections of the granules [figs. 18, 19](#) present the image of white rings in close contact, without, however, to occasion a prismatic structure, a dark interspace lying between each ring. These lenses are filled with a dark mass, and in some the same sort of radiated structure is perceptible as in the lenses of the cephalic eye. In horizontal sections of the cephalic eye the lenses approach the polyedral shape. In another section near to the surface of another specimen the lenses are decidedly hexaedral. The vertical sections, [fig. 14](#), reveal their real nature as lenses where they lie as a string of beads with a dark nucleus enclosed within a thin whitish shell. They are covered by a thin membranous lining. When seen in transmitted light the lenses proper are dark, and the shell white and in reflected light the lenses are lighter than the rest.

Br. laticauda ANGEL.

Pl. II figs. 6-13.

Lower Silurian from Dalecarlia. The hypostoma, [f. 10](#), is of a broad clypeate shape, the anterior margin rounded without any large projecting wings. On the exterior surface there are two grooves parallel with the rounded inferior margin. As in *Br. polyactin* the two elliptic tubercles are situated on the inferior edge or slope of the upper groove and they are deepened by a shallow depression as in *Br. platyactin*. The granulated spot situated along the posterior margin of the macula deviates much in shape from that of the other species. It consists of a long and narrow stripe ending in a fine point outwards and widening inwards, where it is rounded, forming

thus a claviform, curved elevation. Around the granulated area the maculae are quite smooth, and show in a horizontal section an irregular structure of tiny black dots nearly resembling the structure of the hypostomic shell. We have not been able to obtain any good vertical section, but by casts of the interior side of the tubercles it is found that the granules form polyedric facets like those of the eyes though perhaps not so regular. There is a specimen with the granules intact, and the polyedric shape is then not so distinct.

Sections of the eyes elucidate the structure, which partially is obscure in other species of *Bronteus*. There is a thin membranous coating covering the subjacent well formed lenses ([fig. 8](#)). When this membrane in some instances has been peeled off the lenses lie bare ([fig. 9](#)). These have a dark nucleus and in some instances it seems as if there were two. In a vertical section they look spheroidal, in a horizontal section again they are polyedric, especially when taken somewhat below the surface or near the middle line ([fig. 6](#)). In [fig. 7](#), some lenses are delineated in a horizontal section, highly magnified and the corallian appearance is evident. The lenses are well separated by distinct lines, and from their inner tubes whitish reticulations radiate towards the centre.

If we now compare these sections, where the lenses are so distinctly seen, with the more obscure sections of other *Brontei*, it is evident that the dark points in them are nothing but transformed and deteriorated centres of the lenses. In *Br. platyactin* they are a little more distinct than in the other species. And it is with these that the granulated spots on the maculae of the hypostoma are most concordant.

***Bronteus* sp. indet.**

[Pl. II figs. 28-30.](#)

This species, of which only the hypostoma is known, found at Lansa, Gotland, is likely, to judge by that, to be nearly related to *Br. polyactin*. The shape of this hypostoma is exactly the same and the maculae are placed in the same position, but they are larger, a little sunk on the blind surface and the group of the lenses is different. Its superior margin forms an ingoing

arch and the lenses themselves are not convex, but rounded or slightly polyedric, and separated through thick interspaces.

Bumastus MURCHISON.

The only recorded hypostoma with maculæ belongs to *B. insignis* HALL as described by SALTER Monogr. p. 208 pl. 27 fig. 7. He says: »A pair of compressed tubercles occurs at the lower third: they are transverse-ovate, and more than their own diameter apart.» This is nearly in concordance with what is seen on the species to be described below.

Although it can in reality be said that the species of *Bumastus* are the most common of all trilobites in the Silurium of Gotland, in so far that their head pieces and pygidia are met with everywhere in the limestone rocks of the island, the find of entire specimens or of detached hypostomas is amongst the rarest events when collecting there and amongst thousands of fragments, in several places entirely filling large portions of the limestone rock, there have in the Swedish State museum been acquired only three or four hypostomas, which we are now going to describe.

At first we must point out a certain resemblance which prevails between the hypostomas and maculæ in *Bumastus* and *Bronteus*. Compare for instance *Bum. sulcatus* with *Br. platyactin* and *Br. laticauda*. The general form is nearly the same in both and the shape of the maculæ almost identical but for the complete want of granulations on those of *Bumastus*. To this resemblance must be added that there are, as known, forms of trilobites which by some authors have been regarded as *Brontei* and by others, again, ranked amongst the *Illænidæ*.

Three species of *Bumastus* have been recorded as found in the Gotland strata, but of one of these no hypostoma is known. Besides, to judge by the head shields and hypostoma, there are a few new species. To begin with the most common

***B. sulcatus* LDM.**

[Pl. II figs. 33-40.](#)

The hypostoma is broadly shield-formed and the anterior margin is most characteristic being in its central part elongated in a short evenly rounded projection. The side wings are triangular acuminate, bent a little backwards and towards the interior surface. There is only a single rounded field passing into an evenly bent border, in front of which the projecting, elliptic maculæ lie. A second pair of wings stretches backwards near the posterior margin. We have not obtained any sections of the maculæ.

The cephalic eyes [figs. 36-38](#) are covered with an uncommonly thick stratum of homogenous shell, one and a half times as thick as the subjacent lenses. These are comparatively broad, above passing indistinctly in the covering stratum, below, at their basis, convex. In a horizontal section [figs. 33, 34](#) they exhibit the common hexaedral or pentaedric aspect, their exterior sides being white and confluent without any separating lines. Their interior is dark and from the polygonal sides white lines go in giving the whole the appearance as of a starry, composite coral with its septa. The correspondence in structure with *Bronteus laticauda* [pl. II fig. 7](#) is striking. A narrow space separates the surface of the eye from a nearly similar patch below the eye on the cheek [35 b, 38 b](#). But there the position of the composing strata is singularly reversed and the spongy or pseudo-prismatic stratum lies close to the outside and the homogenous stratum on the inside. The white longitudinal lines are well marked out and regular in the vertical section, [fig. 38 b](#), indicating the prismatic structure most clearly.

B. barriensis MURCH.

[Pl. II figs. 31, 32.](#)

The hypostoma [fig. 31](#) is of so transverse a shape that the breadth much surpasses the height, the former being 6 millim., while the latter attains only 3,5 millim. The anterior margin is so much arcuated and sloping backwards that it forms an obtuse angle. The side wings are triangular and acuminate and from them a thick pad runs backwards and forms the posterior margin. Below the largest field of the exterior surface the maculæ rest on a ridge having a much elongated, vermiform shape and sunk in the surface ([fig. 32](#)) a little more than three times longer than broad. In its shape it approaches much to the hypostoma of *Bum. insignis* HALL as delineated by SALTER [pl. 27 fig. 7](#).

In another, undetermined *Bumastus* ([pl. II figs. 47, 48](#))^[40] the hypostoma (from Klints, Othem, isle of Gotland) is of a more elongate obovate form, the anterior margin elevated, rounded, the terrace lines faint, the tubercles are larger than in the former *Bumasti*, oblong, curved, beanshaped, evenly rounded and smooth.

[40] Probably the same as *Bum. sulcatus* pl. XII fig. 12 in my paper on the Trilobites of Gotland 1885.

Another of the new species is one that LILJEVALL has found at Korpklint near Wisby ([Pl. II figs. 41-46](#)). The hypostoma is more elongate than transverse with the anterior margin regularly arcuated, by far not so prominent as in *B. sulcatus* and the former. By a shallow groove the exterior surface is divided in two fields, the anterior one the largest and covered with a few, distantiated, concentric terrace lines. The posterior field is smooth, nearly even and the maculae lie just in the groove, with their axis oblique to the longitudinal axis of the hypostoma, beanshaped and smooth. The cephalic eyes show in a horizontal section hexaedral white walls enclosing a dark space in which some indistinct radii emanate, [fig. 41 a](#). In the vertical sections the prisms are narrow and elongate and the integument by far not so thick as in *Bumastus sulcatus*, [fig. 42, 43](#). The border zone of the eye in this ([fig. 41](#)) as well as in *B. sulcatus*, as seen in horizontal sections, agrees with that of the *Asaphidæ*.

***Bumastus* sp. inlet.**

[Pl. III fig. 1, 2.](#)

In a detached piece of limestone, found in Gotland, at Norderstrand, near Wisby, two hypostomas and a free cheek lay embedded along with a coral of the genus *Acantholithus* and a *Bronteus* probably belonging to a new species. As the coral has not been found above the uppermost beds of the Lower Silurian it is probable that the trilobites also are derived from the same horizon.

The hypostoma, figured [pl. III fig. 1](#), is broadly tongueshaped, with a short, blunt, triangular wing on each side of the anterior margin. The exterior side is nearly completely occupied by a single large field, which is convex and decorated by sparse terrace lines. Below this a smooth, nearly even plane,

reaching to the inferior, elevated border of the hypostoma. The two tubercular maculae are situated exactly on the boundary line between the convex and the plane field. They are elongately ovate, pointed outwards, rounded inwards, with smooth and glossy surface. That portion, on which in the species of *Bronteus* the granular spot rests, is transversally wrinkled by some faintly elevated, sigmoid lines ([fig. 2](#)). It would have been of great interest to investigate the interior structure of these curious maculae, but the scarcity of the material has prevented our doing so.

The other hypostoma, [pl. III fig. 3-5](#), is larger and has the terrace lines more distantiated and forming more open curves, but for the rest it is of the same shape as the smaller, so that we may not ascribe these small deviations to a specific difference. The maculae are more elongated, narrow, curved tubercles, thus differing from the next preceding. A section running through its left tubercle does not show any structure, but the shell of the maculae is extremely thin, thus contrasting strongly with the surrounding thicker shell, [fig. 4 a](#). The cephalic eye, again, remaining on the free cheek found in the same piece of limestone and probably belonging to the same specimen, has its surface well preserved and exhibits semiglobular lenses in low relief, [fig. 5](#).

Calymmene BRONGN.

In the following species of this genus hypostomic tubercles have formerly been recorded.

Cal. Blumenbachi SALTER Monogr. pl. 8, fig. 9. The identity of the species is dubious.

Cal. duplicata SALTER Mon. pl. 9, fig. 22. The tubercles rather more linear and approximated than in the subsequent. SALTER calls them »two small transverse lobes ... forming a nearly continuous ridge«.

Cal. tuberculata ANGEL. Tab. XIX fig. 5 c from the inside.

It is remarkable that the eyes or that part of them where the cornea and the lenses should be found, has been destroyed and entirely lost in all specimens of the *Calymmene* known to us, many hundreds having been searched and every one with a lacuna on that spot. It must have been of an excessive thinness. We know only of one instance, the Bohemian *Calym.*

Arago, in which BARRANDE has seen the eye intact and provided with a small number of about eleven sphaerical lenses.

Calymmene intermedia LDM.

[Pl. III figs. 6, 7.](#)

The clypeiform hypostoma is divided in two fields, one anterior larger, in the centre elevated in a short, blunt knob, and a posterior, bearing the two elevated rounded maculae, just below the semicircular groove which separates it from the anterior field. These fields are surrounded by a large border which is posteriorly emarginated through a short rounded sinus. The surface is entirely covered by small semiglobular wartlets perforated by a straight pore or duct, which being filled with a black mineral, probably iron pyrites, gives a peculiar and characteristic aspect to the sections. In the same manner the whole mass of the hypostoma is pierced by straight, black lines, also filled up pores, starting from the outside and ending before attaining the inner surface.

The maculae are prominent, separated from the hypostoma through narrow, distinct grooves, ovate, with the pointed end directed obliquely outwards, against the margins of the hypostoma. They are covered with wartlets and leaving only a little oblong spot free, quite smooth, this being consequently the macula proper. But in a longitudinal section there is not the least distinction between this macula and the other hypostoma, only, that it is free of pores.

Calymmene tuberculata BRÜNNICH.

[Pl. III figs. 8, 9.](#)

We have succeeded to prepare a horizontal section of the macula. This is oval, perfectly homogenous and without any structure proper, showing only indistinctly a mottled medley of pale, brownish spots in the clear mass. There is in both maculae a small dark, angular spot placed in their inferior part, [fig. 9.](#)

Centropleura ANGELIN.

This author has delineated most distinct maculæ in Centr. Steenstrupi [pl. III fig. 4, b.](#) two, small elongate darkish spots, near the posterior margin. I have not seen any specimen of this trilobite and have consequently not been able to follow up ANGELIN's observation.

Chasmops M'COY.

[Pl. III fig. 10.](#)

In his work on the Russian Phacopidæ FRIEDR. SCHMIDT has given the figures of Chasmops Eichwaldi and Ch. Wesenbergensis showing, as it were, patches of black maculæ. But as there is no mention of them, and as they are quite unlike the macula? which are described here, I am in doubt of their real nature. We have figured a macula from the hypostoma of a Chasmops, probably Ch. macroura, from a drift block near Rostock. It is elongate and narrow, more than four times longer than broad. Its smooth surface is a little scooped out and no structure is observable.

Chirurus BEYRICH.

We find the hypostomic maculæ annotated and delineated as occurring in the following species.

Ch. bimucronatus SALTER pl. 5, fig. 5 as two narrow elongated pits, surface seems intact. No mention of them in the letter-press.

Ch. exsul SCHMIDT pl. VI fig. 9. Indications of the oblong, narrow maculæ.

Ch. macrophthalmus SCHM. pl. VII, figs. 1 c, 2. Faint indications of narrow pits.

Ch. Quenstedti BARR. Tab. 42 fig. 3. Maculæ near lower border as two narrow slits.

Ch. spinulosus SCHM. pl. VII fig. 10. The lateral pits well visible.

Ch. tumidus SCHM. pl. VIII, fig. 22. The pits are short, but present.

The exterior surface of the hypostoma is entirely covered with small wartlets, closely set, of different size in the various species, and spread between them lie a great number of elongate smooth spots dispersed, on the interior side of the hypostoma showing an excavated surface ([pl. III fig. 16](#)).

Chirurus (Cyrtometopus) clavifrons DALM.

[Pl. III figs. 17-21.](#)

The shape of the hypostoma is shown in [fig. 21 pl. III](#). It is regularly shield-formed with posterior margin rounded. A shallow groove running parallel with the margins forms a large, faintly elevated border. The anterior margin projects on both sides in a short, backwards directed, broad and fiat wing, hollowed out with a pit on the front side, and there is a posterior lateral wing, midway between the anterior and posterior margin short, blunt, directed obliquely backwards. The whole exterior surface is covered with diminutive tubercles, the above mentioned smooth spots interspersed on the central disk. A little below the middle there are two lengthened pits, so shallow that they often are not discernible. In a specimen of *Chirurus spinulosus* NIECZKOWSKI from Estland there are two, almost 3 mm. long dots, one on each side above the maculæ as the [fig. 21](#) shows and in several others of this generic group there are also indications of similar. I cannot compare these dots with anything more than the lengthened spots visible in *Acidaspis crenata*.

The maculæ are wanting in this species, but the cephalic eyes are well developed. A section of them [fig. 19](#) shows the enormous difference of the shell in the eye and the surroundings where the shell surpasses it many times in thickness. The eye consists in fact only of the ovate, beadlike lenses of which a string is seen sectioned in the figure mentioned. More enlarged ([fig. 20](#)) a nucleus is visible in each lens, and in a horizontal section ([fig. 18](#)) a little below the surface, where they are more pressed against themselves, they have a polygonal shape. The [figure 17](#) seems to represent lenses that have been much changed interiorly, having only a narrow zone left of the primary structure.

Chirurus glaber ANGELIN.

[Pl. III fig. 11.](#)

As a sample of the shape of the surface of the eyes in this genus, we have given a figure, showing the small rounded lenses.

Chirurus ornatus DALMAN.

[Pl. III figs. 12-14.](#)

The cephalic eyes have globular lenses, of more than double the size of those in *Chir. clavifrons*. The difference between the lenses and the surrounding shell amounts at the highest to thrice the former, while in *Ch. clavifrons* it is at least seven times as much. In a horizontal section the lenses have a little, darkish irregular nucleus surrounded by a radiated structure ([fig. 12](#)).

The maculae ([fig. 14](#)) are slightly concave, oblong, smooth, surrounded by comparatively large granulations.

Chirurus speciosus DALM.

[Pl. IV fig. 1.](#)

The exterior surface of the hypostoma is sparingly covered by granules of larger size than in other *Chiruri*. The maculae as seen in a cast are prominent near the posterior border, sunk in their centre and surrounded by a narrow distinct border line. In a horizontal section it has not been possible to detect any peculiar structure, only a dark border surrounding the interior clear surface.

***Chirurus spinulosus* NIECZKOWSKI.**

[Pl. III fig. 15.](#)

We have had the interior surface of a hypostoma at our disposal from the lower Siluria, of Esthonia (Kuckers C²) belonging to Dr. G. HOLM. The maculae are well preserved, standing out black on the white surface, oblong, 1,5 millim. in greatest length ([fig. 15](#)). There are indications as of a great number of small lenses on these maculae. Remarkable are the two elongated

spots, almost 3 mm. in length, above these maculæ, of which already mention has been made under Ch. clavifrons.

Chirurus sp. indet.

[Pl. III fig. 16.](#)

The hypostoma of an undescribed species from Öland. The maculæ are lengthened, ellipsoid and smooth tubercles lying in a groove. The finely granulated surface of the hypostoma shows rare smooth spots interspersed, being on the inside smooth pits, from which again in casts of the hypostoma larger tubercles are moulded. Of the same nature are those that are visible on the nucleus of Chir. speciosus and others. There are still some undescribed Lower Silurian species, for instance one nearly related to Ch. conformis with tubercular maculæ. A large hypostoma from the Leptæna limestone of Dalecarlia, 33 millim. in length, has a longitudinal macula, 2 millim. in length, as seen on the inside of the fragment.

On [pl. VI fig. 10](#) a little hypostoma is delineated that shows some resemblance with that of the Chirurus, but probably belongs to some other, unknown genus. It is Upper Silurian, found at Mulde in Fröjel, Gotland. It is of an elongated ovate shape, with the anterior border faintly arched, the anterior wings broad, truncate. A narrow elevated border surrounds the lateral and posterior margins. The somewhat vaulted surface is covered by fine granulations, and the macula, situated on equal distance from the anterior and posterior margins are ovate, smooth and directed obliquely inwards. There are two minute pointed processes on each side of the lateral borders and one on each side of the pointed posterior margin.

Ctenopyge LINSN.

Ct. spectabilis BR. (I, pl. XII, fig. 12 a).

A small lengthened, apparently smooth hypostoma with two globular tubercular maculæ near the posterior border. No mention made of them in the description of BRÖGGER. Cambrian. I am not, however, quite sure whether this really belongs to a Ctenopyge, as it rather more resembles the

type of a *Peltura* as shown in *Pelt. scarabæoides*. As *Ctenopyge* is so nearly related to *Sphærophthalmus* it could be expected to see its hypostoma of the same type and the maculæ entirely wanting. The eyes of this genus have been described above at [page 29](#).

Cybele LOVÉN.

Cyb. bellatula DALMAN.

[Pl. IV fig. 2.](#)

We give a new figure of the hypostoma to complete that of FR. SCHMIDT, pl. XIII fig. 9. The anterior wings are larger and obtuse, excavated below and the border rim is not so distinct all round as he has figured. There are two pair of oblique, lengthened grooves above each other. The opposite do not, however, join to form a coherent groove across the median field of the hypostoma, which is level between them. No maculæ. Nearer the anterior border close below the wings, there is a small roundish, dark spot. The surface of the hypostoma is granulated.

Cyphaspis ANG.

Cyph. elegantula ANG.

[Pl. III figs. 22-25.](#)

The hypostoma is elongated, anterior margin evenly arched, anterior wings triangular, lateral margins faintly curved as well as the posterior margin, at both sides of which there is one diminutive spine. The posterior wings are short forming a spiny process from the lateral margin. The surface of the hypostoma is smooth with a shallow transverse groove on the inferior moiety and two corresponding lateral impressions below it, one on each side. The elevated lateral borders are longitudinally striated by a few terrace lines. There are no maculæ visible. The eyes consist of short hexagonal prisms ([figs. 22, 23](#)) nearly resembling those of *Proetus*.

Dalmanites EMMRICH.

In this large genus the hypostoma has been figured in a great number of species. But it is only in the following that we, at least in the figures, if not in the descriptions, are able to detect the maculæ.

D. atavus BARR. Suppl. pl. 5 fig. 14.

D. Calypso J. HALL, vol. VII, pl. XI A. fig. 21 p. 66 »Postero-lateral pits, moderately strong and elongate«. There are casts of the two well distinct maculæ.

D. caudatus BRÜNN.

a. FORBES and SALTER in British Organic remains, Dec. II pl. 1, fig. 3 give a good figure, showing the oblique pitlike maculæ. On page 2 is told that »a pair of lateral strong indentations indicate a second furrow above« (»the transverse furrow between the tip«).

b. SALTER, pl. 3, figs. 7, 8, bad figures, the description verbally the same as above. What ANGELIN pl. VIII, fig. 2 c gives as the hypostoma of »Ph. caudata« is in fact the hypostoma of a species of Lichas.

D. Mac Coyi BARR. Supl., pl. 13 f. 32. This magnificent hypostoma belongs to the same group as *D. micrurus* HALL as to which we may hesitate whether there not be a double pair of maculæ above each other near the posterior border. In this species, however, there is the transverse lower groove and above probably the two maculæ united by a groove so closely as to resemble the lower groove.

D. micrurus J. HALL, Pal. N. Y. III pl. 74 fig. 20. Shows, as it were, a double pair of maculæ, very like *D. spinifer* BARR.

D. rugosus BARR. pl. 24 f. 23. The two maculæ in the ordinary place, without being united by a groove.

D. socialis BARR. pl. 26 f. 21. Rather of an uncommon shape, not so triangular as in the other species, the two maculæ distinct as narrow slits. P. 553 »Vers l'extrémité du corps central, on aperçoit de chaque côté, près du bord, une impression oblique, alongée et arquée«.

D. spinifer BARR. Pl. 25, fig. 20. A large beautifully preserved hypostoma, shows what I think we positively must interpret as a double pair of maculæ

close above each other. As stated above there is reason to believe that in other species of *Dalmanites* there also are indications of four maculæ, though not so evident as in this species. BARRANDE says nothing about this remarkable feature.

***Dalmanites imbricatus* ANGELIN.**

[Pl. III figs. 43, 44.](#)

The interesting image, as given by the agglomerated eyes, is represented in the magnified [figure 43 Plate III.](#) The lenses lie there separated from each other at much varying distance, some in close contact. The presence of the delicate covering membrane is clearly seen, being of white colour contrasting with the black, glossy lens. It is much lacerated and only preserved around the periphery. The size of the ocelli is on an average 0,5 mm., a little one has 0,3 mm. The surface between the ocelli is most finely granulated. The granules scarcely attaining the fourth of the granules in *Dalm. vulgaris*.

In a vertical section the regular biconvex lenses are seen to be covered with the extremely thin integument, which is a direct continuation from the test. The test between the lenses is perforated by some longitudinal tubes, as usual in the skeleton of the trilobites. We have not found any hypostoma with the maculæ.

***Dalmanites obtusus* LDM.**

[Pl. III figs. 45, 46.](#)

We have not succeeded in finding any hypostoma, but as the structure of the cephalic eyes is sufficiently well preserved, we here describe it. The lenses seen from the surface look globular, and in a vertical section they are ovate and in a horizontal circular. They are covered by a very thin and delicate membrane, that envelops their superior moiety completely and between the eyes it is enclosed by the surrounding test and growing out from it. In the horizontal section, [fig. 45](#), it looks as a circular frame around the lenses and the membranes of two contiguous lenses sometimes lie so close, that they entirely fill up the interspace between the lenses, leaving only a narrow slit

between them marking their boundaries. The space between the lenses, fig. 46, is porous, being perforated by tubes, which continue vertically down through the shell of the cheek.

Dalm. sclerops DALMAN.

[Pl. IV fig. 3.](#)

Commonly this species has been ranged with Phacops, but it is evident that it as to the conformation of its glabella and the head in general is highly discrepant, and in these respects is more concordant with Dalmanites, though still deviating as for instance in the pygidium and some details of the head. Therefore it may be justified with FRIEDR. SCHMIDT^[41] to consider it as a generic division, though not, as he has it, as an independent genus, but as a subgenus of Dalmanites.

^[41] Revision, I, p. 76.

The granulated hypostoma differs from that of Dalmanites proper in having the anterior margin strongly arched, the pointed wings almost on the median line of the hypostoma. There is one groove near the posterior border and the maculæ lie as two, narrow long, crescentic grooves close below the wings. The few sections give no clear idea of any structure.

The cephalic eyes are constructed on the same plan as in the previous species, the lenses being surrounded by a frame.

Dalm. vulgaris SALTER.

[Pl. III figs. 47-52.](#)

The hypostoma has the common shield-like shape, [fig. 51](#), anteriorly slightly arcuated, the short obtuse wings curvated towards the interior surface. It is finely granulated. The groove does not go so near the cuspidate posterior margin as in other species. The two maculæ, [fig. 52](#), are narrow and elliptic, somewhat convex and lie in a little concavity, a narrow smooth space around is devoid of the granules, which cover the surface of the rest. They show absolutely no structure, excepting a few indistinct blackish spots.

The cephalic eyes have been often described and delineated. Most detailed are the descriptions and figures given by SALTER in the Memoirs of the Geological Survey Dec. II. pl. 1. In the figure 4 a part of the surface is shown in well preserved state. He says that »the cornea is ... present and distinctly convex over each lens, the intermediate portions being ornamented with tubercles and granules». He thinks also that there are undeveloped or small lenses between the larger. His fig. 5 represents the frames around the lenses, these being probably lost. His explanation of fig. 6 I cannot understand. There are weathered or spoilt lenses figured.

There is no doubt that the granulated surface of the head continues between the more or less free lying lenses as shown here in a new figure taken from a specimen from Dudley, [fig. 47](#). The granules of the surface are larger and more rare between the ocelli than for instance in *Dalm. imbricatus*. Small lenses, undeveloped as SALTER calls them, may in fact only be such granules of somewhat larger size than usual. Lenses having the appearance as in SALTER's fig. 6, as if a covering was partially destroyed and the lens visible below it, I have also found in Swedish specimens, as represented in [fig. 47](#). In a vertical section, [fig. 50](#), a delicate covering integument is seen and the oval lenses lie regularly with their frames around them quite as in *Dalmanites obtusus* and in horizontal sections, [figs. 48, 49](#), there are also the same sort of black, irregular dots, being the sectioned tubes of the surface between the lenses. But in another feature there is great interest. Beneath each lens there are fascicules of tiny rods, twice as long as the lenses. They are represented in a vertical section in [fig. 50](#) and in a tangential section in [fig. 49](#).

As our imperfect knowledge of so delicate anatomical structures in fossil crustaceans does not admit of secure comparisons with the visual organs of recent crustaceans, no suggestion can be given to interpret their nature. Probably they have no connection with the structure of the eyes and it is not even certain that they are of organic origin.

Dysplanus BURM.

If there were no other characters to distinguish this genus from *Illænus* its strangely deviating hypostoma must do it. We have examined two species and in both the hypostoma is almost oval, evenly rounded both anteriorly

and posteriorly. Near the anterior border two narrow, long hornlike wings project. They are flattened, thin and lamellar from the basis and end in a hollow, acuminate point.

D. centrotus DALM.

[Pl. III figs. 53-56.](#)

This species has the maculae placed near the posterior border, in a semilunar flat, devoid of the terrace-lines. The maculae are elongated tubercles and do not show any structure when sectioned. The cephalic eyes have a thin membrane covering the short cupshaped lenses, which, closely packed, in the transverse section give the usual polygonal mosaic of hexaeders. They have a radiate structure, when decayed.

D. ladogensis HOLM.

[Pl. III figs. 57, 58.](#)

A specimen from Öland has the maculae lying in the large field of terrace lines above the smooth zone at the posterior margin of the hypostoma. They are larger than in the former species, pearshaped, smooth and no structure visible.

Encrinurus EMMR.

I cannot find any former figure indicating the maculae of the hypostoma in this genus than the only one of ANGELIN, Pal. Scand. pl. IV fig. 6.

As to the shape of the singularly formed hypostoma in the three Swedish species *Encr. punctatus*, *obtus* and *laevis* we have found the anterior border so characteristic in them that it is sufficient to distinguish them from each other and to annul the doubts concerning the distinction of *Encr. laevis* from *Encr. punctatus*. We therefore in [pl. IV figs. 12-17](#) give the anterior margins of these three species, along with the profiles.

Encrin. punctatus WAHLENBERG.

[Pl. IV figs. 5-9, 12, 13.](#)

The surface of the hypostoma is so finely and obscurely granulated that it seems to be almost smooth. It is in the shallow groove above the lamellar tongueshaped posterior border that the elongated tubercular maculae are situated. They are prominent and in horizontal sections they have a surface mottled by irregular black and white specks, the whole surrounded by a white border, [fig. 6](#). This border and the other surface is pierced by series of parallel slitlike pores, [fig. 7](#), a feature peculiar to this genus. The vertical sections indicate a chain of indistinct prisms with a black, rounded central hollow between thick strata of clear whitish shell substance, [figs. 8, 9](#). The shell substance on the sides is pierced by the straight, black tubes.

The cephalic eyes have the prisms indistinctly developed or badly preserved with a diameter of 0,04 millim. and height of 0,07 millim. In *Encr. lævis* the prisms are more distinct and larger with a diameter of 0,00 millim. ([pl. IV figs. 10-11](#)).

The maculae are well developed, ovate, in *E. lævis* and *obtus* and in the same position as in *E. punctatus*.

Griffithides PORTLOCK.

The hypostoma of *Griff. globiceps* PHILL. shows according to the figure given in H. WOODWARD'S work pl. VI fig. 5 two somewhat obscure maculae, but no mention is made of them in the description.

Harpes GOLDF.

Harpes d'Orbigny BARR. NOVÁK II Taf. I fig. 4, with small black maculae. *Harpes venulosus* CORDA. NOVÁK II, Taf. I figs. 1, 2. Two narrow, horizontally placed maculae.

Harpina BARR.

H. prima BARR. NOVÁK II Taf. I fig. 5. Two maculae in continuation of a narrow groove, at the base of the central globosity.

Holmia WALCOTT.

Holmia Lundgreni MOBERG, Sveriges äldsta trilobiter, Tafl. 14, fig. 10, 11, 12. The maculæ are much prominent and tubercular, elongate narrow, oblique but not mentioned in the description. Hypostoma probably belonging to the terrace-line group. As well seen in original specimens and in the figures in Dr. HOLM's paper on *Olenellus* (*Holmia*) *Kjerulfi* there are also in this species two distinct macula; near the posterior margin of the hypostoma, above the marginal groove.

Homalonotus KÖNIG.

Maculæ have been previously observed in the following species.

Hom. delphinocephalus GREEN, HALL Pal. N. York vol. II pl. 68 fig. 11, incomplete figure with two well marked tubercular maculæ. No description.

Hom. Knighti KÖNIG. SALTER II, pl. 12 fig. 10. Maculæ well expressed as oblique tubercles. On page 120 it is said »with a pair of lateral tubercles well developed (as in *Asaphus* ...)».

This species is identical with ANGELIN's *Homalon. rhinotropis*, Pal. Sc., pl. XX fig. 1 e and his figure has two hollow maculæ seen from the inside of the hypostoma.

Homalonotus Knighti KÖNIG (= *Homal. rhinotropis* ANGELIN).

[Pl. IV figs. 20, 21.](#)

The maculæ lie as oblong, smooth tubercles surrounded by the irregularly grown granules of the surface, just below the central globosity of the hypostoma. In a preparation from the interior surface of the macula, as seen in transmitted light, [fig. 21](#), this is white with only a few grey spots and on the superior border black streaks, lying obliquely. The substance of the shell around the macula is perforated by the minutest pores amongst which a few larger are intermingled. The granules are also perforated as those in

Calymmene and the tubes continue through the shell, visible by their black colour.

Hysterolenus MOBERG.

Hyst. Törnquisti MBG., En trilobit från Dictyograptus-skiffern p. 320. In the descriptive letter-press, there is said, that in the exterior (lateral) parts of the anterior groove of the hypostoma a distinct tubercle is to be seen on each side. But none of the figures given, Pl. 17 figs. 6, 7, shows them or at least very indistinctly.

Illænus DALMAN.

It is remarkable, considering the great number of species in this almost exclusively Lower Silurian species, that there is none but a single species in which the maculæ hitherto have been delineated. This is *Illænus angustifrons* var. *depressus* HOLM I, pl. VIII fig. 18. A little, fragmentary hypostoma with globular maculæ.

Illænus Chiron HOLM.

[Pl. IV figs. 22-25.](#)

The hypostoma has the shape characteristic of all true *Illæni*, viz. a straight anterior margin with two large flat rectangular or nearly quadratic wings, a globular or spherical median field, forming the chief portion of the exterior surface, which is covered by some rare, fine, terrace lines. In some other *Illæni* the surface is smooth. At the posterior base of the median field the two maculæ are placed, tiny, oblong, but lying on the same level, at right angles to the median longitudinal axis of the hypostoma. They are covered with lenses of irregular and indistinct shape [fig. 25](#). I do not think that they in this and the other species of *Tiberius* are lenses of the same conformation as in *Bronteus*, they are rather the tops of the subjacent lenses which are of the prismatic form. The cephalic eyes do not show any lenses on the outside. They have a narrow, opaque exterior integument, hiding hexaedral, thickly packed, straight prisms, forming a stratum in thickness of 0,2 millim.

The hypostoma consists of several thin layers of superposed calcite, which easily peel off, so that it is seldom that the true exterior sculptured surface remains.

Illænus Esmarki SCHLOTH.

[Pl. IV figs. 26-33.](#)

This the most common of all the Illæni has in several specimens shown the hypostoma with the maculæ in situ. The hypostoma is smooth excepting on the marginal ridges where there are some terrace lines. On the median axis of the central gibbosity ([fig. 30](#)) five faintly visible transverse folds are situated, a curious feature which we have not observed in any other trilobite. The maculæ are exceedingly small, [fig. 31](#), elongate, spindleshaped and acuminate in both ends. On their surface there are accumulated several lenslike globules. As the horizontal sections show there are, however, no regular lenses, but a network of polygonal meshes with radii in the hollows, [fig. 32](#). The perfect identity of this structure with that of certain states of the cephalic eyes will at once be perceived by comparing this figure with [figure 26](#) representing the same sort of polyedric meshes, though a little more regular in the cephalic eye, where, however, the irregularity also prevails near the periphery of the eye. If we now compare the sections of the cephalic eye of Illænus Chiron and the section of the regular and nearly unchanged eyes of Ill. Esmarki with the second section of the latter, we cannot avoid the conclusion that we in those of Ill. Chiron, [figs. 22, 23](#), and the first of Ill. Esmarki, [fig. 27](#), see the primitive and intact state of the lenses and in the last sections of Ill. Esmarki, [figs. 26, 28](#), the changed and deteriorated state of the same prismatic lenses.

The vertical section of the hypostomic eyes [fig. 33](#) shows in their present changed condition an evident longitudinal direction of the chief elements, quite as the lenses of the eyes. But in how far there ever has existed such prismatic lenses, now changed, we have no evidence definitely to ascertain, but the probability seems to be great that this was the case.

Illænus gigas HOLM.

[Pl. IV figs. 34-37.](#)

The maculæ in this as well as the other Illæni are so faintly elevated and so inconspicuous that the greatest attention is needed to find them.

The maculæ are elongated, narrow, elliptic on the same level and parallel with the straight anterior margin of the hypostoma. A cast of the left macula is pitted by numerous marks of the lenticular globules of the surface, [fig. 37](#). In the cephalic eye the prisms are seen translucent beneath the cornea. Near the superior border they are larger, [fig. 34](#), almost the double size of those near the inferior border, [fig. 35](#).

Illænus sphæricus HOLM.

[Pl. IV figs. 42, 43.](#)

Comes near to the previous species. The globules on the surface of the elliptic maculæ are not, however, spread over the entire macula, leaving only a narrow border all around free.

The lenses in the cephalic eyes [fig. 42](#) are globular as seen on the surface, alike those of the maculæ.

Illænus Roemeri VOLLBORTH.

[Pl. IV figs. 38-41.](#)

The maculæ lie on elevated tubercles and are of a strange shape, being sharply pointed outwards, rounded inwards. That in [fig. 39](#) is still more peculiar with a narrow stripe on the interior and upper margin. The lenses are larger and more distinct than in any other species of this genus.

Beside the species now mentioned we have also observed hypostomas with maculæ in *Illænus laticlavus* EICHWALD from Estland, in *Illænus Linnarssoni* HOLM, and in an unnamed species from Dalecarlia figured without macula by Dr. HOLM in his memoir on the Illænidæ pl. III fig. 22. We have not seen any lenses on the maculæ of these three species.

Lichas DALMAN.

We have not been able to find any notice about the exceptional maculæ of this genus. There is only a figure of the hypostoma of *Lichas hylæus* HALL Pal. of N. York vol. VII pl. XXV fig. 5 which would seem to show a pair of narrow, oblique maculæ, but these are quite without resemblance with the maculæ of the true *Lichas*.

The hypostoma of *Lichas* is of a characteristic type, broad, with the incised posterior margin and its two lappets reminding of *Asaphus*, with its large rounded, faintly elevated median part near the anterior margin reminding of *Illænus*, as also do the lenticles of the maculæ. The maculæ are of a minimal size, scarcely 1 millim. in diameter and thence easily overlooked. [Pl. IV figs. 46, 48.](#)

As the hypostomas in this genus generally are found detached it is often difficult to decide to which species they have belonged. So it is with that delineated in [Pl. IV](#) fig. 44 natural size. The two maculæ are situated near the centre of the hypostoma, just below the grooves which surround the large, globular disc. They are sunk in a little cavity, rounded and entirely covered by globular lenses ([pl. IV fig. 46](#)).

In another detached hypostoma of an unknown species the macula is smaller and the few lenticular globules larger than in the other, [fig. 48](#).

***Lichas latifrons* ANGELIN.**

[Pl. IV fig. 49.](#)

The maculæ, seen in a thin section from the inside, are reticulate, [fig. 49](#), or of a pattern exactly like that in the changed eyes or maculæ of *Illænus* and even *Asaphus*, and thus indicating that it was composed of short prismatic lenses. The vertical section of a macula of a specimen from another locality rather suggests the presence of large sphærical lenses than elongated prisms. The great thinness of the macula in contrast to the excessive thickness of the shell around the eye is seldom so evident as here.

A specimen of an unknown species, found detached, has the macula? unlike the other species, oblique with the inferior end directed inwards, placed on the top of a sort of a little elevation, [fig. 47](#).

Megalaspides BRÖGGER.

This author has a figure of *M. dalecarlicus* in II pl. 1 fig. 19 and also of an unnamed species fig. 20, both showing globular maculæ. In HOLM's original description of his *Megalaspis dalecarlicus* (Trilobiten des Phyllograptusschiefers Dalecarliens figs. 8, 9) the same hypostoma is also figured. This genus seems to have a closer affinity with *Ptychopyge* than with *Megalaspis*.

Megalaspis ANG.

The following figures of hypostomas with maculæ have been published.

Megal. limbata by BRÖGGER I, tab. XII, fig. 10, and in II pl. 2, fig. 22, somewhat differing from the first figure.

Meg. planilimbata BRÖGGER II tab. 2 figs. 21, 21 a. Both figures incomplete and incorrect.

The hypostoma has a large central ovate gibbosity with the maculæ near its basis, surrounded by the winglike expanded limbus.

Megal. attenuata WAHLENBERG.

[Pl. V figs. 1-6.](#)

The long and narrow maculæ are crescent-shaped and project beyond the hypostoma supported on the superior surface of a sharp extenuated edge where they rest as on a shelf. They consist, as seen in vertical sections, [figs. 5, 6](#), of alternating rows of light and black streaks, and in a horizontal section the spongy network appears, as it is found in *Asaphus* and others and thus indicating the tendency of this organ to assume a prismatic structure though undeveloped.

The cephalic eyes in *M. attenuata* form the most regular hexaedra figs. [1 a](#), short and covered by a thin cornea. The border zone of the eyes has a structure exactly like that of the maculæ, the same as in the spongy zone of *Asaphus*.

Megal. limbata BOECK.

[Pl. V fig. 7.](#)

We give a new figure of BRÖGGER's original specimens, as his figure 23, Tab. II is incomplete and incorrect. The maculæ are situated a little higher up on the sides of the central gibbosity.

Megal. planilimbata ANG.

[Pl. V fig. 8.](#)

The maculæ are crescent-shaped and not so prominent as in *Meg. attenuata* and the visual spot in the same position as there.

To observe is that the general structure of the hypostomic shell in this and many other species is prismatic.

Nileus DALMAN.

Hypostoma with maculæ figured in *N. armadillo* by ANGELIN pl. XVI fig. 5 c, and BR. II pl. 3 fig. 40.

The hypostoma in this genus is transversal with large lateral expansions, the central convexity only faintly elevated and almost coherent with the posterior part. The maculæ lie exactly on the horizontal median line of the hypostoma. The anterior wings are well developed like slightly bent acuminate horns and the posterior wings are conical points. The terrace lines are transverse and a little wavy.

Nileus armadillo DALM.

[Pl. V figs. 10-15.](#)

There is some variability in the hypostoma of specimens from different localities. The most known and numerous specimens are from the renowned localities of Östergötland. They are decidedly transverse, the proportion of the height to breadth is as 3 to 5 and the narrow border runs all round. In specimens from Kongslena, Vestergötland, the proportion is as 15 to 22, in

specimens from Sandvik, Öland, again, we have 13 to 23 or 6 to 11. The maculæ, [fig. 14](#), are oblong, inwardly pointed, quite smooth, with a little concavity in their centre, and their surface is slanting obliquely towards the surface of the hypostoma, almost as in Niobe. There is no trace of any structure.

The cephalic eyes are covered by a comparatively thick membrane, a direct continuation of the general test, [fig. 12](#), hiding a row of prismatic lenses. In some specimens elongate, clear, crystalline rods continue downwards from the bases of the lenses and are probably nothing but inorganic crystals of calcareous spar, [fig. 11](#).—The eyes are larger comparatively than in other species.

Nileus (Symphysurus) palpebrosus DALMAN.

[Pl. V figs. 16-18.](#)

As to its general form the hypostoma resembles that of the preceding species, but there is no border line, which is so characteristic to the former. The maculæ have the same form and are likewise situated on the median line. The cephalic eyes have prismatic lenses with much convex bases, in a transverse section they are hexaedral and of internal radiate structure.

Niobe ANGELIN.

The following figures have formerly been given of its hypostoma with maculæ.

N. emarginula BR. II tab. 2 fig. 33.

N. explanata BR. II tab. 2 fig. 35.

N. frontalis BR. II fig. 37.

N. insignis BR. I tab. IV fig. 1 d, 11 tab. 2 fig. 28.

N. læviceps BR. II pl. 2 fig. 34.

The maculæ are distinguished from all other sorts of maculæ, excepting those of Megalaspis, in being, as it were, shelved on a sort of support, formed through the bulging of the hypostomic surface just below them, one

such console for each. Their surface lies consequently almost rectangularly to the adjoining surface of the hypostoma. They are white and smooth.

Niobe frontalis ANG.

[Pl. V figs. 19-21.](#)

The pocketlike consoles are shorter and broader than in *N. læviceps* and the maculæ also larger and of an elliptic shape. In a horizontal section near the surface the macula has a speckled appearance of black dots, indicating an undeveloped prismatic structure, much more primitive than in the *Asaphidæ*.

Niobe læviceps ANG.

[Pl. V figs. 22-26.](#)

It differs from the preceding through longer consoles below the maculæ, which are oval and in a horizontal section exhibit the same spongy structure, [pl. V fig. 25](#). The cephalic eyes consist of polyedric prisms, which through deterioration show a central cylindrical core surrounded by darkish matter like *Asaphus*. The passage from solid, homogenous prisms to changed ones is most evident in the [figure 22](#).

Ogygia BRONGN.

Figures of the maculæ are found in

O. Buchi SALTER, Mem. Geol. Surv., Dec. II pl. VI fig. 3 p. 2, where it is stated that »there are two transverse furrows near the apex, with compressed tubercles between them» There are two narrow crescentic maculæ and between them a little lower a longer, crescentic ridge, which may be the ridge that in other species connects the maculæ. Moreover, the shape of the hypostoma is not concordant with that of the other species. In Monogr. Brit Trilob. pl. 15 figs. 2, 3 there is only a reproduction of the former figure.

O. corndensis MURCH. SALTER in Monogr. pl. 16 fig. 10 interior side of the hypostoma with two lateral maculæ. In the description two pairs of furrows the uppermost is the maculæ. H. WYATT-EDGELL »On the Genera of Trilobites Asaphus and Ogygia and the Subgenus Ptychopyge» in Geol. Magaz. 1867, p. 14, 15 fig. 2 probably a cast with two oblique impressions of maculæ.

O. dilatata (Asaphus) SARS var. Sarsi ANG. SARS in Isis 1835 p. 342, pl. IX fig. 11. The figure has two small maculæ near the posterior margin and the author says »mit einer starken Vertiefung oder Einschnitt an jeder Seite des Endes». ANGELIN pl. XLII fig. 1 *b*. BRÖGGER's fig. 38 pl. 3, in II, collies near to the specimen, which is described below, also belonging to Dr. G. HOLM.

***O. dilatata* var. *Sarsi* ANG.**

[Pl. V figs. 27, 28.](#)

The hypostoma is in so far deviating that the terrace lines have a nearly vertical direction and in this respect much resemble those of the Phillipsiæ and the Proeti. The maculæ, near the exterior angles of the border groove are pear- or spoon-shaped with the apex directed downwards and outwards, and in consequence the longitudinal axis quite opposite to that of the usual direction. The surface is smooth, a little concave. No structure has been discovered. From the pointed apex a slightly elevated ridge runs along the posterior border of the hypostoma and joins the apex of the other macula.

Paradoxides BRONGN.

The oldest record of any hypostoma at all amongst the trilobites is that given by WAHLENBERG in his Petref. Suecana (1818) p. 37 Tab. 1, fig. 6, which he, however, considered as the head of the trilobite which he named Entomostracites bucephalus. Figures with maculæ have been given of

Par. Davidis SALTER Mem. Geol. Survey Dec. XI pl. X, fig. 3, a fine figure of a hypostoma with two large oblique maculæ. LINNARSSON in »de undre Paradoxideslagren vid Andrarum» pl. II fig. 2 delineates a gigantic hypostoma with two crescentic tubercular maculæ, having 10 mms. in length.

Par. Forchhammeri ANGELIN Pl. II fig. 3. BRÖGGER, Paradoxidesskiffr. vid Krækling tab. II fig. 10 two marks somewhat different from ANGELIN's figure. LINNARSSON 1. c. pl. I figs. 9, 10 two hypostomas with maculæ.

Par. Tessini, var. *Wahlenbergi* ANGEL. Pl. I a, fig. 1 b.

Par. Tessini, var. *oelandicus* ANG. Tab. I a, fig. 2 b. ANGELIN's figure of this hypostoma is incomplete in so far that he had not found the peculiar falciform horns on both sides of the posterior border. Pl. V, fig. 33.

Par. rugulosus CORDA has, according to BRÖGGER, Krækling, pl. II fig. 2 two long, straight callosities, probably corresponding to the maculæ in other Paradoxidæ.

CHR. BOECK gives in Mag. for Naturvidenskaberne Bd 8, 1828, in his paper on the Trilobites on the plate fig. 16 the hypostoma of a Bohemian species with well developed concave maculæ, probably a cast. He compares it with the Entom. bucephalus of WAHLENBERG, but says he cannot explain its nature.

The maculæ of *P. oelandicus*, pl. V, fig. 34, are tubercles, oblong, smooth, and have invariably an oblong scar along the centre, where probably once a thinner membrane, as in the cephalic eyes of Calymmene, contained the lenses.

Peltura M. EDW.

Peltura scarabæoides WAHLENB.

[Pl. III fig. 42.](#)

We have given a figure of the hypostoma, showing two large, elongated tubercular maculæ or rather resemblances of such, as they are covered with terrace lines as well as the rest of the surface. They have, however, the same position as real maculæ might have, but are directed in a line parallel with the longitudinal axis of the hypostoma. The cephalic eyes have been described above at [page 29.](#)

Phacops EMMRICH.

Maculæ figured of Phac. (Acaste) Downingiæ. SALTER Monogr. pl. 2 fig. 34 b, p. 25 »high up on each side a small tubercle«.

Phac. rana HALL Pal. N. Y. vol. VII, pl. VIII A, fig. 18. Hypostoma with the large commalike maculæ high up.

Of the following species *Ph. cephalotes* and *Ph. macrophthalmus* belong to the genus *Phacops* proper, while *Phac. quadrilineata*, as well as *Ph. Downingiæ* are of a quite different generic type. The name *Acaste* proposed for them is not good as LEACH in 1817 named a *Cirrhipe* as *Acasta*.

Ph. cephalotes BARR.

[Pl. V figs. 35, 36.](#)

The hypostoma is elongate, almost triangular, tricuspidate at the posterior border, a little above this there is a crescent-shaped ridge, and still a little higher above this and the median horizontal line, the two macula, as diminutive tubercles. Seen with magnifying powers they are slightly oblong or reniform.

Ph. macrophthalmus BURM.

[Pl. VI figs. 1-9.](#)

The triangular hypostoma has a little below the median horizontal diameter two shallow grooves, and a little above the same diameter the extraordinary minute maculæ may with some care be observed, [figs. 8, 9.](#) They are visible as a small bare, smooth spot amongst the curious elevated and incoherent terrace lines of the surface. The exceedingly small size of these maculæ naturally does not allow any research of their structure. They are rather in this species and many of the congeneric to be regarded as small ocelli. In some specimens these spots are quite rudimentary and are reduced to the smallest area possible. The anterior wings are of a peculiar earlike shape, finely striate, enlarged and flat near the hypostoma and ending in a little hollow horn ([figs. 6, 7.](#)). A little below it, the posterior wing emerges as a small angular knob. The aggregate cephalic eyes are more distinct.

As the vertical section learn us the regular ovate lenses consist generally of clear, crystalline calcareous spar and are on the surface covered by an extremely thin membrane, not a common one for all, but a separate cornea for each lens, which envelopes its superior moiety and continues down as a thickened appendix between the lens and the interstitial test ([figs. 3-5](#)). In the horizontal sections it surrounds the lenses as an annular wall. Around some lenses there is as if still another ring lay outside this, but more indistinct, probably only clearer portions of the test surrounding them. The cheeks close to the lenses are indistinctly perforated by longitudinal tubes.

Phacops (Acaste) quadrilineata ANGELIN.

[Pl. V figs. 37, 38.](#)

We have not succeeded in finding any well preserved hypostoma.

The lenses of the cephalic eyes resemble much those of the preceding species though more elongate and ovate. They are covered by a membrane which surrounds their upper moiety and between the lenses joins the interstitial test. In the horizontal section these membranes surround the lenses as an annular wall.

Phaëtonides BARR.

Ph. Stokesi MURCH.

[Pl. VI fig. 11.](#)

The hypostoma somewhat reminds of that of Calymmene, has projecting angles and a prominent knob below the straight anterior border. The two oblique and oblong macula are tubercular and placed near the lateral borders. This hypostoma is upon the whole much related to that of Phillipsia and Proetus.

Phillipsia PORTLOCK.

DE KONINCK figured a hypostoma of *Phillipsia* in his »Description des Animaux Foss. de Belgique« as *Cyclus Brongniartianus* pl. LII fig. J and NOVÁK in III figs. 6, 7 copied it and corrected the error. It is provided with two globular maculæ united by a curved ridge.

Phillipsia Eichwaldi WOODW. NOVÁK III fig. 5 a hypostoma with globular maculæ. We have been able to examine three different, unnamed or undetermined species, which we distinguish by numbering them.

Phill. No. 1 from the Keokuk group of Crawfordsville, N. America. The hypostoma is broader than in the other species. The nearly horizontal elongated maculæ placed near the median line, close to the lateral margins.

Phill. No. 2 ([pl. VI, figs. 15-18](#)) probably from the Carboniferous formation of Belgium is of a lengthened form. The elliptic maculæ in the groove, below the central elevation. They are smooth and surrounded by a flat border.

The cephalic eyes are much destroyed, but seem to have consisted of somewhat semi-prismatic lenses or prisms of the same type as *Dysplanus*.

Phillipsia No. 3 ([pl. VI figs. 12-14](#)) from the Carboniferous strata of Beeren Eiland communicated by Hr J. G. ANDERSSON. It comes near the preceding, is lengthened, but has broader and shorter anterior wings and the inferior surface just above the posterior border peculiarly pitted. The macula lying above this pitted field are ovate. The lenses of the cephalic eye are hemispheric on the surface.

Platymetopus ANGELIN.

Pl. planifrons ANGELIN.

[Pl. IV figs. 50, 51.](#)

This genus may well be kept distinguished from *Lichas* in consequence of its peculiar hypostoma and its still more peculiar maculæ, not to remind of the other deviating features. The maculæ lie at the end of narrow, sigmoid grooves of their own above the long transverse groove, separating the posterior border from the anterior elevated moiety. They are elongated,

curved and entirely covered with a great number of irregular lenticles, translucent through the thin covering membrane.

Pliomera ANG.

Pliomera Törnquisti HOLM in Trilobit. Dalecarliens p.
5 fig. 2.

While Pl. Fischeri and Pl. actinura not have any hypostomic maculæ, this species, which Dr. HOLM with some doubt joins in the same genus, has a hypostoma, which as to its general shape resembles that of both the mentioned species, evenly rounded at its posterior border, with a shallow groove encircling the central field. A little below the horizontal diameter of this the maculæ stretch obliquely inwards and downwards. They rest on a little elevation and are elongate, anteriorly acuminate. The figure on HOLM's plate does not show the maculæ.

Proetus STEININGER.

There are some indistinct indications of maculæ in several of the hypostomas figured in BARRANDE's Système pl. 15.

Proet. signatus LINDSTR., Gotl. Trilob. Pl. XV fig. 17, shows very distinctly the two maculæ.

Proet. concinnus DALM.

[Pl. VI figs. 19-23.](#)

In this, as well as in the few other species of Proetus which we have studied, there is that peculiarity of the surface of the hypostoma that two more or less distinct ridges form an acute angle just below the faintly arcuated anterior border. The terrace lines are exactly alike those of the Phillipsiæ, only visible on the central field and ceasing just above the maculæ, where there is a transverse smooth field. The posterior margin has two pair of very short points, one on each side. The macula are lengthened

tubercles, somewhat bent and have an oblique position. They are surrounded by tiny prickles and as seen in [fig. 23](#) the macula proper forms a white, elliptic spot on the tubercle. The cephalic eyes are lengthened prisms covered by a thick membrane. They are convex at their base and in a transverse section they show a radiate structure of their interior.

Proet. conspersus ANGELIN.

[Pl. VI figs. 24-26.](#)

We have figured a decorticated hypostoma with impressions of two reniform maculæ. The vertical sections of the cephalic eyes show a thin stratum where the lenses are much shorter than in the previous species and covered by a thin film of the common test. The same is also the case in a species which probably is *Proet. verrucosus*.

Proet. signatus LINDSTRÖM.

[Pl. VI figs. 27-30.](#)

On the lateral borders of the hypostoma there are short spines, two on each side, but they are placed differently on the specimens and even wanting in some. The elongated, elliptical maculæ bear in their lower end a white spot on which is seen a little cluster of a varying number of small, segregated ocelli, from three to five, according to the different specimens, like so many black points. The maculæ attain the exceptional length of 4,7 millim. in some specimens, but commonly only 2 millim. We have as yet not been able to detect these remarkable ocelli in any other species of *Proetus*.

Ptychoparia CORDA.

WALCOTT in »Palæontology of the Eureka District», gives on plate X fig. 21 a representation of a hypostoma which near the posterior margin shows two maculæ as narrow, crescentlike ridges and above them two longer and thicker ridges directed obliquely toward the central axis of the hypostoma.

Ptychopyge ANG.

Previous figures of hypostomas with maculæ are the following:

Pt. aciculata BR. II, pl. 1 fig. 13.

Pt. glabrata BR. II, pl. 1 fig. 14.

Pt. aciculata ANG.

[Pl. VI figs. 40, 41](#)

The broad hypostoma bears the elongated, acuminate maculæ in an oblique direction, contrary to what is seen in most other species. The anterior end slopes inwards and the lower end outwards, the surface, which in the cast drawn is surrounded by a marginal line, is directed sideways against the broad lateral wings and is obscured by the declivity formed by the groove. The same disposition of the mature also occurs in *Ptych. glabrata*. These two species consequently in this respect differ essentially from the other four species which we have examined, but there is for the rest no reason to make these form different genera or subgenera.

Pt. angustifrons DALM.

[Pl. VI figs. 31-36.](#)

There are two varieties of hypostomas: one broad [fig. 32](#) and one more elongated [fig. 33](#). The maculæ having a convex surface lie at the inferior end of the lateral grooves prominent on a semicylindrical support, slightly inclining outwards. In the broad variety they are oblique ([fig. 34](#)) and in the elongated variety they are horizontal ([fig. 35](#)). In a vertical section [fig. 36](#) we have the same indications of a prismatic arrangement in the maculæ as in the *Asaphi*. The horizontal section does not represent so clear and distinct a reticulate structure as in the *Asaphi*, the prisms are much smaller.

The cephalic eyes are composed of closely packed hexaedral prisms, the gradual change from solid homogenous ones to those which are interiorly destroyed, only with the exterior form preserved is easily seen. The figure of a horizontal section ([fig. 31](#)) gives a good idea of the peculiar destruction of the interior of these prisms. In the vertical section the prisms are not discernible.

Pt. cincta BRÖGGER? according to FRIEDR. SCHMIDT in Mus. Holm.

[Pl. VI figs. 37-39.](#)

The hypostoma which BRÖGGER figures (II, pl. I fig. 2 *a*) as belonging to this species is completely identical with that of *Pt. angustifrons*, while the specimens which FR. SCHMIDT has marked out as being the true *Pt. cincta* have a different hypostoma. The maculæ vary in the specimens, being in some placed horizontally or in a right angle to the longitudinal axis of the hypostoma, in others, again, a little obliquely outwards. They form prominent, free lying tubercles. It is from the superior face that the rudimentary lenses have left their traces, as seen in a longitudinal section ([fig. 39](#)). It is consequently this superior surface which is the macula proper.

Pt. glabrata ANG.

[Pl. VI fig. 42.](#)

We have given a figure of the elliptic acuminate maculæ as seen on a cast of the inside of a hypostoma, the same which is the original to fig. 14 in BR. II.

Schmidtia MARCOU.

MOBERG has in his paper on »Sveriges äldsta Trilobiter», pl. 15, fig. 6, drawn a hypostoma of *Schmidtia Torelli* with scars of maculæ, placed like that of the *Paradoxidæ*. There is no mention made of them in the descriptions.

Trochurus BEYRICH.

Tr. pusillus ANG.

[Pl. IV figs. 52, 53.](#)

The strange transverse, bipartite hypostoma carries the small, round, tuberculose maculæ on the superior moiety. As seen from the interior ([fig.](#)

[53](#)) they are obscurely maculate, but it is doubtful whether these black dots are derived from lenses, as the surrounding hypostomic surface also has the same mottled aspect.

Trochurus sp.

[Pl. IV figs. 54, 55.](#)

This hypostoma belongs to an unknown species of Trochurus from the Southern Gotland (Burs). Its maculae are placed almost as in the preceding species, but are much larger. They are surrounded by prominent margins. No lenses have been discovered on them.

CONCLUSIONS.

In reviewing the genera and species enumerated above, as to their number in which the structure of the macula has been observed, we arrive at the following results.

	Genera.	Number of species with maculæ.	Number of species in the maculæ of which the structure has been examined.
1	Acaste	1	—
2	Acidaspis	1 ?	—
3	Asaphus	18	7
4	Barrandia	1	—
5	Bronteus	9	5
6	Bumastus	7	1
7	Calymmene	4	2
8	Centropleura	1	—
9	Chasmops	1	—
10	Chirurus	9	2
11	Ctenopyge?	1	—
12	Dalmanites	10	2
13	Dysplanus	2	2
14	Encrinurus	3	1
15	Griffithides	1	—
16	Herpes	2	—
17	Harpina	1	—
18	Holmia	2	—

19	Homalonotus	2	1
20	Hysterolenus	1	—
21	Illænus	9	5
22	Lichas	4	1
23	Megalaspides	2	—
24	Megalaspis	3	1
25	Nileus	2	2
26	Niobe	5	2
27	Ogygia	3	—
28	Paradoxides ^[42]	8	—
29	Peltura	1	—
30	Phacops	3	—
31	Phaëtonides	1	—
32	Phillipsia	5	—
33	Platymetopus	1	—
34	Pliomera	1	—
35	Proetus	3	—
36	Ptychoparia	1	—
37	Ptychopyge	4	2
38	Schmidtia	1	—
39	Trochurus	2	—
Total		136	36

^[42] To these have here been added two Bohemian species *P. bohemicus* and *P. spinosus* omitted above at page 64.

We have thus 136 species of 39 genera in which hypostomic maculæ have been found and only 36 species, in which it has been possible to study the structure of the maculæ through sections.

Common for a great number of maculæ in various groups, whether they show any organic structure or not, is the excessive thinness of their shell in comparison with that of the surrounding hypostoma. This is also in accordance with the tenuity of the cephalic eyes in relation to the test of the cheeks.

The structure which characterizes the macula as a visual organ, although often so rudimentary, is not in all instances spread through the whole substance of the macula. This structure has in several species been restricted to a narrow circumference of the body of the macula, to its interior apex as in *Bronteus* and *Proetus*. Only in the *Asaphidæ*, in *Illænus* and *Lichas* the entire macula shows this structure. Perhaps, to judge by certain indications in *Bronteus*, once in a larval or preceding stage of evolution the whole surface of the macula was also in that genus covered with lenses, which have been reduced.

The different groups in which the examined genera may be divided are the following seven, showing the remarkably great diversity of these organs. There is even in the same genus so great a variability that species with structure in the macula occur along with those devoid of any structure or also, as in *Lichas*, with a different structure. It must, however, be remembered, that the species of such genera may not be coeval.

Group 1. In the sectioned macula there is no trace whatever of any structure. The test of the macula considerably thinner than that of the hypostoma. According to the affinities of the genera they may be subdivided as follows. *a* *Bumastus*, *Dysplanus*. *b* *Nileus*, *Symphysurus*, ? *Ogygia*. *c* *Calymmene*, *Homalonotus*. *d* *Chirurus* pro parte.

Group 2. The whole macula is of a spongy or irregularly polyedric structure, showing prisms in vertical sections. Its concordance with the structure of the deteriorated cephalic eyes or with the so called border zone is complete. If a supposition may be hazarded, I think that the spongy or reticulate structure in the maculæ is their real and original state, a lower stage of development of the visual organs, out of which the prisms in the cephalic eyes have been formed, and that the prismatic lenses in their decay, as seen in the *Asaphidæ*, reveal to us their original state and structure, and thus, as it were, return to the primary stage in the maculæ.

Asaphus, Isotelus, Megalaspis, Ptychopyge, Niobe, ? Megalaspides, ? Barrandia.

Illænus, Lichas, Trochurus, ? Platymetopus.

Encrinurus also probably belongs to this group.

Group 3. Maculæ with well developed globular lenses on the interior third of the macula. The blank part of the macula without any structure. In one instance (Chirurus spinulosus) the whole macula covered with lenses.

Bronteus, Chirurus spinulosus.

Group 4. The macula form sunk pits with smooth bottom. Structure unknown. In a few species there seems to be no less than four maculæ in two pair above each other.

Dalmanites.

Group 6. The elongate straight maculæ carrying on their innermost point from three to five diminutive, segregate ocelli situated on a clear, white patch. Proetus.

As yet these ocelli have not been observed on the related genera Cyphaspsis, Phaëtonides, Phillipsia and Griffithides, but they may be preliminarily ranked here in consequence of their close conformity.

Group 7. The maculæ have in this group been reduced to a pair of tiny ocelli situated high up on the hypostoma, near its anterior margin.

Phacops, Acaste.

The position of the hypostomic eyes on the ventral surface of the trilobites is not quite so abnormous, nor so isolated a feature amongst the crustaceans as might at first be supposed. Amongst the recent crustacea there is probably none, which as to the conformation of its labrum or hypostoma resembles the trilobites so closely as the species of the genus Apus, however different they may else be in all other respects. We have studied chiefly the hypostoma of Apus cancriformis. In its shape as seen from the outside and as to its outlines it reminds strongly of the trilobite hypostoma. It is nearly square ([pl. VI figs. 46, 47](#)) with rounded corners. The anterior margin is arched and in the middle it has a broad tonguelike prolongation. On the sides of the anterior margin there are two short, sharply pointed

wings. The lateral margins are thick, almost as doubled, and bent forward so that a groove is formed alongside them and the large, evenly vaulted median surface of the hypostoma. These grooves are continued in a short groove parallel with the posterior margin. From this margin a narrow oblique surface is slanting backwards, as it were, doubling that margin. On its surface there are two small, resplendent, white or clear spots, almost of the shape of a crescent, though more irregular when seen in higher enlargement ([pl. VI fig. 49](#)). They are translucent as to be visible on the interior side of the hypostoma. In *Apus glacialis* there are also similar spots in the same position, but they form short narrow, straight stripes, parallel with the border of the posterior margin. I have no opinion as to the nature or function of these white spots. I do not think that they at all are homologous with the macula; of the trilobites. On the other hand we see a little higher up on the exterior side of the hypostoma two faintly elevated elliptic tubercles which as to their position and shape highly remind of the maculae of the trilobites. They are hollow on the inside and we have not been able to find any peculiar structure in them. That they have any visual function is precluded through the circumstance that the whole exterior surface is densely covered with microscopic bristles ([pl. VI fig. 49](#)) which are especially remarkable along the posterior margin.

In the class of the Cirrhipedia a sufficient number of facts is known to ascertain us of the presence of eye spots on the hypostomas of the larva. The researches of SPENCE BATE^[43] show in the larva of *Balanus balanoides*, on the hypostoma or labrum, which he calls proboscis, an eye, a little from the posterior margin of the labrum. It seems during the various moults of the animal to have changed its place, being in the two first moults situated near the anterior margin of the hypostoma. In *Clitia Strömi* it lies at the anterior part of the labrum ([figs. 9, 10](#)). SPENCE BATE does not accept the opinion of their nature as eyes, but the following authors as DARWIN and CLAPARÈDE agree in considering these spots as eyes. DARWIN^[44] describes two such eyes in *Lepas australis* and, if I catch his meaning rightly, one on each side of the labrum. CLAPARÈDE has given a figure of the larva of *Lepas anatifera*^[45] in the hypostoma of which (by him called »Rüssel«, proboscis) a single eye is seen, having at the sides two round problematic organs, that he at first considered as auditory cells, but later as eyes. HESSE^[46] has also in a species of *Scalpellum* seen several coloured spots on

the inferior surface of the carapace, below the antennæ and he expressly says that they are eyes and a special pair of supplemental eyes that are becoming functional after the first moult.

[43] On the development of the Cirripedia, Ann. Mag. Nat. list. 1851, p. 324, pl. VI figs. 1, 2, 3, pl. VII fig. 10.

[44] Monogr. of Cirripedia I p. 17.

[45] Beobachtungen ... über wirbellose Thiere, 1863, pl. XVII fig. 22.

[46] Mémoire sur les métamorphoses que subissent les Anatifes appelés Scalpels obliques. Annales des Sciences nat. 1859. Tome XI p. 163.

These sternal eyes in the larva are previous to the eyes in the dorsal side of the later stages and whenever such are discovered in other groups of adult animals as the trilobites they must of course be considered as remnants from the larval stage. As a rule, in several tribes of the crustacea it has been observed that the eyes originate on the ventral or sternal side of the body and in the advancement of growth move to the superior side. So according to PACKARD and others the embryo of *Limulus* has originally the median eyes on the ventral side. In the same way the cephalic eyes of *Dalmanites* move, as BARRANDE has shown, from the ventral side to their permanent place on the superior side of the head shield. But while in the Cirripedia the larval or hypostomic eyes are lost in the adult they remain on the hypostoma of a great number of the adult trilobites and have probably left their mark behind in *A. pus*. No wonder then that we find these maculæ developed as precursors of the hypostomic eyes in a number of the oldest and in other respects blind trilobites as the *Olenellidæ* and *Paradoxidæ* to which they may have been of some use as a sort of visual organs, though probably very imperfect.

We find the maculæ of the trilobites present from the oldest Cambrian times and we find also in them a progressive evolution, in some to a high degree, lenses and facets, perfectly identical with those of the eyes on the head shield, converting them into true eyes. It was in the Lower Silurian times, in the Lower grey *Orthoceratite* limestone of Sweden, that *Illænus Esmarki* appeared provided with actual prismatic lenses on its maculæ and probably also *Lichas*. Still higher at the top of the Lower Silurians, in the *Leptæna* limestone of Dalecarlia the genus *Bronteus* has appeared with those highly developed hypostomic eyes.

But there are, no doubt, still more facts to adduce for filling up extant lacunæ in the knowledge of these matters. The material now at hand has, however, not been sufficient for doing this. At present I must leave this interesting field of investigation and be contented if I have succeeded in showing where rich harvests await every one who will make a search for them.

Explanation

of

the plates.

All specimens figured belong to the Swedish State Museum,
unless otherwise stated.

Pl. I.

Acidaspis crenata EMMR. p. 85.

Fig. 1 & 2. Vertical sections of the eye 60/1.

- » 3. The outside of the hypostoma 6/1.
 - » 4. The anterior margin of the hypostoma, showing one of the clubshaped spots in transmitted light 20/1.
 - » 5. The granulations of the hypostoma in transmitted light 80/1.
 - » 6. Vertical section of the hypostoma with granules 60/1.
- Specimens from Djupvik, Eksta.

Agnostus glandiformis ANGELIN p. 35.

Fig. 7. Vertical section of a part of a rolled up specimen 8/1.
Specimen signed CQ. Andrarum, Scania.

Asaphus expansus L. p. 39.

Fig. 8. Horizontal section of the eye near the upper margin.

- a.* The surrounding spongy zone,
 - b.* The surface of the eye with lenses 60/1.
- Spec. CG.
- » 9. Horizontal section from the centrum of the eye 60/1.
Spec. CF.
 - » 10. Horizontal section near the inferior margin,
a the eye, *b* marginal zone, *c* a portion of the cheek 60/1. Specim. CF.
 - » 11. Vertical section near the inferior margin 60/1,
same specimen as in fig. 10.
 - » 12. Eye fractured vertically showing regular prisms in the eye (*a*) and irregular ones in the marginal zone (*b*) 30/1.
 - » 13. Part of the hypostoma with the macula 12/1.
 - » 14. The right macula from the inside 12/1.

- » 15. A vertical section across the hypostoma,
aa the maculæ 8/1.
- » 16. A similar one of the macula at left in the
preceding figure 60/1.
- » 17. The right macula from the outside in transmitted
light 60/1. Sp. BQ.

Asaphus fallax DALM.? p. 41.

Fig. 18. Horizontal section of the eye 60/1. Spec. CD.

- » 19. Vertical section of the same passing obliquely
through two rows of prisms 60/1.
- » 20. Fragment of the same eye with entire prisms 30/1.
- » 21. The hypostoma seen from the inside 3/1 from
Sandvik, Öland.
- » 22. The left macula 12/1. Spec. BO. From Vestanå,
Östergötland.

Asaphus raniceps DALM. p. 40.

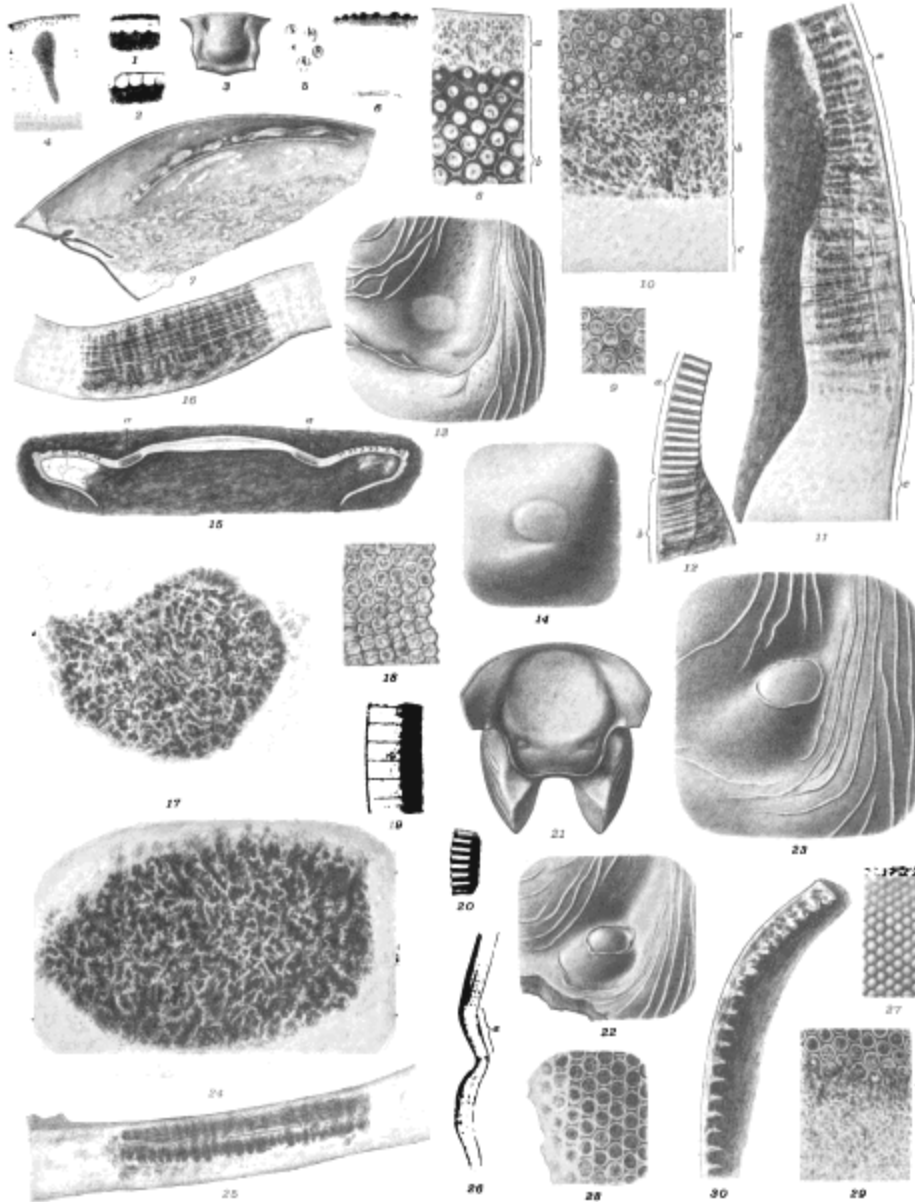
Fig. 23. The left macula 12/1. Spec. BV.

- » 24. Horizontal section of right macula 60/1. Spec. BY.
- » 25. Vertical section of a macula 60/1.
- » 26. A similar section of a macula (*a*) in the
longitudinal axis of the hypostoma 12/1.
Exterior surface at right.
All from Östergötland.

Asaphus sp. p. 41.

Fig. 27. A portion of the eye near the superior margin 30/1.

- » 28. Horizontal section of the eye 60/1.
 - » 29. Horizontal section near the inferior margin 60/1.
 - » 30. Vertical section near the superior margin 60/1.
- All figures from specimen CE. From Gotska Sandön.



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G. Liljevall del.

Ljustryck af Chr. Westphal Stockholm.

Pl. II.

Asaphus (*Isotelus*) **gigas** J. HALL p. 42.

Fig. 1. The right macula surrounded by terrace lines 3/1.

- » 2. Horizontal section of a macula with a part of the test 60/1.
- » 3. Vertical section of the same, the surface of the outside upwards 60/1. All specimens from Cincinnati, Ohio, Hudson River group. Spec. BU.

Bronteus irradians LINDSTRÖM. p. 43.

Fig. 4. The hypostoma 4/1.

- » 5. The left macula 12/1. Specimen from Gotland, Lönsta.

Bronteus laticauda ANG. p. 44.

Fig. 6. Horizontal section of the cephalic eye 20/1.

- » 7. The same highly magnified 250/1.
- » 8 & 9. Vertical sections of the cephalic eyes 20/1.
In fig. 8 with integument, in fig. 9 decorticated.
- » 10. Hypostoma. Specim. N. 2/1.
- » 11. The right macula, from the outside. Spec. O. 12/1.
- » 12 & 13. Left maculae, casts of the interior surface 12/1.
Specimens from Dalecarlia.

Bronteus platyactin ANG. p. 43.

Fig. 14. Vertical section of cephalic eye, *a* the eye.
b, cheek 20/1.

- » 15. Hypostoma 2/1.
- » 16. The same from the interior surface. Sockets of the maculae sunk in the shell 2/1.
- » 17. The right macula 12/1.
- » 18. Portion of the eye on the macula in transmitted light 20/1.
- » 19. The same more magnified 60/1. Specimens from the stratum *b*. Wisby, Norderstrand.

Bronteus polyactin ANG. p. 8.

Fig. 20. Horizontal section of cephalic eye 20/1.

- » 21. Vertical section of the same 60/1.
- » 22. Hypostoma 4/1.
- » 23-25. Samples of right maculæ 20/1.
- » 26. Horizontal section of macula, the portion bearing the lenses 20/1.
- » 27. Vertical section of the same 60/1, *a* the lenses. Specimen from Stora Carlsö.
- » 28. Hypostoma 4/1.
- » 29. The right macula 12/1.
- » 30. A portion of the lenses on the macula in reflected light 30/1. The lenses are only faintly convex and encased in a white frame. The diameter of the largest lenses 0,06 millim. Specimen from Lansa, Fårö.

Bumastus barriensis MURCH. p. 46.

Fig. 31. Hypostoma 4/1.

- » 32. The right macula 12/1. Specim. CU from Wisby.

Bumastus sulcatus LINDST. p. 46.

Fig. 33. Horizontal Section of a cephalic eye 20/1.

- » 34. The same magnified 60/1.
- » 35. A similar, near the inferior margin and close to the surface, *a* the eye, *b* cheek 60/1. Compare fig. 38.
- » 36. Vertical section of the cephalic eye near the suture 20/1.
- » 37. The same, more magnified 60/1.
- » 38. A similar, near the inferior margin, *a* eye, *b* cheek 60/1.
- » 39. Cast of the interior surface of a hypostoma 2/1.
- » 40. The same seen from the side.

Originals to figs. 33-38 from Samsugu, Othem, Gotland, figs. 39 & 40 from Stora Vede, Follingbo, Gotland.

Bumastus sp. p. 47.

Fig. 41. Horizontal section of cephalic eye, near the inferior margin,

a eye, *b* cheek, 60/1.

» 42. Vertical section of the same 60/1.

» 43. A similar near the inferior margin, *a* eye,
b cheek, 60/1.

» 44. Hypostoma 4/1.

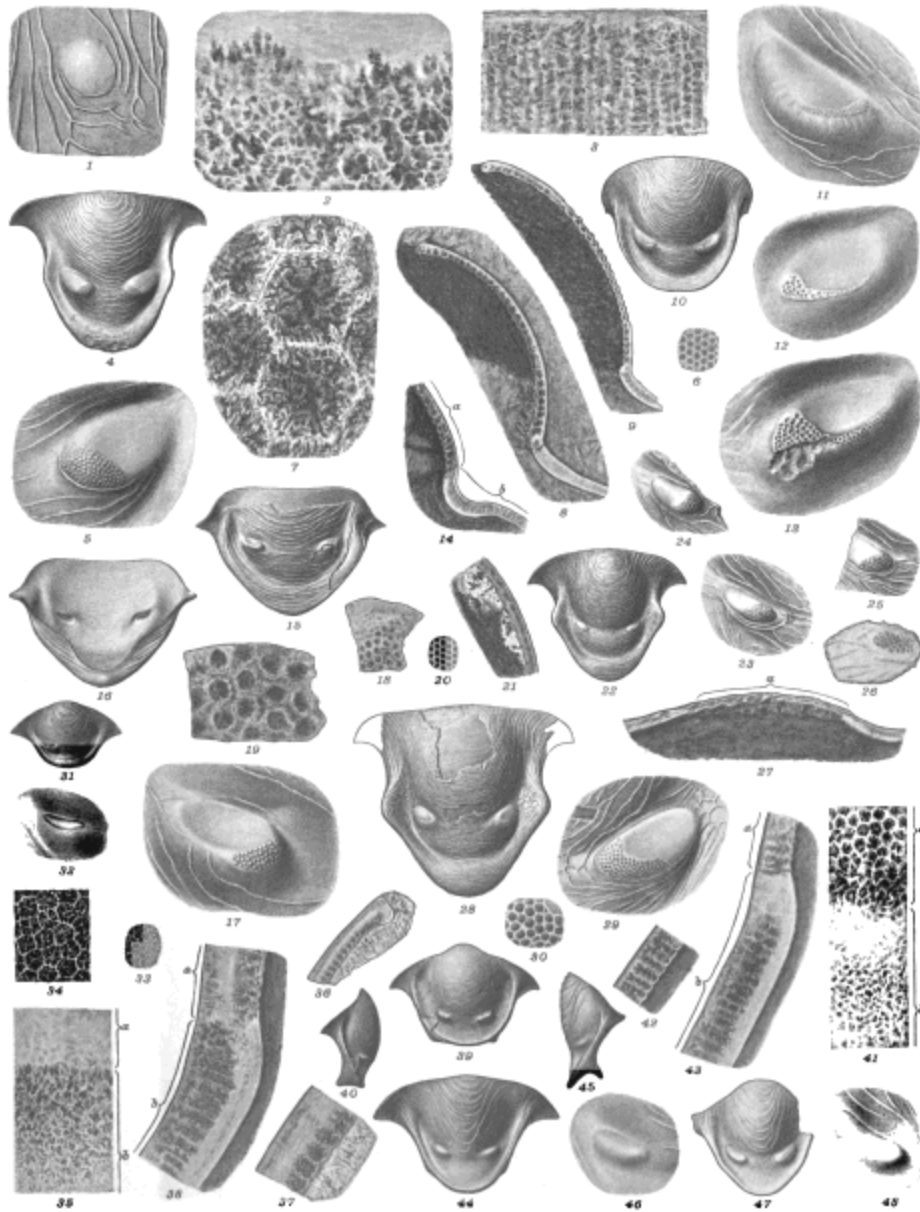
» 45. Side view of the same.

» 46. The right macula 12/1. Specimens from Korpklint near Wisby.

Bumastus sp. p. 46.

Fig. 47. Hypostoma 4/1.

» 48. The right macula 12/1. Specimen from Klints, Othem, Gotland.



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Pl. III.

Bumastus n. sp. p. 47.

Fig. 1. Hypostoma 4/1. Specimen AY. From Wisby, Norderstrand.

» 2. Its left macula 12/1.

Bumastus n. sp. (different from the preceding) p. 47.

Fig. 3. Hypostoma 4/1. Spec. AT from Wisby, Norderstrand.

» 4. Vertical section of the macula *a* in its longest axis.

» 5. A portion of the cephalic eye, the facets very low.

Calymmene intermedia LINDSTR. p. 48.

Fig. 6. Hypostoma 4/1. Spec. BG from Gotland.

» 7. The left macula 12/1.

Calymmene tuberculata BRÜNNICH p. 48

Fig. 8. The inside of the hypostoma 3/1. Spec. BA from Gotland.

» 9. Horizontal section near the surface, seen from the inside 20/1.

Chasmops macroura (?) p. 49.

Fig. 10. The left macula, cast of the inside 12/1. Rostock.

Chirurus glaber ANG. p. 50.

Fig. 11. A portion of the surface of the cephalic eye 12/1.

From Dalecarlia.

Chirurus ornatus DALM. p. 50.

Fig. 12. Horizontal section of a Cephalic eye near the surface 60/1.

» 13. Vertical section of the same, 20/1, at the inferior margin. A line added to show the supposed size of the fragmentary lenses.

- » 14. The left macula of the hypostoma 12/1.
From Östergötland, Westanå.

Chirurus spinulosus NIECZKOWSKI p. 51.

Fig. 15. The right macula of the hypostoma from the
inside 12/1. Belongs to Dr. G. HOLM, from
Esthonia, Kuckers C².

Chirurus sp. p. 51.

Fig. 16. The left hypostomic macula 12/1. Öland, Brunsby
kanal.

Cyrtometopus clavifrons DALM. p. 49.

Fig. 17 & 18. Horizontal sections of cephalic eyes 60/1.

- » 19. Vertical sections of cephalic eyes and a part
of the cheek 20/1.
- » 20. A highly magnified portion of the eye in vertical section
60/1.
- » 21. Hypostoma, 3/1 the anterior wings completed after
another
specimen. From Östergötland.

Cyphaspis elegantula ANGELIN p. 52.

Fig. 22 & 23. Horizontal and Vertical sections of cephalic eyes
60/1.

Diameter of facets 0,03 mm.

- » 24 & 25. Hypostomas from the front side and
in side view 8/1. Upper Silurian, Gotland.

Ctenopyge pecten SALTER p. 29.

Fig. 26. The head shield, the free cheeks added from other
specimens 4/1.

- » 27. A free cheek, viewed without foreshortening, as in its
natural position 4/1. Alumschists of Andrarum,
Scania

Ctenopyge n. sp. (LINNARSSON, Geol. För. Förhandl.
1880, pl. VI, figs. 14, 15) p. 29.

Fig. 28. A free cheek 6/1.

» 29. Surface of the eyes, near the suture 30/1.

» 30. The same next the free cheek 30/1. Andrarum.

Sphærophthalmus alatus ANG. p. 29.

Fig. 31. The head shield; the free cheeks have been added 8/1.

» 32. A free cheek without foreshortening 8/1.

» 33 & 34. Horizontal and Vertical sections of cephalic eyes
100/1. The cornea nearly 0,01 mm. in thickness.

Andrarum.

Peltura scarabæoides WAHLENB. p. 29.

Fig. 35. The head shield; the free cheeks added 2/1.

» 36. A free cheek, not foreshortened 4/1.

» 37. A cast of the inside of the cephalic eye 20/1.

» 38. A portion of the cephalic eye from the inside 60/1.

» 39. A portion of the same in transmitted light 100/1.

» 40. Vertical section of an eye 100/1. The rock on the inside.

» 41. A similar one passing through two rows of lenses, the one

behind the other. The rock on the outside 100/1.

» 42. A hypostoma 8/1. Andrarum.

Dalmanites imbricatus ANGELIN p. 53.

Fig. 43. The cephalic eye near the inferior margin 20/1.

» 44. Vertical section of the same and a portion of the free
cheek 60/1. The lenses are almost contiguous and
the

interspace between them is formed by the
common test of

the body, which is perforated by longitudinal
tubes.

Dalmanites obtusus LINDSTR. p. 53.

Fig. 45. Horizontal section of cephalic eye near the inferior
margin 30/1.

» 46. Vertical section, the interspaces being in this wider than
in the preceding 60/1. From Gotland.

Dalmanites vulgaris SALTER P. 54.

Fig. 47. Part of the cephalic eye near the inferior margin 20/1.

- » 48. Horizontal section near the surface at the inferior margin 30/1.
- » 49. A tangential section, so that the inferior part is below the lenses where the peculiar radiating structure is seen in horizontal section 30/1.
- » 50. Vertical section of a small Specimen, fibrous appendices below the lenses 30/1.
- » 51. Hypostoma 2/1.
- » 52. The right macula 12/1. English specimens from Dudley, Shropshire.

Dysplanus centrotus DALMAN p. 55.

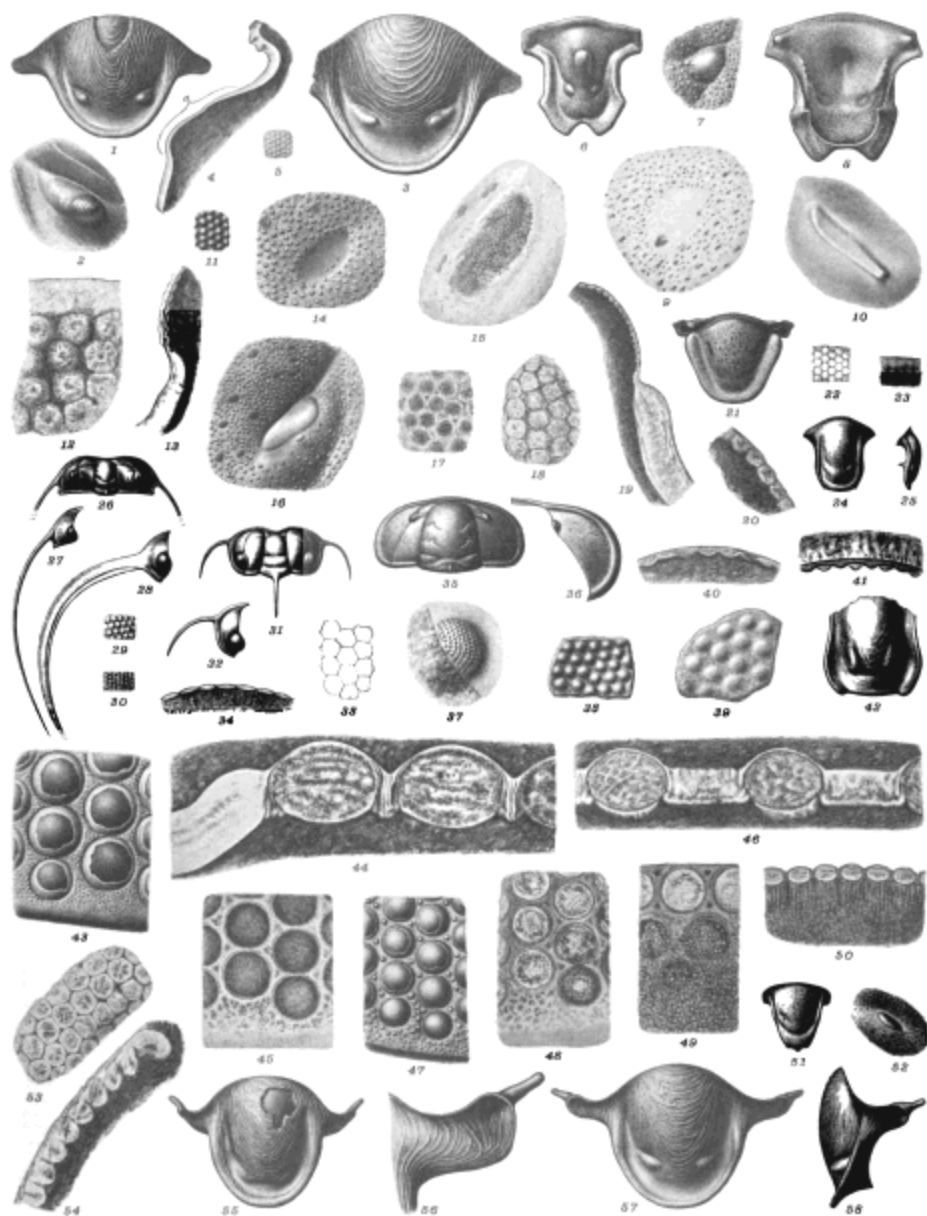
Fig. 53 & 54. Horizontal and vertical sections of an eye 60/1.

- » 55. Hypostoma 3/1.
- » 56. The anterior wing of the hypostoma seen in foreshortening 6/1. Specimen AX from Westanå, Östergötland.

Dysplanus ladogensis HOLM p. 55.

Fig. 57. Hypostoma 3/1.

- » 58. The same, seen from the side 3/1. Specimen AI from Östergötland.



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Pl. IV.

Chirurus speciosus HIS.

Fig. 1. The right macula, cast of the interior side 12/1.

Specimen from Grumpevik, Vamlingbo, Gotland.

Cybele bellatula DALM. p. 52.

Fig. 2. The hypostoma 3/1. Vestanå, Östergötland.

Dalmanites sclerops DALMAN p. 54.

Fig. 3. The hypostoma 2/1.

Encrinurus punctatus WAHLENB. p. 56.

Fig. 4. Horizontal Section of an eye 69/1. Diameter
of the prisms 0,04 mill., their length 0,07 millim.

» 5. The hypostoma 3/1.

» 6. The left macula. Reflected light 12/1.

» 7. Part of the left macula in transmitted light 20/1.

The black streaks are sectioned pores. Spec. BH.

» 8. Vertical section of the right macula, a sagittal
section 20/1. BH.

» 9. A similar one *a* along the shorter diameter 20/1.

Specimens 5, 6 from Vikers Boge, 4, 9 Medebys,
Hall,

7, 8 Svarfvarehuk. All from Gotland.

Encrinurus lævis ANG. p. 56.

Fig. 10. Horizontal section of the cephalic eye.

» 11. Vertical section of the same. Both 60/1.

Prisms 0,05 in breadth, 0,06 in length. Specimen
from Wisby.

» 12-17. Longitudinal profiles and anterior margins of the
hypostomas of Encr. punctatus (figs. 12, 13),

E. obtusus (figs. 14, 15) and E. lævis (figs. 16, 17).

Harpes vittatus BARR. p. 31.

Fig. 18. The two ocelli at right 8/1.

» 19. Sagittal Section of two ocelli 60/1. Specimen from Lochkow

E, e², Bohemia.

Homalonotus Knighti KÖNIG p. 57.

Fig. 20. The hypostoma 3/2.

» 21. The right macula, from the interior side in transmitted light 20/1. From Scania.

Illænus chiron HOLM p. 58.

Fig. 22. Horizontal Section of an eye.

» 23. Vertical section of the same, both 60/1. Spec. AG.

» 24. Hypostoma 2/1. Spec. AF.

» 25. The right macula seen in reflected light. The lenses translucent 12/1. Ex. AA. All from Persnäs, Öland.

Illænus Esmarki SCHLOTH. p. 511.

Fig. 26 & 27. Horizontal sections of the eyes in two different specimens 60/1.

» 28. Vertical section of an eye 60/1.

» 29. Likewise, *a* the eye, *b* a portion of the cheek 12/1.

» 30. Hypostoma 2/1.

» 31. The inside of the right macula 12/1. Spec. AR.

» 32. The outside of the left macula in transmitted light 60/1.

» 33. Sagittal section of the macula (*a*) 60/1.

All specimens from Vestanå, Östergötland, excepting

fig. 30 belonging to Dr. HOLM, from Norway, Bjerkaas.

Illænus gigas HOLM P. 59.

Fig. 34 & 35. Surface views of the prisms of the eyes, fig. 34 at the superior margin and 35 at the inferior one 12/1.

» 36. Cast of the inside of a hypostoma 1/1. Same original as HOLM's fig. 6 Tafl. I, Illænider.

» 37. Cast of the inside of the left macula. Same specimen 12/1.

Illænus Roemeri VOLLBORTH (Ill. vivax HOLM) p. 59.

Fig. 38. Hypostoma 1/1. Spec. AO.

» 39. The right macula; the lenses translucent in reflected light 12/1. Spec. AO.

» 40, 41. Both maculæ of another, larger specimen 12/1. Sp. AP.

From the Leptæna limestone of Dalecarlia.

Illænus sphæricus HOLM p. 59.

Fig. 42. The prisms of cephalic eye 12/1. }

» 43. The right maculæ, transparent in }
transmitted light 12/1. Sollerön, } Spec. AH.
Dalecarlia. }

Lichas sp. p. 60.

Fig. 44. The hypostoma 1/1.

» 45. Cast of the inside of the right macula 12/1. Spec. T.

» 46. Cast in wax of the preceding 12/1.

» 47. Cast of the inside of the hypostoma. The central portion with the maculæ 3/1. Spec. V.

» 48. Inside of the right macula 12/1. Spec. Y.

From a quarry north of the gallows near Wisby.

Lichas latifrons ANG. p. 60.

Fig. 49. The right macula, transmitted light, seen from the inside 60/1. From Kyrkberget, Wisby.

Platymetopus planifrons ANG. p. 67.

Fig. 50. The hypostoma 1/1.

» 51. The left macula. Transparent lenses? 12/1.

From the Leptæna limestone of Dalecarlia.

Trochurus pusillus ANG. p. 70.

Fig. 52. The hypostoma 5/1.

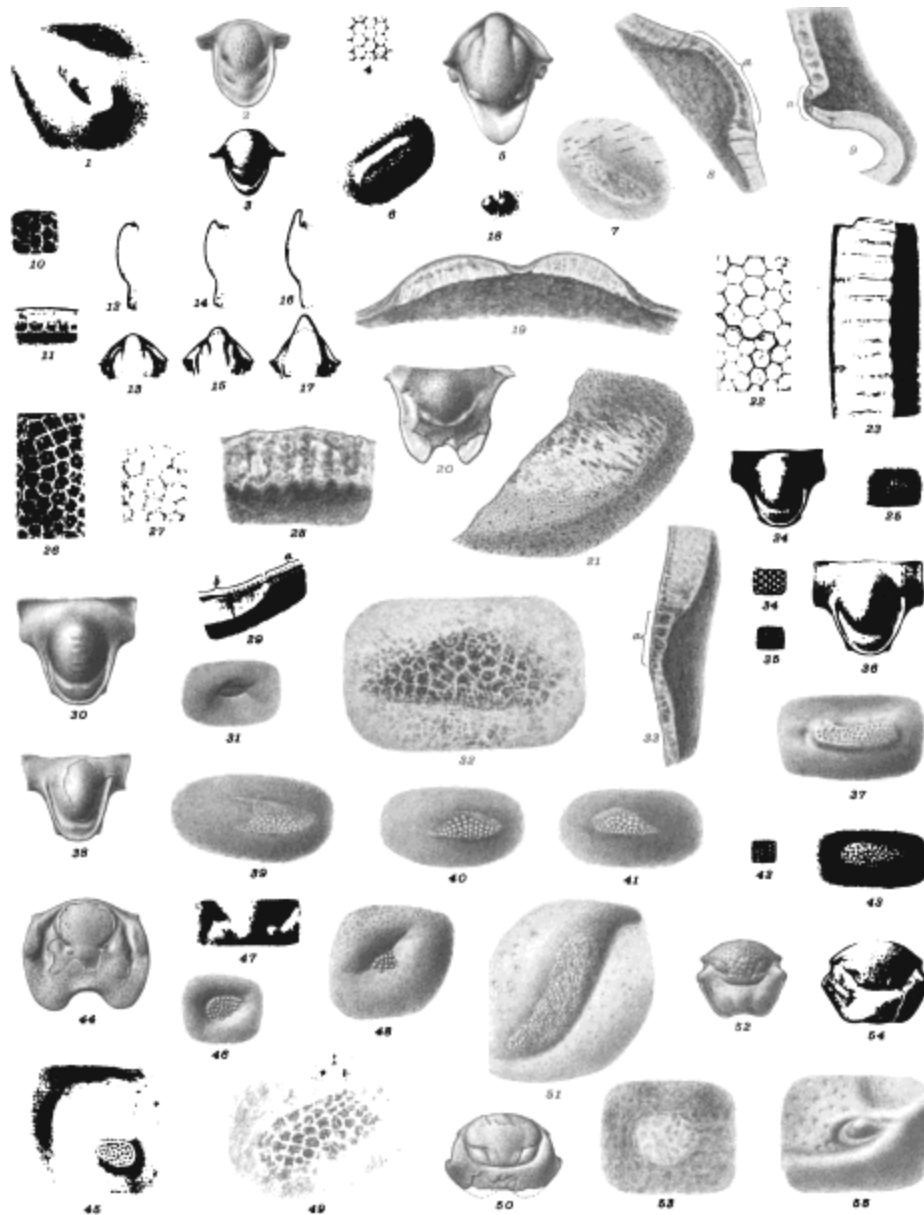
» 53. The right macula in transmitted light. Seen from the inside

60/1. From Fröjel.

Trochurus? p. 70.

Fig. 54. The hypostoma 3/1.

» 55. The right macula from the inside 12/1. From När Gotland.



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Pl. V.

Megalaspis attenuata WAHLENB.

Fig. 1. Horizontal section of cephalic eye near the inferior margin 60/1.

a surface of the eye, *b* frame with irregular or reserve prisms, *c* part of the cheek with normal structure. Spec. DP.

- » 2. Hypostoma 2/1. Spec. BM.
- » 3. The right macula 12/1.
- » 4. Part of the same in transmitted light 60/1. Spec. DR.
- » 5. Vertical (sagittal) section of the inferior part of the hypostoma, crossing the macula *a* 12/1.
- » 6. Vertical section of the same macula 60/1. All spec. from Westanå, Östergötland.

Megalaspis limbata BOECK p. 61.

Fig. 7. Hypostoma, mostly decorticated 2/1. Specimen from Fogelsång,

Scania, the same as BRÖGGER's original to his fig. 23.

Megalaspis planilimbata ANGEL. p. 61.

Fig. 8. Hypostoma, Öland, Äleklinta 2/1.

Megalaspis sp.

Fig. 9. Vertical section of macula (*a*) 12/1. Öland, Lower grey Orthoc. limestone.

Nileus armadillo DALM. p. 62.

Fig. 10. Horizontal section of cephalic eye 60/1.

- » 11. Vertical section of the same 60/1.
- » 12. A similar section 60/1. *a* the prisms of clear calcareous spar, *b* the surface of the eye and

the free cheek of yellow spar.

- » 13. The hypostoma 2/1.
- » 14. The right macula 12/1.
- » 15. Vertical section of macula (*a*) 12/1. All specimens from Persnäs, Öland, excepting fig. 13 from Westanå, Östergötland.

Symphysurus palpebrosus DALMAN p. 62.

Fig. 16. Horizontal section of cephalic eye 60/1.

- » 17. Vertical section of the same 60/1.
- » 18. Hypostoma 2/1. From Westanå, Östergötland.

Niobe frontalis ANG. p. 63.

Fig. 19. The hypostoma 2/1.

- » 20. The right macula 10/1.
- » 21. Part of the same in transmitted light 60/1.
Specimen from Kungs Norrby, Östergötland.

Niobe læviceps ANG. p. 63.

Fig. 22. Horizontal section of cephalic eye near the inferior margin,

a cheek 60/1.

- » 23. Hypostoma 2/1.
- » 24. The right macula 10/1.
- » 23. Part of the same in transmitted light 60/1.
- » 26. Vertical section of macula *a* 12/1.

Ogygia dilatata var. **Sarsi** ANGEL. p. 63.

Fig. 27. Hypostoma 2/1.

- » 28. The right macula 10/1. From Christiania, Haakavik, Collection
of Dr. G. HOLM.

Olenus sp. p. 33.

Fig. 29. A part of the free cheek with the scallop 6/1.

- » 30. Profile of the scallop. Andrarum, Scania.

Parabolina spinulosa WAHLENB. p. 33.

Fig. 31. The free cheek with the scallop 3/1.

» 32. Profile of the scallop. From Jemtland.

Paradoxides oelandicus ANGEL. p. 64.

Fig. 33. Hypostoma 2/1.

» 34. The left macula 12/1. Öland, Torp.

Phacops cephalotes BARR. p. 65.

Fig. 35. The hypostoma; the shell mostly decorticated and the maculae visible as casts on the inside.

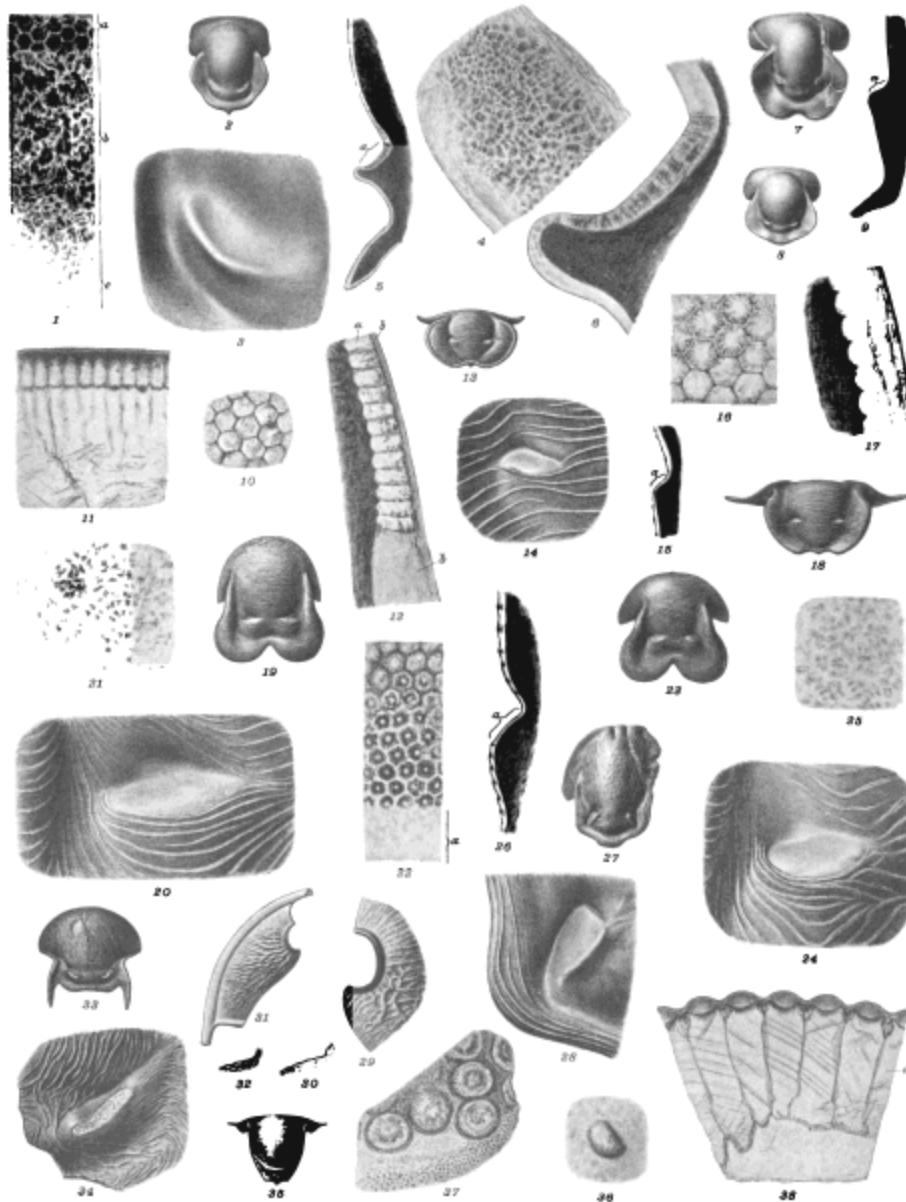
» 36. Cast of the inside of the right macula 12/1. Spec. from Lochkow, Bohemia.

Phacops quadrilineata ANGEL. p. 66.

Fig. 37. Horizontal section of cephalic eye near the inferior margin

30/1. The dark points lie in the outermost stratum of the cheek.

» 38. Vertical section of cephalic eye 20/1. An irregularly hexaedral crystal below each lens. Specimen from Wisby a.



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G. Liljevall del.

Ljustryck af Chr. Westphal Stockholm.

Pl. VI.

Phacops macrophthalmus BURM. p. 65

Fig. 1. Horizontal section of the cephalic eye 30/1.

- » 2. A similar one near the inferior margin, *a* cheek 30/1.
- » 3. Vertical section of a cephalic eye 30/1.
- » 4. A similar, twice as much enlarged 60/1.
- » 5. Also a similar, *a* cheek 30/1.
- » 6. Hypostoma 2/1. Spec. EA.
- » 7. The wings of the hypostoma, seen without foreshortening
4/1. EA.
- » 8. The diminutive right macula 12/1, surrounded by
incoherent
terrace lines.
- » 9. The left macula 12/1. Specimens from Gerolstein Eifel,
Devon.

Acaste sp.? p. 51.

Fig. 10. The hypostoma 8/1. From Mulde, Fröjel, Gotland.

Phaëtonides Stokesi MURCH. p. 66.

Fig. 11. Hypostoma 4/1. Gotland, Eksta, Djupvik.

Phillipsia sp. p. 66.

Fig. 12. A portion of the cephalic eye 20/1.

- » 13. The hypostoma 4/1.
- » 14. The right macula 12/1. Specimen from Beeren Eiland,
Oswalds

förberg, communicated by Hr. J. G. ANDERSSON.

Phillipsia sp.? p. 66.

Fig. 16. The hypostoma 3/1.

- » 16. The right macula 12/1.
- » 17. Horizontal section of the cephalic eye 60/1.

- » 18. Vertical section of the cephalic eye, *a* cheek 60/1.
Probably from the Carbon of Belgium.

Proetus concinnus DALM. p. 67.

Fig. 19. Horizontal section of the cephalic eye, a little oblique, at right the innermost stratum with the rock 60/1.

- » 20. Vertical section of the cephalic eye. The diameter of the prisms 0,03 millim.
- » 21. Hypostoma 4/1.
- » 22. The right macula 12/1.
- » 23. The same with a light, transparent spot 12/1. Specimen from Djupvik, Eksta.

Proetus conspersus ANG. p. 68.

Fig. 24. A portion of the cephalic eye 20/1.

- » 25. The hypostoma. Cast of the interior side 4/1.
- » 26. Cast of the interior side of the left macula 12/1.

Fig. 24 from Öja, Gotland, the others from Kräklingbo.

Proetus signatus LDM. p. 68.

Fig. 27. Hypostoma 4/1.

- » 28-30. Three pair of maculae with dark ocelli in a white spot.

Ptychopyge angustifrons DALM. p. 69.

Fig. 31. Horizontal section of the cephalic eye 250/1.

- » 32. Hypostoma, a broad variety 1/1.
- » 33. Another, a narrow variety 1/1.
- » 34. The left macula of the hypostoma fig. 32 4/1.
- » 35. A similar of the narrow variety 4/1.
- » 36. Vertical section of a macula. *a* the macula 20/1.

Specimens from Vestanå and Kungs Norrby, Östergötland.

Ptychopyge cincta BRÖGGER p. 69

(according to FR. SCHMIDT. The figure 12 *a* of BRÖGGER resembles, however, the hypostoma of *Pt. angustifrons*, but not the hypostomata which have been prepared from the *Ptych. cincta*).

Fig. 37. A hypostoma; the maculae are directed obliquely 2/1.

» 38. The right macula of another specimen 4/1.

» 39. Vertical section of a macula 60/1. Specimens from Kungs

Norrby, Östergötland.

Ptychopyge aciculata ANG. p. 68.

(Compare BRÖGGER II Taf. I, fig. 13.)

Fig. 40. The hypostoma, cast of the interior surface 1/1.

» 41. Cast of the right macula 4/1. Specimen from Öland, Brunsby.

Ptychopyge glabrata ANG. p. 69.

Fig. 42. Original of BRÖGGER's fig. 14. Cast of the inside of the right macula 4/1. Älleberg.

Elyx laticeps ANG. p. 19.

Fig. 43. The interior surface of the right fixed cheek and the glabella. Magnified. Specimen from Andrarum. 4/1.

Conocoryphe p. 19.

Fig. 44. Cast of the interior surface of the left cheek. Magnified. From Andrarum. 4/1.

Solenopleura brachymetopa ANGEL. p. 19.

Fig. 45. The head shield, cast of the interior surface. Magnified. Specimen from Andrarum, Scania. 4/1.

Apus cancriformis SCHÄFFER p. 73.

Fig. 46. The hypostoma from the outside 8/1.

» 47. The same from the inside 8/1.

» 48. A side view of the same, 8/1. *a* the duplicature of the head shield, *b* posterior integument of the hypostoma, *c* the ventricle, *d* the mouth.

» 49. The right white spot at the posterior margin of the

hypostoma,

seen from the outside in transmitted light 150/1.

K. Sv. Vet. Akad. Handl., Bd 34, No 8.

Lindström: Trilobites. Pl. VI.



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G. Liljevall del.

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Transcriber Note

Minor Typos were corrected. Inconsistencies in capitalization and abbreviations were left as printed.

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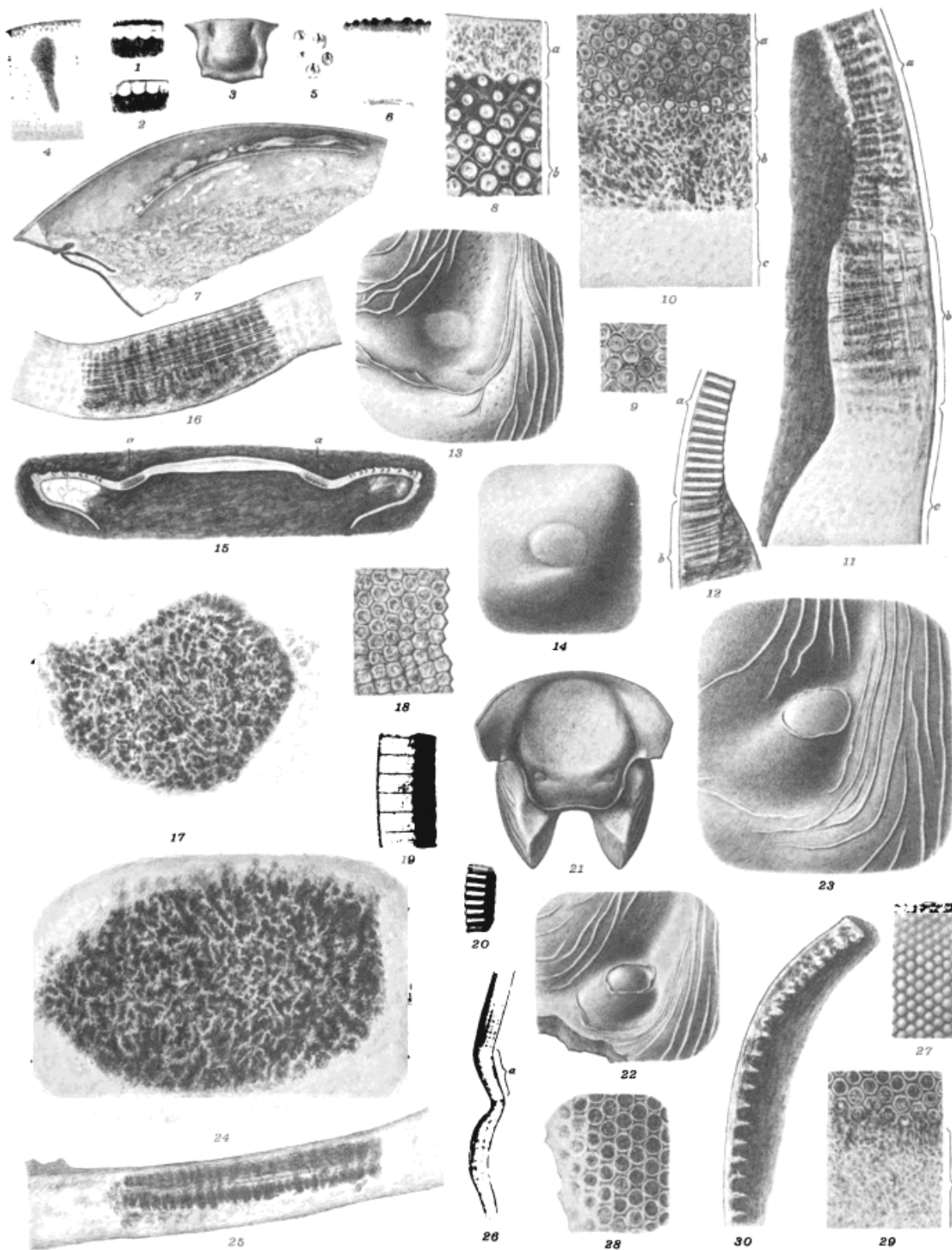
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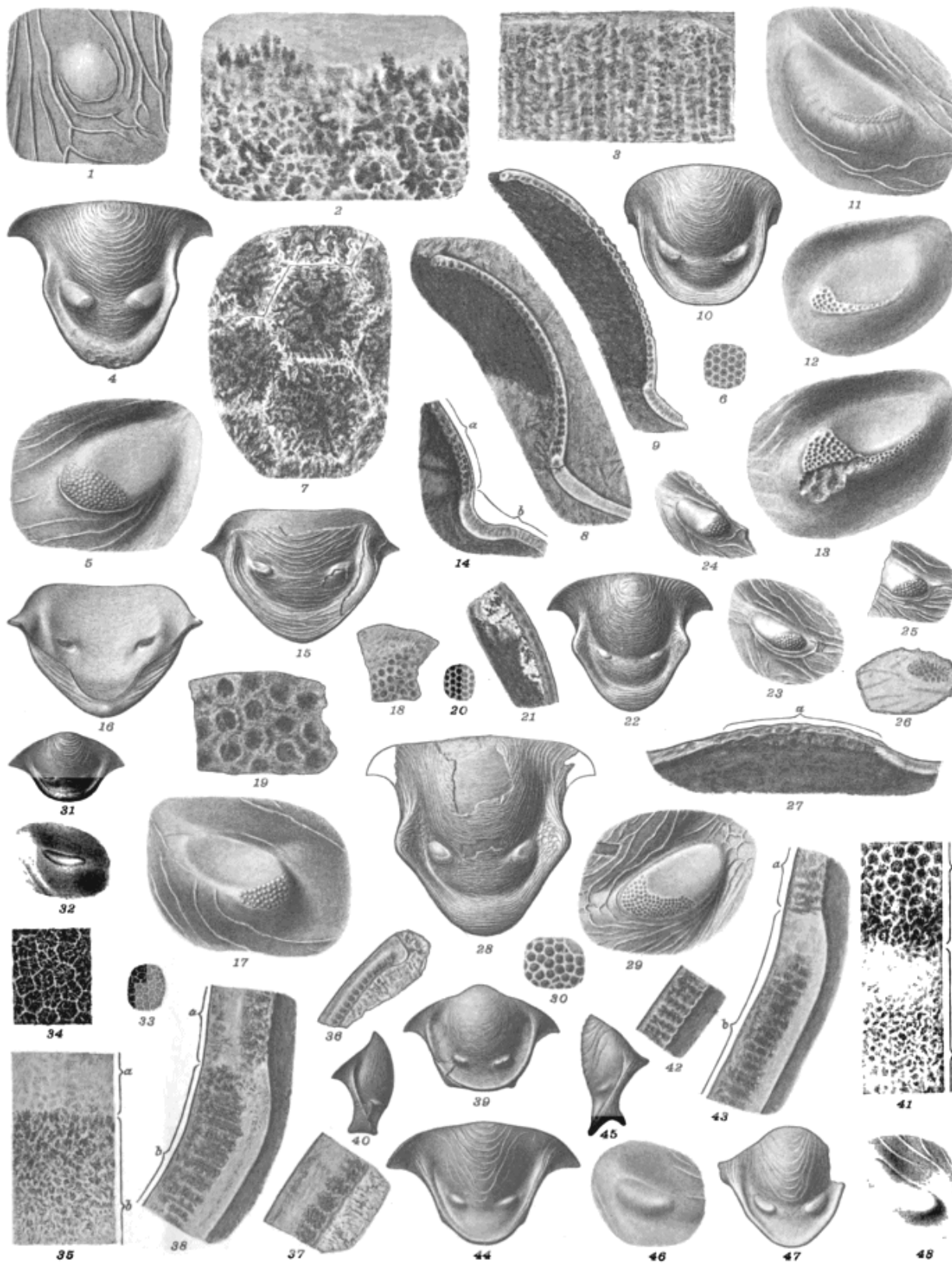
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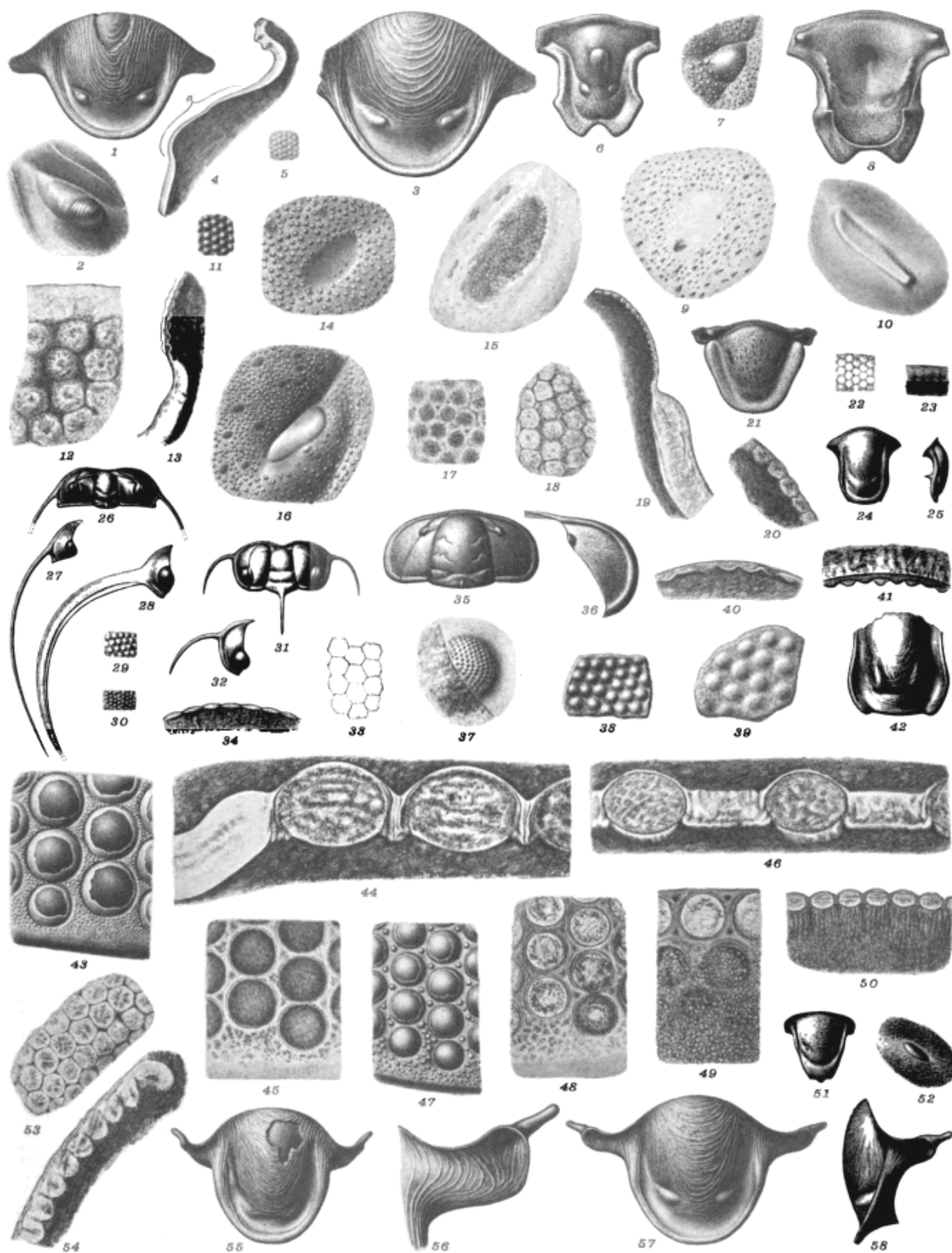
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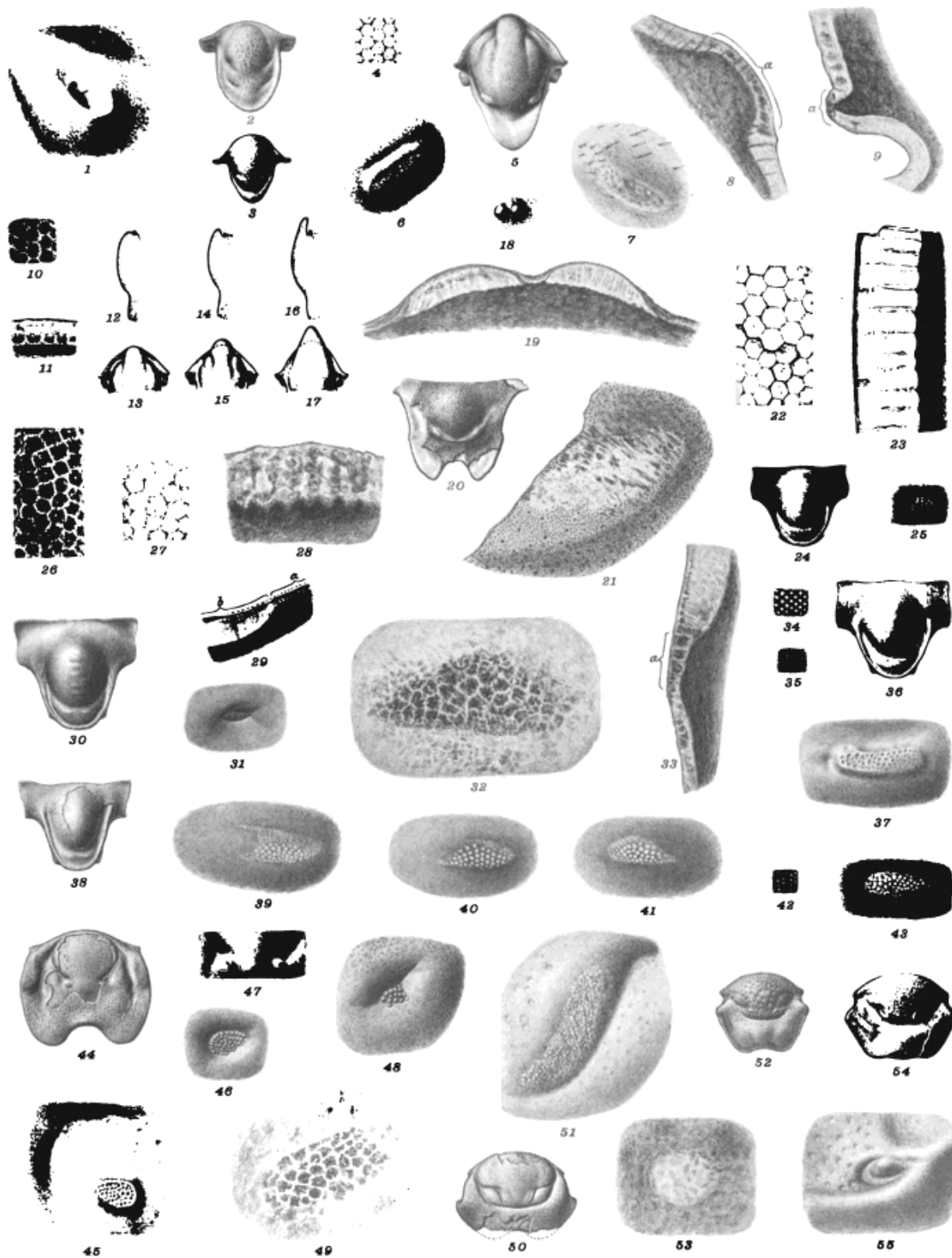
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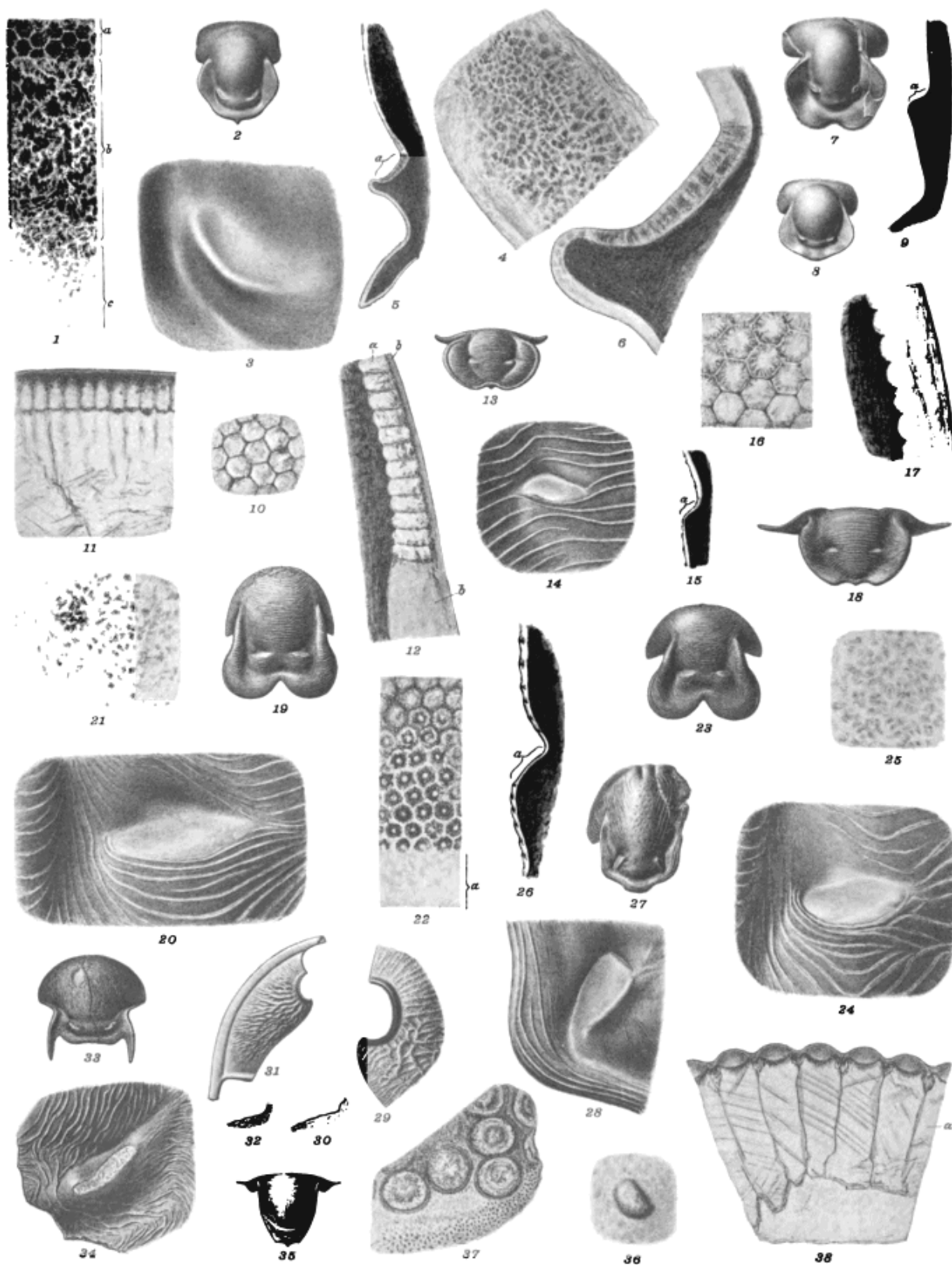
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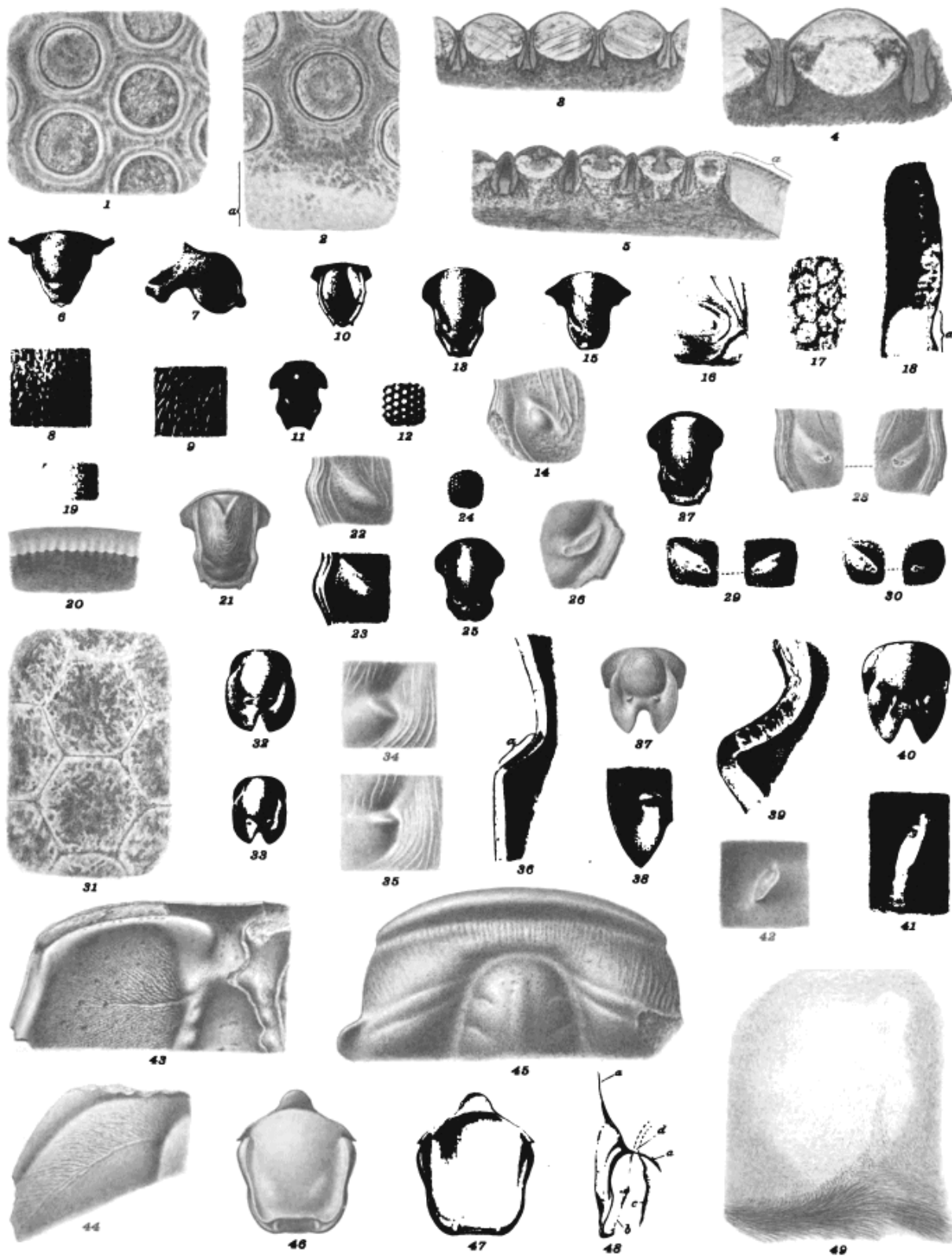
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